Assetto Corsa EVO

Game Version : **0.7.1**
Assets Type : **Cars**
Subtype : **Car Physics Assets**

1. Brake System [.brakesystem]

A. Description

I. General Description

The BrakeSystem asset is the core configuration file that defines the entire physical, mechanical, and electronic behavior of a vehicle's braking system within the simulation engine.

Rather than just setting a raw stopping power value, this asset acts as a central hub. It governs how braking force is distributed between the front and rear axles (static and dynamic bias), how secondary systems like the handbrake behave, and how advanced modern electronic driving aids —such as Electronic Brake Balance (EBB) and brake-based torque vectoring systems—alter braking pressure in real time based on telemetry.

II. Area of Influence / Impact on Vehicle Dynamics

The parameters configured in this asset directly impact how the vehicle handles during one of the most critical phases of driving:

- **Stopping Distance (Braking Power):** Dictated by the total available torque, defining how quickly the vehicle can decelerate.
- **Braking Stability (Brake Bias):** Controls the balance of force. Too much front bias leads to understeer or front-wheel lockup; too much rear bias causes instability, potentially throwing the car into a spin when braking heavily into a corner.
- **Cornering Agility & Trajectory Control:** Through dynamic controllers, the system can modulate brake pressure on individual wheels or axles based on steering angle, lateral G-forces, or wheel slip, acting like a performance Stability Control (ESP) or electronic torque vectoring system.
- **Energy Recovery Systems (Hybrid/EV Integration):** Variables linked to the ERS (Energy Recovery System) interact with regenerative braking, managing how hydraulic pressure scales alongside electric motor resistance (ErsCoastTorque) and battery state-of-charge.

III. Key Architecture & Data Fields Explained

The schema is organized into three main categories: **Mechanical Constants**, **Electronic Subsystems**, and **Algorithmic Logic (Stages)**.

1 - BASE MECHANICAL & PHYSICAL PARAMETERS

- **Total Torque:** The absolute maximum braking torque (usually in *Nm*) applied across the entire vehicle when the brake pedal is depressed 100%.
- **Front Bias:** The baseline, static percentage of braking force sent to the front axle (*e.g. 0.60 = 60% front, 40% rear*).
- **Hand Brake Torque:** Dedicated torque value applied exclusively by the handbrake (typically impacting only the rear wheels).

- **Has Cockpit Bias & Bias Step:** Determines whether the driver can adjust the brake balance on the fly from inside the cockpit, and by what percentage increment each button press alters the bias (*e.g. 0.5%*).

- **Front/Rear Compound Path:** File paths linking the brakes to external thermal and friction physics models (governing brake wear, fade, and optimal operating temperature windows).

2 - DYNAMIC CONTROL UNITS (EBB & STEER BRAKE)

Advanced vehicles use automated controllers to alter braking performance based on real-time physics data:

1. **Controller EBB / Controllers EBB:** Electronic Brake Balance controllers. They continuously shift the brake balance forward or backward depending on live driving conditions.

2. **Steer Brake Controller:** An agile-handling system that applies micro-braking forces to specific wheels based on steering inputs (SteerDEG) or yaw rates (SteerYawDelta). It mimics modern electronic differentials or stability management.

3. **Torque Controller EBB:** Dynamically scales total braking torque output to prevent global lockups or optimize regenerative braking efficiency.

3 - CONTROL STAGES & ALGORITHMIC LOGIC

Every dynamic controller processes data through one or multiple **Stages**. Each stage functions as an automated logical loop:

- **Input Var (Input Variable):** The live telemetry channel the game reads (e.g., Brake pedal pressure, wheel SlipRatio, lateral forces LatG, or suspension/weight transfer data like LoadSpreadLF).

- **LUT (Look-Up Table):** A path to a external .curve file. This data curve acts as an instruction set: *"If the input variable is X, apply a modifier value of Y."*

- **Combinator Mode:** Dictates how this specific stage interacts with others in the pipeline (Add to sum the values together, or Mult to multiply them).

- **Filter Gain:** A smoothing coefficient that dampens rapid telemetry spikes, preventing jittery or unnaturally sudden braking corrections.

- **Up / Down Limit:** Standard clamping values ensuring the controller output stays within safe, physically realistic boundaries.

4 - EBB MODES

ebbDisabled: Electronic brake distribution is off. The car relies entirely on its static physical brake bias.

ebbInternal: Uses hardcoded engine-level logic to distribute braking.

ebbDynamicControllerAbsolute / Relative: Fully activates the custom look-up tables and stages defined in the asset to modulate balance absolutely or relative to the base setup.

B. Schema

```
├ 1. Total Torque : float
├ 2. Front Bias : float
├ 3. Hand Brake Torque : float
├ 4. Has Cockpit Bias : boolean
├ 5. Bias Step : float
├ 6. Front Compound Path [x] : float | can have multiple Front Compound
Path
├ 7. Rear Compound path [x] : float | can have multiple Rear Compound
Path
├ 8. Controller EBB : object with an array of stages within
│  ├ 8a. Name : string
│  ├ 8b. Stages [x] : object | Controller EBB can have multiple stages
│  │  ├ 8b1. Input Var : enum
│  │  ├ 8b2. Combinator Mode : enum
│  │  ├ 8b3. Lut : string – path
│  │  ├ 8b4. Filter Gain : float
│  │  ├ 8b5. Up Limit : float
│  │  ├ 8b6. Down Limit : float
│  │  ├ 8b7. Current Value : float
│  └ └ 8b8. Const Value : float
├ 9. Controllers EBB [x] : object with an array of stages within
│  ├ 8a. Name : string
│  ├ 8b. Stages [x] : object | Controllers EBB [x] can have multiple
stages
│  │  ├ 8b1. Input Var : enum
│  │  ├ 8b2. Combinator Mode : enum
│  │  ├ 8b3. Lut : string – path
│  │  ├ 8b4. Filter Gain : float
│  │  ├ 8b5. Up Limit : float
│  │  ├ 8b6. Down Limit : float
│  │  ├ 8b7. Current Value : float
│  └ └ 8b8. Const Value : float
├ 10. Steer Brake Controller : object with an array of stages within
│  ├ 8a. Name : string
│  ├ 8b. Stages [x] : object | Steer Brake Controller can have multiple
stages
│  │  ├ 8b1. Input Var : enum
│  │  ├ 8b2. Combinator Mode : enum
│  │  ├ 8b3. Lut : string – path
│  │  ├ 8b4. Filter Gain : float
│  │  ├ 8b5. Up Limit : float
│  │  ├ 8b6. Down Limit : float
│  │  ├ 8b7. Current Value : float
│  └ └ 8b8. Const Value : float
├ 11. Troque Controller EBB : object with an array of stages within
│  ├ 8a. Name : string
│  ├ 8b. Stages [x] : object | Torque Controller EBB can have multiple
stages
│  │  ├ 8b1. Input Var : enum
│  │  ├ 8b2. Combinator Mode : enum
│  │  ├ 8b3. Lut : string – path
```

```
│    │  ├ 8b4. Filter Gain : float
│    │  ├ 8b5. Up Limit : float
│    │  ├ 8b6. Down Limit : float
│    │  ├ 8b7. Current Value : float
│    └  └ 8b8. Const Value : float
├ 12. EBB Mode : enum
├ 13. EBB Front Multiplier : float
└ 14. EBB Min Speed : float
```

Descriptions - Brake System

Key	Description
1	The absolute maximum braking torque applied across the whole vehicle at 100% pedal pressure.
2	The static baseline percentage of braking force assigned to the front axle
3	The dedicated torque capacity assigned exclusively to the handbrake system
4	Boolean flag determining if the driver can manually shift the brake balance from the cockpit during gameplay.
5	The percentage increment by which the brake bias changes per adjustment button press
6	File path structure linking front wheels to specific physical hardware assets (e.g., .BRAKES files)
7	File path structure linking rear wheels to specific physical hardware assets (e.g., .BRAKES files).
8	
8a	Unique identifier string for the specific controller instance
8b	An array of sequential evaluation steps containing individual logic filters within a controller.
8b1	The specific real-time telemetry channel polled as the source signal for the stage (e.g., BRAKE, LATG, SLIPRATIOFRONTMAX, ERSCHARGELEVEL).
8b2	Dictates mathematical interaction with neighboring stages—either summing (ADD) or multiplying (MULT) outputs
8b3	File path to the custom data curve determining the specific input/output ratio mappings
8b4	A smoothing value used to responsive-dampen raw input signals and avoid erratic controller corrections
8b5	The maximum upper ceiling clamping limit allowed for the calculated stage value
8b6	The minimum lower floor clamping limit allowed for the calculated stage value
8b7	Live monitoring variable showcasing the real-time calculated value during calculations
8b8	A baseline fallback mathematical constant utilized if no external curve mapping evaluates

Key	Description
9	
10	
11	
12	State enum determining if the system is off (ebbDisabled), hardcoded (ebbInternal), or running curves absolutely or relatively
13	Scaling factor adjusting front-end braking bias responsiveness when EBB calculations are live
14	Minimum vehicle velocity threshold required before the electronic brake balance calculations become active.

Enum List - Brake System

Enum	Values
Input Var	UndefinedInput, Brake, Gas, LatG, LonG, Steer, Speed, Gear, SlipRatioFrontAVG, SlipRatioRearAVG, SlipRatioFrontMAX, SlipRatioRearMAX, SlipAngleFrontAVG, SlipAngleRearAVG, SlipAngleFrontMAX, SlipAngleRearMAX, OversteerFactor, RearSpeedRatio, SteerDEG, Const, RPMS, WheelSteerDEG, LoadSpreadLF, LoadSpreadRF, AvgTravelRear, SusTravelLR, SusTravelRR, SteerYawDeltaLeft, SteerYawDeltaRight, ErsChargeLevel, ErsCoastTorque
CombinatorMode	UndefinedMode, Add, Mult
EBB Mode	ebbDisabled, ebbInternal, ebbDynamicControllerAbsolute, ebbDynamicControllerRelative

C. Example data

I. Chosen Cars for Example

- Alfa Romeo Giulia GTAm (slug : ks_alfa_romeo_giulia_gtam)
- Lancia Delta HF Integrale EVO II (slug : ks_lancia_delta_hf_integrale_evo_ii)
- Ferrari 296 GT3 (slug : ks_ferrari_296_gt3)
- Ferrari SF25 (slug : ks_ferrari_sf_25)

II. Example

Alfa Romeo Giulia GTAm

```
├ 1. Total Torque : 4100.00000
├ 2. Front Bias : 0.75000
```

```
├ 3. Hand Brake Torque : 200.00000
├ 4. Has Cockpit Bias : false
├ 5. Bias Step : 0.50000
├ 6. Front Compound Path : None
├ 7. Rear Compound path : None
├ 8. Controller EBB
│ ├ 8a. Name : None
│ ├ 8b. Stages 1
│ │ ├ 8b1. Input Var : LoadSpreadLF
│ │ ├ 8b2. Combinator Mode : Add
│ │ ├ 8b3. Lut :
content\cars\ks_alfa_romeo_giulia_gtam\data\ctrl_ebbCONTROLLER_0.curve
│ │ ├ 8b4. Filter Gain : 0.01000
│ │ ├ 8b5. Up Limit : 1.00000
│ │ ├ 8b6. Down Limit : 0.00000
│ │ ├ 8b7. Current Value : 0.00000
│ │ └ 8b8. Const Value : 0.00000
│ ├ 8b. Stages 2
│ │ ├ 8b1. Input Var : SlipAngleRearMAX
│ │ ├ 8b2. Combinator Mode : Mult
│ │ ├ 8b3. Lut :
content\cars\ks_alfa_romeo_giulia_gtam\data\ctrl_ebbCONTROLLER_1.curve
│ │ ├ 8b4. Filter Gain : 0.95000
│ │ ├ 8b5. Up Limit : 1.20000
│ │ ├ 8b6. Down Limit : 0.00000
│ │ ├ 8b7. Current Value : 0.00000
│ │ └ 8b8. Const Value : 0.00000
│ ├ 8b. Stages 3
│ │ ├ 8b1. Input Var : Brake
│ │ ├ 8b2. Combinator Mode : Mult
│ │ ├ 8b3. Lut :
content\cars\ks_alfa_romeo_giulia_gtam\data\ctrl_ebbCONTROLLER_2.curve
│ │ ├ 8b4. Filter Gain : 0.95000
│ │ ├ 8b5. Up Limit : 1.20000
│ │ ├ 8b6. Down Limit : 0.00000
│ │ ├ 8b7. Current Value : 0.00000
│ └ └ 8b8. Const Value : 0.00000
├ 9. Controllers EBB [x] : None
├ 10. Steer Brake Controller : None
├ 11. Troque Controller EBB : None
├ 12. EBB Mode : ebbDisabled
├ 13. EBB Front Multiplier  1.10000
└ 14. EBB Min Speed : 0.00000
```

Lancia Delta HF Integrale EVO II (slug : ks_lancia_delta_hf_integrale_evo_ii)

```
├ 1. Total Torque : 2800.00000
├ 2. Front Bias : 0.78000
├ 3. Hand Brake Torque : 1300.00000
├ 4. Has Cockpit Bias : false
├ 5. Bias Step : 0.00000
├ 6. Front Compound Path : None
├ 7. Rear Compound path : None
├ 8. Controller EBB
```

```
│ │ ├ 8a. Name : None
│ │ ├ 8b. Stages 1
│ │ │ ├ 8b1. Input Var : LoadSpreadLF
│ │ │ ├ 8b2. Combinator Mode : Add
│ │ │ ├ 8b3. Lut :
content\cars\ks_lancia_delta_hf_integrale_evo_ii\data\ctrl_ebbCONTROLLER
_0.curve
│ │ │ ├ 8b4. Filter Gain : 0.01000
│ │ │ ├ 8b5. Up Limit : 1.00000
│ │ │ ├ 8b6. Down Limit : 0.00000
│ │ │ ├ 8b7. Current Value : 0.00000
│ │ │ └ 8b8. Const Value : 0.00000
│ │ ├ 8b. Stages 2
│ │ │ ├ 8b1. Input Var : Brake
│ │ │ ├ 8b2. Combinator Mode : Mult
│ │ │ ├ 8b3. Lut :
content\cars\ks_lancia_delta_hf_integrale_evo_ii\data\ctrl_ebbCONTROLLER
_1.curve
│ │ │ ├ 8b4. Filter Gain : 0.95000
│ │ │ ├ 8b5. Up Limit : 1.20000
│ │ │ ├ 8b6. Down Limit : 0.00000
│ │ │ ├ 8b7. Current Value : 0.00000
│ │ │ └ 8b8. Const Value : 0.00000
├ 9. Controllers EBB [x] : None
├ 10. Steer Brake Controller : None
├ 11. Troque Controller EBB : None
├ 12. EBB Mode : ebbDisabled
├ 13. EBB Front Multiplier  1.20000
└ 14. EBB Min Speed : 0.00000
```

Ferrari 296 GT3

```
├ 1. Total Torque : 4300.00000
├ 2. Front Bias : 65.00000
├ 3. Hand Brake Torque : 200.00000
├ 4. Has Cockpit Bias : true
├ 5. Bias Step : 0.20000
├ 6. Front Compound Path 1 :
content\cars\common_phsx\brakes\racing\racing_GT3_pad2.brakes
├ 7. Rear Compound path 2 :
content\cars\common_phsx\brakes\racing\racing_GT3_pad2.brakes
├ 8. Controllers EBB [x] : None
├ 9. Controller EBB : None
├ 10. Steer Brake Controller : None
├ 11. Troque Controller EBB : None
├ 12. EBB Mode : ebbDisabled
├ 13. EBB Front Multiplier  0.00000
└ 14. EBB Min Speed : 0.00000
```

Ferrari SF25

```
├ 1. Total Torque : 5500.00000
├ 2. Front Bias : 54.00000
```

```
├ 3. Hand Brake Torque : 0.00000
├ 4. Has Cockpit Bias : true
├ 5. Bias Step : 0.10000
├ 6. Front Compound Path : None
├ 7. Rear Compound path : None
├ 8. Controller EBB : None
├ 9. Controllers EBB 1
│ ├ 8a. Name : Mig0
│ ├ 8b. Stages 1
│ │ ├ 8b1. Input Var : Brake
│ │ ├ 8b2. Combinator Mode : Add
│ │ ├ 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_0.c
urve
│ │ ├ 8b4. Filter Gain : 0.90000
│ │ ├ 8b5. Up Limit : 10.00000
│ │ ├ 8b6. Down Limit : 0.00000
│ │ ├ 8b7. Current Value : 0.00000
│ │ └ 8b8. Const Value : 0.00000
│ ├ 8b. Stages 2
│ │ ├ 8b1. Input Var : ErsCoastTorque
│ │ ├ 8b2. Combinator Mode : Add
│ │ ├ 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve
│ │ ├ 8b4. Filter Gain : 0.90000
│ │ ├ 8b5. Up Limit : 10.00000
│ │ ├ 8b6. Down Limit : 0.00000
│ │ ├ 8b7. Current Value : 0.00000
│ └ └ 8b8. Const Value : 0.00000
├ 8. Controllers EBB 2
│ ├ 8a. Name : Mig2
│ ├ 8b. Stages 1
│ │ ├ 8b1. Input Var : Brake
│ │ ├ 8b2. Combinator Mode : Add
│ │ ├ 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_2.c
urve
│ │ ├ 8b4. Filter Gain : 0.90000
│ │ ├ 8b5. Up Limit : 10.00000
│ │ ├ 8b6. Down Limit : 0.00000
│ │ ├ 8b7. Current Value : 0.00000
│ │ └ 8b8. Const Value : 0.00000
│ ├ 8b. Stages 2
│ │ ├ 8b1. Input Var : ErsCoastTorque
│ │ ├ 8b2. Combinator Mode : Add
│ │ ├ 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve
│ │ ├ 8b4. Filter Gain : 0.90000
│ │ ├ 8b5. Up Limit : 10.00000
│ │ ├ 8b6. Down Limit : 0.00000
│ │ ├ 8b7. Current Value : 0.00000
│ └ └ 8b8. Const Value : 0.00000
├ 8. Controllers EBB 3
│ ├ 8a. Name : Mig4
```

```
│ ├ 8b. Stages 1
│ │ ├ 8b1. Input Var : Brake
│ │ ├ 8b2. Combinator Mode : Add
│ │ ├ 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_4.c
urve
│ │ ├ 8b4. Filter Gain : 0.90000
│ │ ├ 8b5. Up Limit : 10.00000
│ │ ├ 8b6. Down Limit : 0.00000
│ │ ├ 8b7. Current Value : 0.00000
│ │ └ 8b8. Const Value : 0.00000
│ ├ 8b. Stages 2
│ │ ├ 8b1. Input Var : ErsCoastTorque
│ │ ├ 8b2. Combinator Mode : Add
│ │ ├ 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve
│ │ ├ 8b4. Filter Gain : 0.90000
│ │ ├ 8b5. Up Limit : 10.00000
│ │ ├ 8b6. Down Limit : 0.00000
│ │ ├ 8b7. Current Value : 0.00000
│ └ └ 8b8. Const Value : 0.00000
├ 8. Controllers EBB 4
│ ├ 8a. Name : Mig6
│ ├ 8b. Stages 1
│ │ ├ 8b1. Input Var : Brake
│ │ ├ 8b2. Combinator Mode : Add
│ │ ├ 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_6.c
urve
│ │ ├ 8b4. Filter Gain : 0.90000
│ │ ├ 8b5. Up Limit : 10.00000
│ │ ├ 8b6. Down Limit : 0.00000
│ │ ├ 8b7. Current Value : 0.00000
│ │ └ 8b8. Const Value : 0.00000
│ ├ 8b. Stages 2
│ │ ├ 8b1. Input Var : ErsCoastTorque
│ │ ├ 8b2. Combinator Mode : Add
│ │ ├ 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve
│ │ ├ 8b4. Filter Gain : 0.90000
│ │ ├ 8b5. Up Limit : 10.00000
│ │ ├ 8b6. Down Limit : 0.00000
│ │ ├ 8b7. Current Value : 0.00000
│ └ └ 8b8. Const Value : 0.00000
├ 8. Controllers EBB 5
│ ├ 8a. Name : Mig8
│ ├ 8b. Stages 1
│ │ ├ 8b1. Input Var : Brake
│ │ ├ 8b2. Combinator Mode : Add
│ │ ├ 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_8.c
urve
│ │ ├ 8b4. Filter Gain : 0.90000
│ │ ├ 8b5. Up Limit : 10.00000
```

```
│  │  ├ 8b6. Down Limit : 0.00000
│  │  ├ 8b7. Current Value : 0.00000
│  │  └ 8b8. Const Value : 0.00000
│  ├ 8b. Stages 2
│  │  ├ 8b1. Input Var : ErsCoastTorque
│  │  ├ 8b2. Combinator Mode : Add
│  │  ├ 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve
│  │  ├ 8b4. Filter Gain : 0.90000
│  │  ├ 8b5. Up Limit : 10.00000
│  │  ├ 8b6. Down Limit : 0.00000
│  │  ├ 8b7. Current Value : 0.00000
│  └ └ 8b8. Const Value : 0.00000
├ 10. Steer Brake Controller : None
├ 11. Troque Controller EBB : None
├ 12. EBB Mode : ebbDynamicControllerRelative
├ 13. EBB Front Multiplier  0.00000
└ 14. EBB Min Speed : 0.00000
```

2. Brakes [.brakes]

A. Description

I. General Description

The **Brakes** asset defines the microscopic physical properties, thermal dynamics, and wear characteristics of the individual hardware components (disks/rotors and pads) on a specific wheel or axle.

While the previous BrakeSystem asset manages macroscopic commands, bias, and electronics, the Brakes asset handles the actual physics of friction and heat. It simulates how brake pads grab the disks, how heat builds up during intense deceleration, how ambient airflow and rain cool the components down, and how the braking performance degrades due to overheating (fade) or physical wear over time.

II. Area of Influence / Impact on Vehicle Dynamics

The parameters in this file dictate the endurance, consistency, and tactile feel of the brakes under racing conditions:

- **Brake Fade & Optimal Temperature Window:** Defines how temperature fluctuations change the friction coefficient. Brakes that are too cold or too hot will lose stopping power, drastically altering stopping distances.
- **Thermal Management & Cooling Strategy:** Controls how fast the brakes heat up and shed energy. This directly influences how hard a driver can push lap after lap, and dictates if cooling ducts need adjustment.
- **Brake Wear & Lifespan:** Determines the consumption rate of pads and disks, which is vital for endurance racing strategy where brake swaps might be necessary.
- **Weather Adaptability:** Governs how environmental factors like rain disrupt both the cooling rate and the friction surface.

III. Key Architecture & Data Fields Explained

The data in this schema is structured into three fundamental categories: **Thermal Modeling**, **Wear & Dimensions**, and the **Friction Performance Curve**.

1 - <u>THERMAL MODELING & HEAT DISSIPATION</u>

- **Cool Transfer & Cool Speed Factor:** Base coefficients for convective cooling. Cool Transfer sets the ambient cooling rate, while Cool Speed Factor dictates how much faster the brakes cool as vehicle velocity increases (forcing more air through the brake ducts).
- **Rain Cool Factor:** Modifies the cooling rate when the track is wet, as water spraying onto the brakes drastically increases heat dissipation.

- **Emissivity:** The radiant heat efficiency of the brake material, simulating how much heat energy is emitted as infrared radiation (crucial at glowing, high-temperature states).
- **Surface:** The physical surface area of the brake assembly exposed to airflow.
- **Thermal Capacity & Core Thermal Capacity:** The amount of heat energy required to raise the temperature of the brake surface and the internal "core" mass. Higher capacity means the brakes heat up more slowly but take longer to cool down.
- **Thermal Conductivity & Conduction Thickness:** Controls the rate of heat transfer from the friction surface into the core of the brake disk.

2 - <u>DIMENSIONS, WEAR, AND DEGRADATION (M M = PER MILLIMETER)</u>

- **Disk / Pad Thickness:** The initial, brand-new thickness of the brake disk and pads (typically in millimeters).
- **Disk / Pad Consumption Rate:** The speed at which material is worn away relative to friction and heat cycles.
- **T Reference Wear:** The reference temperature at which normal wear calculations occur. Exceeding this temperature usually spikes the wear exponentially.
- **Perf Decrease M M:** The percentage drop in braking performance or friction for every millimeter of pad/disk material lost to wear.
- **Mu Reduction M M:** The literal decrease of the friction coefficient (μ) as the pads and disks get thinner.
- **Area Reduction M M:** The reduction of effective pad contact surface area as the components wear down.
- **Gamma Correction M M:** A scaling factor that non-linearly adjusts how wear alters the brake feel and performance curves over time.

3 - <u>PERF CURVE (FRICTION COEFFICIENT VS. TEMPERATURE)</u>

This is a look-up table mapping Temperature (X-axis in °C) to Performance Efficiency (Y-axis where 1.000 = 100% target friction). It dictates the "sweet spot" of the brake compound.

IV. Interpretation of Asset Implementation & Data Profiles

Depending on the vehicle category and simulation depth, the deployment of .BRAKES files generally follows one of two implementation profiles:

PROFILE A: SPLIT FRONT/REAR AXLE CONFIGURATIONS (E.G., VINTAGE/ ROAD CARS)

In setups where front and rear hardware are separated into distinct files, the simulation treats the axles as independent thermal entities.

- **Axle Balancing:** Developers can mirror the properties exactly (as seen in some classic vehicles where front and rear drums or basic disks share structural dimensions) or modify them to reflect different rotor sizes.

- **Thermal Synchronization:** Separate files allow the front brakes to face realistic, intense thermal loading (due to forward weight transfer) while the rear brakes are modeled with lower heat accumulation or distinct cooling properties.

PROFILE B: SINGLE SHARED COMPOUND PROFILES & ENDURANCE PADS (E.G., RACING GT3)

For modern racing environments—often denoted by specific naming conventions like _PAD1, _PAD2 (representing alternative endurance, sprint, or wet-weather friction compounds)—the simulation frequently utilizes a single master .BRAKES asset applied globally across the vehicle.

- **Compound Swaps:** In this scenario, switching files changes the fundamental chemistry of the pads vehicle-wide. For example, a _PAD2 compound might represent a long-distance endurance pad characterized by lower consumption rates (Pad Consumption Rate) and a wider, flatter thermal efficiency window, albeit at the cost of a slightly lower maximum friction coefficient (μ).

- **Unified Thermal Performance:** A shared file applies identical material limits (such as T Reference Wear and Thermal Capacity) globally, leaving the volumetric difference in heat handling to be modulated purely by the car's distinct front/rear brake duct settings.

READING THE PERFORMANCE CURVE TREND

The performance curve look-up table translates directly to driver feel and pedal consistency on track:

- **The Bite Threshold (0°C - 200°C):** A high efficiency value (*0.90+*) at low temperatures indicates a street-friendly or versatile compound that works instantly without requiring a warm-up phase. Modern carbon-ceramic or pure racing pads will often show a massive dip here (0.40 - 0.60), requiring aggressive warming tactics.

- **The Sweet Spot (300°C - 600°C):** The plateau where the curve hits its absolute maximum (*1.000*). A wider plateau indicates an easy-to-manage brake system that provides a consistent stopping distance over a broad operational window.

- **Thermal Fade Zone (700°C+):** The rate at which the curve drops at extreme temperatures defines the severity of the brake fade. A gradual taper allows the driver to feel the car losing stopping power safely, while a sharp drop-off causes catastrophic brake failure under heavy racing conditions.

B. Schema

```
├ 1. Cool Transfer : float
├ 2. Torque K : float
├ 3. Cool Speed Factor : float
├ 4. Rain Cool Factor : float
├ 5. Emissivity : float
```

```
├ 6. Surface : float
├ 7. Thermal Capacity : float
├ 8. Core Thermal Capacity : float
├ 9. Thermal Conductivity : float
├ 10. Conduction Thickness : float
├ 11. Disk Consumption Rate : float
├ 12. Pad Consumption Rate : float
├ 13. Disk Thickness : float
├ 14. Pad Thickness : float
├ 15. Perf Decrease M M : float
├ 16. Gamma Correction M M : float
├ 17. Mu Reduction M M : float
├ 18. Area Reduction M M : float
├ 19. T Reference Wear : float
└ 20. Perf Curve : string – path
```

Descriptions - Brakes

Key	Description
1	Base ambient cooling rate; how fast the brakes shed heat to the surrounding air when the car is stationary
2	Mechanical torque conversion factor; scales how efficiently hydraulic pressure translates into actual stopping torque
3	Airflow cooling multiplier; dictates how much faster the brakes cool down as the vehicle's speed increases
4	Wet-weather cooling modifier; increases the heat dissipation rate to simulate rain and water spray hitting the brakes
5	Thermal radiation efficiency; how effectively the brake material radiates infrared heat, especially when glowing hot
6	Total exposed surface area of the brake assembly; larger surfaces naturally dissipate heat faster
7	Heat storage of the surface; how much energy is needed to raise the temperature of the outer friction layer
8	Heat storage of the core; how much energy the interior mass of the brake disc can absorb
9	Internal heat transfer rate; how fast heat moves from the hot friction surface into the cooler core.
10	Physical distance/depth for internal heat transfer between the surface layer and the core
11	Wear rate of the brake disc/rotor relative to temperature and friction cycles
12	Wear rate of the brake pads relative to temperature and friction cycles
13	Initial, brand-new thickness of the brake disc (usually in millimeters)
14	Initial, brand-new thickness of the brake pads (usually in millimeters)
15	Global percentage drop in overall braking performance for every millimeter of material lost
16	Non-linear scaling factor adjusting how wear progressively alters the pedal feel and friction behavior over time

Key	Description
17	Literal reduction value applied to the friction coefficient (μ) for every millimeter of pad/disk wear
18	Loss of effective pad contact surface area as the components wear down and warp/taper
19	Baseline temperature threshold; exceeding this thermal limit causes brake wear to accelerate exponentially
20	File path to the look-up table curve mapping brake temperature (X-axis) to friction efficiency (Y-axis)

C. Example data

I. Chosen Brakes for Example

- Vintage Road Front (slug : vintage_road_front) / Vintage Road Rear (vintage_road_rear)
- Racing GT3 [Pad 2] (slug : racing_gt3_pad2)

II. Example

Vintage Road [Front]

```
├ 1. Cool Transfer : 1.30000
├ 2. Torque K : 0.70000
├ 3. Cool Speed Factor : 1.50000
├ 4. Rain Cool Factor : 0.80000
├ 5. Emissivity : 0.70000
├ 6. Surface : 0.20000
├ 7. Thermal Capacity : 100.00000
├ 8. Core Thermal Capacity : 1600.00000
├ 9. Thermal Conductivity : 250.00000
├ 10. Conduction Thickness : 0.00500
├ 11. Disk Consumption Rate : 0.18000
├ 12. Pad Consumption Rate : 0.19800
├ 13. Disk Thickness : 32.00000
├ 14. Pad Thickness : 29.00000
├ 15. Perf Decrease M M : 15.00000
├ 16. Gamma Correction M M : 1.50000
├ 17. Mu Reduction M M : 0.01000
├ 18. Area Reduction M M : 0.16000
├ 19. T Reference Wear : 600.00000
└ 20. Perf Curve :
content\cars\common_phsx\brakes\vintage\tcurve_vintage_front.curve
```

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P0	0.000	0.900
P1	100.000	0.950
P2	500.000	0.980

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P3	600.000	1.000
P4	800.000	0.980
P5	900.000	0.950
P6	1000.000	0.850
P7	1200.000	0.800

Friction Coefficient Modifier = Performance Efficiency where 1.000 = 100%

Vintage Road [Rear]

```
├ 1. Cool Transfer : 1.30000
├ 2. Torque K : 0.70000
├ 3. Cool Speed Factor : 1.50000
├ 4. Rain Cool Factor : 0.80000
├ 5. Emissivity : 0.70000
├ 6. Surface : 0.20000
├ 7. Thermal Capacity : 100.00000
├ 8. Core Thermal Capacity : 1600.00000
├ 9. Thermal Conductivity : 250.00000
├ 10. Conduction Thickness : 0.00500
├ 11. Disk Consumption Rate : 0.18000
├ 12. Pad Consumption Rate : 0.19800
├ 13. Disk Thickness : 32.00000
├ 14. Pad Thickness : 29.00000
├ 15. Perf Decrease M M : 15.00000
├ 16. Gamma Correction M M : 1.50000
├ 17. Mu Reduction M M : 0.01000
├ 18. Area Reduction M M : 0.16000
├ 19. T Reference Wear : 600.00000
└ 20. Perf Curve :
content\cars\common_phsx\brakes\vintage\tcurve_vintage_rear.curve
```

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P0	0.000	0.900
P1	100.000	0.950
P2	500.000	0.980
P3	600.000	1.000
P4	800.000	0.980
P5	900.000	0.950
P6	1000.000	0.850

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P7	1200.000	0.800

Friction Coefficient Modifier = Performance Efficiency where 1.000 = 100%

Racing GT3 [Pad 2]

```
├ 1. Cool Transfer : 0.90000
├ 2. Torque K : 0.70000
├ 3. Cool Speed Factor : 1.60000
├ 4. Rain Cool Factor : 0.80000
├ 5. Emissivity : 0.70000
├ 6. Surface : 0.40000
├ 7. Thermal Capacity : 100.00000
├ 8. Core Thermal Capacity : 1600.00000
├ 9. Thermal Conductivity : 250.00000
├ 10. Conduction Thickness : 0.00500
├ 11. Disk Consumption Rate : 0.01050
├ 12. Pad Consumption Rate : 0.02100
├ 13. Disk Thickness : 32.00000
├ 14. Pad Thickness : 29.00000
├ 15. Perf Decrease M M : 13.00000
├ 16. Gamma Correction M M : 1.00000
├ 17. Mu Reduction M M : 0.01000
├ 18. Area Reduction M M : 0.08000
├ 19. T Reference Wear : 500.00000
└ 20. Perf Curve :
content\cars\common_phsx\brakes\racing\tcurve_racing_GT3_pad2
```

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P0	-84.050	0.774
P1	-0.905	0.839
P2	39.477	0.914
P3	88.596	0.952
P4	121.867	0.967
P5	156.821	0.974
P6	203.872	0.978
P7	252.036	0.980
P8	303.284	0.980
P9	365.345	0.980
P10	380.803	0.979
P11	443.598	0.977

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P12	526.801	0.973
P13	599.594	0.966
P14	722.684	0.945
P15	781.584	0.936
P16	849.036	0.923
P17	908.993	0.898
P18	972.698	0.854
P19	1047.645	0.734
P20	1245.218	0.567

Friction Coefficient Modifier = Performance Efficiency where 1.000 = 100%

3. Car Data [.car]

A. Description

I. General Description

The **Car Data** asset is the primary foundational master file for a vehicle's physics package in the simulation engine. If BrakeSystem and Brakes represent localized hardware components, Car Data represents the physical carcass, weight distribution, structural constraints, and regulatory rules of the entire vehicle.

It acts as the structural blueprint that holds all other modular components (engine, tires, suspension, and brakes) together by defining how the global mass moves, how the center of gravity shifts, and how the physical dimensions interact with the environment.

II. Area of Influence / Impact on Vehicle Dynamics

The properties configured within the Car Data asset dictate the base handling DNA and fundamental physical limits of the vehicle before any specific aerodynamic downforce or mechanical grip is added:

- **Weight Distribution & Inertia:** Governs the vehicle's resistance to rotational changes (pitch, roll, and yaw). It directly dictates how quick the car changes direction and how much weight transfers under braking, acceleration, or cornering.
- **Dimensional Footprint (Track & Wheelbase):** Determines the baseline mechanical stability. A wider track increases cornering stability, while a longer wheelbase stabilizes the car at high speeds but reduces low-speed agility.
- **Driver Environment & Ergonomics:** Configures the driver's perspective (camera placement, cockpit nodes) and controls the operational boundaries of steering inputs.
- **Fuel Weight Dynamics:** Simulates dynamic mass changes over a race stint. As fuel burns off, the overall vehicle weight drops and the center of gravity shifts, drastically altering handling characteristics over time.

III. Key Architecture & Data Fields Explained

The parameters in this schema are universally categorized into **Mass & Inertia**, **Dimensions & Offsets**, and **Fuel & Regulation Logistics**.

1 - GLOBAL MASS & INERTIA PROPERTIES

- **Total Mass / Dry Weight:** The baseline mass of the vehicle without fuel, driver, or fluids (expressed in kg). This is the fundamental variable for F=ma calculations.
- **Inertia (Pitch, Roll, Yaw Vectors):** Critical tensors (I_x, I_y, I_z) defining how mass is distributed away from the center of gravity. High polar inertia means a car resists spinning but is hard to catch once it breaks traction.

- **Center of Gravity Height (CG Height):** The vertical position of the balance point. Lower CG decreases body roll and load transfer across tires, significantly increasing overall cornering potential.
- **Weight Bias (Front/Rear %):** The static balance of the car at rest (e.g. 45/55 for mid-engine layouts). It underpins the car's natural handling bias (understeer vs. oversteer tendency).

2 - DIMENSIONS, TRACK AND ALIGNEMENT COORDINATES

- **Wheelbase:** The linear distance between the center lines of the front and rear axles.
- **Track Width (Front/Rear):** The lateral distance between the center points of the left and right tires on an axle.
- **Graphics / Collision Mesh Offsets:** Coordinates that align the structural 3D model and physical hitbox with the theoretical center of gravity, ensuring visual impacts and ground scraping clip accurately.
- **Steer Lock / Rack Ratio:** Max steering angle of the front wheels and the corresponding ratio to the steering wheel column. Defines the vehicle's turning circle and quickness of steering inputs.

3 - FUEL MANAGEMENT & CONSUMABLES

- **Fuel Tank Capacity:** The maximum volume of fuel the car can physically hold (typically in Liters or Gallons).
- **Fuel Tank Position (X, Y, Z Coordinates):** The exact physical location of the fuel payload inside the chassis. Crucial for calculating how the center of gravity and weight balance shift as the fuel burns down.
- **Driver Mass Toggle:** Dictates whether the weight of an average driver (e.g., standard regulatory 75 kg - 80 kg) is automatically appended to the physical mass calculations.

IV. Short Interpretation of Asset Implementation

When analyzing a vehicle's physics using the Car Data profile, you can immediately predict its handling characteristics through basic parameter archetypes:

- **The Prototype/Open-Wheel Profile:** Characterized by ultra-low total mass (< 800 kg), a wide track width, and an incredibly low Center of Gravity Height. The inertia tensors are compressed closely to the center, yielding instantaneous directional response.
- **The GT3/Production Profile:** Features higher total mass (1200 kg - 1400 kg), higher CG positions, and a sprawling wheelbase. These vehicles experience heavy weight transfer, demanding careful management of pitch and roll stiffness elsewhere in the setup.
- **The Endurance Strategy Vector:** By looking at the Fuel Tank Capacity paired with its structural 3D coordinates, you can see how heavily a full stint will punish tire wear. If the tank is behind the rear axle, a full fuel load will temporarily shift the weight bias

significantly rearward, introducing entry understeer that slowly fades into snappier oversteer as the tank empties.

B. Schema

```
├ 1. Screen Name : string
├ 2. General : object
│ ├ 2a. Screen Name : string
│ ├ 2b. Total Mass : float
│ ├ 2c. Tank Position : x, y, z float
│ ├ 2d. Fuel : float
│ ├ 2e. Max Fuel : float
│ ├ 2f. Efficiency : float
│ ├ 2g. Kg Per Liter : float
│ ├ 2h. Body Box Sizes : x, y, z float
│ ├ 2i. Pickup Front Height : float
│ ├ 2j. Pickup Rear Height : float
│ ├ 2k. Check Rules : boolean
│ ├ 2l. Minimum Height : float
│ ├ 2m. Torsional Stiffness : float
│ ├ 2n. Torsional Damping : float
│ ├ 2o. Body Mesh Offset : object
│ │ ├ 2o1. Position : x, y, z float
│ │ ├ 2o2. Rotation : x, y, z, float
│ └ └ 2o3. Scale : x, y, z float
├ 3. General Path : string – path
├ 4. Suspensions : object
│ ├ 4a. Wheel Base : float
│ ├ 4b. Longitudinal Cg Location : float
│ ├ 4c. Base Y Front : float
│ ├ 4d. Base Y Rear : float
│ ├ 4e. Track Front : float
│ ├ 4f. Track Rear : float
│ ├ 4g. Damage : object
│ │ ├ 4g1. Min Velocity : float
│ │ ├ 4g2. Gain : float
│ │ ├ 4g3. Max Damage : float
│ │ └ 4g4. Debug Log : boolean
│ ├ 4h. Coilover Front path : string – path
│ ├ 4i. Coilover Rear Path : string – path
│ ├ 4j. Front Suspension Path : string – path
│ ├ 4k. Rear Suspension Path : string – path
│ ├ 4l. Heavy Springs [x] : object | Can have multiple Heavy Springs,
where x can be 1, 2, 3, …
│ │ ├ 4l1. Spring Rate : float
│ │ ├ 4l2. Progressive Sprint Rate : float
│ │ ├ 4l3. Bump Stop Up : object
│ │ │ ├ 4l3a. Range : float
│ │ │ ├ 4l3b. Reference : float
│ │ │ ├ 4l3c. Force : float
│ │ │ ├ 4l3d. Gamma : float
│ │ │ ├ 4l3e. Length : float
│ │ │ └ 4l3f. Damping : float
│ │ ├ 4l4. Bump Stop Down : object
```

```
│ │ │ ├ 4l4a. Range : float
│ │ │ ├ 4l4b. Reference : float
│ │ │ ├ 4l4c. Force : float
│ │ │ ├ 4l4d. Gamma : float
│ │ │ ├ 4l4e. Length : float
│ │ │ └ 4l4f. Damping : float
│ │ ├ 4l5. Collar Position : float
│ │ ├ 4l6. Damper : object
│ │ │ ├ 4l6a. Fast : object
│ │ │ │ ├ 4l6a1. Bump : float
│ │ │ │ └ 4l6a2. Rebound : float
│ │ │ ├ 4l6b. Slow : object
│ │ │ │ ├ 4l6b1. Bump : float
│ │ │ │ └ 4l6b2. Rebound : float
│ │ │ ├ 4l6c. Fast Threshold Bump : float
│ │ │ ├ 4l6d. Fast Threshold Rebound : float
│ │ │ ├ 4l6e. Cooling Surface : float
│ │ │ ├ 4l6f. Nominal Force : float
│ │ │ ├ 4l6g. Min Stress Fatigue : float
│ │ │ ├ 4l6h. Max Stress Fatigue : float
│ │ │ ├ 4l6i. Thermal Capacity : float
│ │ │ ├ 4l6j. Heat Transfer Coefficient : float
│ │ │ ├ 4l6k. Lut List : string – path
│ │ │ └ 4l6l. Damper Lut Scale : string – path
│ │ ├ 4l7. Helper K : float
│ │ ├ 4l8. Helper Range : float
│ │ ├ 4l9. Rod Controllers : object with an array of stages within
│ │ │ ├ 4l9a. Name : string
│ │ │ ├ 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
│ │ │ │ ├ 4l9b1. Input Var : enum
│ │ │ │ ├ 4l9b2. Combinator Mode : enum
│ │ │ │ ├ 4l9b3. Lut : string – path
│ │ │ │ ├ 4l9b4. Filter Gain : float
│ │ │ │ ├ 4l9b5. Up Limit : float
│ │ │ │ ├ 4l9b6. Down Limit : float
│ │ │ │ ├ 4l9b7. Current Value : float
│ │ └ └ └ 4l9b8. Const Value : float
│ ├ 4m. Arb Front : object
│ │ ├ 4m1. Stiffness : float
│ │ ├ 4m2. Controller : object
│ │ │ ├ 4l9a. Name : string
│ │ │ ├ 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
│ │ │ │ ├ 4l9b1. Input Var : enum
│ │ │ │ ├ 4l9b2. Combinator Mode : enum
│ │ │ │ ├ 4l9b3. Lut : string – path
│ │ │ │ ├ 4l9b4. Filter Gain : float
│ │ │ │ ├ 4l9b5. Up Limit : float
│ │ │ │ ├ 4l9b6. Down Limit : float
│ │ │ │ ├ 4l9b7. Current Value : float
│ │ └ └ └ 4l9b8. Const Value : float
│ ├ 4n. Arb Rear : object
│ │ ├ 4n1. Stiffness : float
│ │ ├ 4n2. Controller : object
│ │ │ ├ 4l9a. Name : string
```

```
│ │ │ ├ 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
│ │ │ │ ├ 4l9b1. Input Var : enum
│ │ │ │ ├ 4l9b2. Combinator Mode : enum
│ │ │ │ ├ 4l9b3. Lut : string - path
│ │ │ │ ├ 4l9b4. Filter Gain : float
│ │ │ │ ├ 4l9b5. Up Limit : float
│ │ │ │ ├ 4l9b6. Down Limit : float
│ │ │ │ ├ 4l9b7. Current Value : float
│ │ └ └ └ 4l9b8. Const Value : float
│ ├ 4o. Flex Bar Front : object
│ │ ├ 4o1. Stiffness : float
│ │ ├ 4o2. Controller : object
│ │ │ ├ 4l9a. Name : string
│ │ │ ├ 4l9b. Stages [x] : float | Rod Controllers can have multiple
stages
│ │ │ │ ├ 4l9b1. Input Var : enum
│ │ │ │ ├ 4l9b2. Combinator Mode : enum
│ │ │ │ ├ 4l9b3. Lut : string - path
│ │ │ │ ├ 4l9b4. Filter Gain : float
│ │ │ │ ├ 4l9b5. Up Limit : float
│ │ │ │ ├ 4l9b6. Down Limit : float
│ │ │ │ ├ 4l9b7. Current Value : float
│ │ └ └ └ 4l9b8. Const Value : float
│ ├ 4p. Flex Bar Rear : object
│ │ ├ 4p1. Stiffness : float
│ │ ├ 4p2. Controller : object with an array of stages within
│ │ │ ├ 4l9a. Name : string
│ │ │ ├ 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
│ │ │ │ ├ 4l9b1. Input Var : enum
│ │ │ │ ├ 4l9b2. Combinator Mode : enum
│ │ │ │ ├ 4l9b3. Lut : string - path
│ │ │ │ ├ 4l9b4. Filter Gain : float
│ │ │ │ ├ 4l9b5. Up Limit : float
│ │ │ │ ├ 4l9b6. Down Limit : float
│ │ │ │ ├ 4l9b7. Current Value : float
│ │ └ └ └ 4l9b8. Const Value : float
│ ├ 4q. Dampers Controller : object
│ │ ├ 4q1. Wheel Gain : float
│ │ ├ 4q2. Heave Gain : float
│ │ ├ 4q3. Pitch Gain : float
│ │ ├ 4q4. Roll Gain : float
│ │ ├ 4q5. Front : object
│ │ │ ├ 4q5a. Base : object
│ │ │ │ └ 4q5a1. Type : enum
│ │ │ ├ 4q5b. Heave : object
│ │ │ │ └ 4q5a1. Type : enum
│ │ │ ├ 4q5c. Roll : object
│ │ │ │ └ 4q5a1. Type : enum
│ │ │ ├ 4q5d. Pitch : object
│ │ │ │ └ 4q5a1. Type : enum
│ │ │ ├ 4q5e. Max : object
│ │ │ │ └ 4q5a1. Type : enum
│ │ │ ├ 4q5f. Min : object
│ │ │ └ └ 4q5a1. Type : enum
```

```
│  │  ├ 4q6. Rear : object
│  │  │  ├ 4q5a. Base : object
│  │  │  │  └ 4q5a1. Type : enum
│  │  │  ├ 4q5b. Heave : object
│  │  │  │  └ 4q5a1. Type : enum
│  │  │  ├ 4q5c. Roll : object
│  │  │  │  └ 4q5a1. Type : enum
│  │  │  ├ 4q5d. Pitch : object
│  │  │  │  └ 4q5a1. Type : enum
│  │  │  ├ 4q5e. Max : object
│  │  │  │  └ 4q5a1. Type : enum
│  │  │  ├ 4q5f. Min : object
│  │  │  └ └ 4q5a1. Type : enum
│  └ 4r. Has Dampers Cockpit Settings : boolean
├ 5. Drivetrain Path : string - path
├ 6. Gearbox Path : string - path
├ 7. Clutch Path : string - path
├ 8. Engine Path : string - path
├ 9. Brakes Path : string - path
├ 10. Steering System : object
│  ├ 10a. Four W S Controllers : object with an array of stages within
│  │  ├ 4l9a. Name : string
│  │  ├ 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
│  │  │  ├ 4l9b1. Input Var : enum
│  │  │  ├ 4l9b2. Combinator Mode : enum
│  │  │  ├ 4l9b3. Lut : string - path
│  │  │  ├ 4l9b4. Filter Gain : float
│  │  │  ├ 4l9b5. Up Limit : float
│  │  │  ├ 4l9b6. Down Limit : float
│  │  │  ├ 4l9b7. Current Value : float
│  └ └ └ 4l9b8. Const Value : float
├ 11. Electronics : object
│  ├ 11a. T C : object
│  │  ├ 11a1. Has T C2 : boolean
│  │  ├ 11a2. Frequency Hz : float
│  │  ├ 11a3. Min Speed Kmh : float
│  │  ├ 11a4. Gear Change Time : float
│  │  ├ 11a5. Min Cut Level : float
│  │  ├ 11a6. Max Cut Level : float
│  │  ├ 11a7. Settings [x] : object | T C can have multiple Settings
│  │  │  ├ 11a7a. Min Slip Ratio : float
│  │  │  ├ 11a7b. Max Slip Ratio : float
│  │  │  ├ 11a7c. Ref Slip Angle Deg : float
│  │  │  ├ 11a7d. Engine Cut Level : float
│  │  │  ├ 11a7e. Angular A C Cgain : float
│  │  │  ├ 11a7f. Oversteer Gain : float
│  │  │  ├ 11a7g. Slip Angle Activation Deg : float
│  ├ 11b. A B S : object
│  │  ├ 11b1. Settings [x] : object | A B S can have multiple Settings
│  │  │  ├ 11b1a. Min Slip Ratio : float
│  │  │  ├ 11b1b. Max Slip Ratio : float
│  │  │  ├ 11b1c. Ref Slip Angle Deg : float
│  │  │  ├ 11b1d. Cut Level : float
│  │  │  ├ 11b1f. Max Torque Variation : float
│  │  ├ Frequency : float
```

```
│ │ ├ Channels : integer
│ │ ├ Min Speed Kmh : float
│ ├ 11c. E D L : object
│ │ ├ 11c1. Active : boolean
│ │ ├ 11c2. Brake Torque Power : float
│ │ ├ 11c3. Brake Torque Coast : float
│ │ ├ 11c4. Dead Zone Coast : float
│ │ ├ 11c5. Dead Zone Power : float
│ │ ├ 11c6. Max Spin Power : float
│ │ ├ 11c7. Max Spin Coaster : float
│ │ ├ 11c8. Min Speed : float
│ ├ 11d. E S P : object
│ │ ├ 11d1. Frequency Hz : float
│ │ ├ 11d2. Min Speed Kmh : float
│ │ ├ 11d3. Settings [x] : object | E S P can have multiple settings
│ │ │ ├ 11d3a. Gain : float
│ │ │ ├ 11d3b. Steer Gain : float
│ │ │ ├ 11d3c. Min Steer Gain : float
│ │ │ ├ 11d3d. Steer Gain Max Speed : float
│ │ │ ├ 11d3e. Oversteer Gain : float
│ │ │ ├ 11d3f. Understeer Gain : float
│ │ │ ├ 11d3g. Max Slip Ratio : float
│ │ │ ├ 11d3h. Dead Zone : float
│ │ │ ├ 11d3i. Filter Gain : float
│ │ │ ├ 11d3j. Brake Perc : float
│ └ └ └ 11d3k. Brake Perc Activation : float
├ 12. Electronics Path : string - path
├ 13. Controls : object
│ ├ 13a. Ff Mult : float
│ ├ 13b. Steer Lock : float
│ ├ 13c. Steer Ratio : float
│ ├ 13d. Linear Steer Rod Ratio : float
│ └ 13e. Steer Assist : float
├ 14. Box Colliders [x] : object | can have multiple Box Colliders
│ ├ 14a. Center : x, y, z float
│ ├ 14b. Size : x, y, z float
│ └ 14c. Pitch Rotation Deg : float
├ 15. Front Tyre Compounds [x] : string - path | can have multiple Front
Tyre Compounds path
├ 16. Rear Tyre Compounds [x] : string - path | can have multiple Rear
Tyre Compounds path
├ 17. Aero : object
│ ├ 17a. Slip Gain Multiple : float
│ ├ 17b. Speed Factor Mult : float
│ ├ 17c. Downforces [x] : object | can have multiple Downforces
│ │ ├ 17c1. Position : x, y, z float
│ │ ├ 17c2. Cl Gain : float
│ │ ├ 17c3. Cd Gain : float
│ │ ├ 17c4. Yaw Gain : float
│ │ ├ 17c5. Drag Per Cool Transfer : float
│ │ ├ 17c6. Damage C L [x] : string | can have multiple Damage C L
│ │ ├ 17c7. Damage C D [x] : string | can have multiple Damage C D
│ │ ├ 17c8. Downforce Controllers [x] : object | can have multiple
Downforce Controllers
│ │ │ ├ 17c8a. Combinator Mode : enum
│ │ │ ├ 17c8b. Input : enum
```

```
│  │  │  ├ 17c8c. Filter : float
│  │  │  ├ 17c8d. Up Limit : float
│  │  │  ├ 17c8e. Down Limit : float
│  │  │  └ 17c8f. Lut : string - path
│  │  ├ 17c9. Lift Per Front Angle : float
│  │  ├ 17c10. Lift Per Rear Angle : float
│  │  ├ 17c11. Drag Per Front Angle : float
│  │  ├ 17c12. Drag Per Rear Angle : float
│  │  ├ 17c13. Default Front Angle : float
│  │  └ 17c14. Default Rear Angle : float
│  ├ 17d. Front Lift : string - path
│  ├ 17e. Rear Lift : string - path
│  ├ 17f. Drag : string - path
│  └ 17g. Wings Path [x] : string - path | can have multiple Wings Path
├ 18. Drs : object
│  ├ 18a. Ignore Zones : boolean
│  ├ 18b. Limit G : float
│  ├ 18c. Wing Connections [x] : object | can have multiple Wing
Connections
│  │  ├ 18c1. Mode : enum
│  │  ├ 18c2. Connected Wing : integer
│  │  ├ 18c3. Effect : float
│  │  ├ 18c4. Angle : float
├ 19. Ers : object
│  ├ 19a. Torque Lut : string - path
│  ├ 19b. Coast Lut : string - path
│  ├ 19c. Battery Charge Kj : float
│  ├ 19d. Has Button Override : boolean
│  ├ 19e. Max Kj Per Lap : float
│  ├ 19f. Max Charge Kj Per Lap : float
│  ├ 19g. Heat Charge Perc : float
│  ├ 19h. Heat Power Kw : float
│  ├ 19i. Default Power Controller Index : integer
│  ├ 19j. Power Controllers Front [x] : object | can have multiple Power
Controllers Front
│  │  ├ 4l9a. Name : string
│  │  ├ 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
│  │  │  ├ 4l9b1. Input Var : enum
│  │  │  ├ 4l9b2. Combinator Mode : enum
│  │  │  ├ 4l9b3. Lut : string - path
│  │  │  ├ 4l9b4. Filter Gain : float
│  │  │  ├ 4l9b5. Up Limit : float
│  │  │  ├ 4l9b6. Down Limit : float
│  │  │  ├ 4l9b7. Current Value : float
│  │  └ └ 4l9b8. Const Value : float
│  ├ 19k. Power Controllers Rear [x] : object | can have multiple Power
Controllers Rear
│  │  ├ 4l9a. Name : string
│  │  ├ 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
│  │  │  ├ 4l9b1. Input Var : enum
│  │  │  ├ 4l9b2. Combinator Mode : enum
│  │  │  ├ 4l9b3. Lut : string - path
│  │  │  ├ 4l9b4. Filter Gain : float
│  │  │  ├ 4l9b5. Up Limit : float
```

```
│  │  │  ├ 4l9b6. Down Limit : float
│  │  │  ├ 4l9b7. Current Value : float
│  │  └ └ 4l9b8. Const Value : float
│  ├ 19l. Brake Rear Correction : float
│  ├ 19m. Has Cockpit Controls : boolean
│  ├ 19n. Cockpit Controls : object
│  │  ├ 19n1. Delivery Profile : boolean
│  │  ├ 19n2. Mgu H Mode : boolean
│  │  └ 19n3. Recovery : boolean
│  ├ 19o. Has Front Motors : boolean
│  ├ 19p. Front Motors : object
│  │  ├ 19p1. Torque Lut : string - path
│  │  ├ 19p2 Discharge Time : float
│  │  ├ 19p3. Torque Vectoring Bias : float
├ 20. Setup Limits : string path
├ 21. Collider Mesh : string path
├ 22. Body Mesh Offset : object
│  ├ 22a. Position : x, y, z float
│  ├ 22b. Rotation: x, y, z float
│  └ 22c. Scale : x, y, z float
├ 23. Stock Setup : string path
├ 24. Ai Setup : string path
├ 25. Wet Setup : string path
├ 26. Performance Modes [x] : object | can have multiple Performance
Modes
│  ├ 26a. Performance Mode Name : string
│  ├ 26b. Electronics Settings : object
│  │  ├ 26b1. Tc1 : float
│  │  ├ 26b2. Tc2 : float
│  │  ├ 26b3. Abs : float
│  │  ├ 26b4. Esc : float
│  │  ├ 26b5. Ebb : float
│  │  ├ 26b6. Engine Map : float
│  │  ├ 26b7. Telemetry laps To Record : float
│  │  ├ 26b8. Turbo Boost Lv : float
│  │  ├ 26b9. Ers Deployment Map : float
│  │  ├ 26b10. Ers Recharge Lv : float
│  │  └ 26b11. Ers Heat Charging : float
│  ├ 26c. Brakes Settings : object
│  │  ├ 26c1. Front Bias : float
│  │  ├ 26c2. Torque Multiplier : float
│  │  ├ 26c3. Brake Ducts [x] : float | can have multiple Brake Ducts
│  ├ 26d. Damper Settings [x] : object | can have multiple Damper
Settings
│  │  ├ 26d1. Slow Bump : float
│  │  ├ 26d2. Fast Bump : float
│  │  ├ 26d3. Slow Rebound : float
│  │  └ 26d4. Fast Rebound : float
│  ├ 26e. Differential Data : object
│  │  ├ 26e1. Type : enum
│  │  ├ 26e2. Power : float
│  │  ├ 26e3. Coast : float
│  │  ├ 26e4. Preload : float
│  │  ├ 26e5. Front Share : float
│  │  ├ 26e6. Torque Bias Ratio Power : float
│  │  ├ 26e7. Torque Bias Ratio Coast : float
```

```
│ │ ├ 26e8. Thermal Capacity : float
│ │ ├ 26e9. Surface : float
│ │ ├ 26e10. Heat Transfer Coeff : float
│ │ ├ 26e11. Wear Factor : float
│ │ ├ 26e12. Friction Reduction With T : float
│ │ └ 26e13. Friction Ref T : float
│ ├ 26f. Four W D Differentials : object
│ │ ├ 26f1. Front Diff : object
│ │ │ ├ 26e1. Type : enum
│ │ │ ├ 26e2. Power : float
│ │ │ ├ 26e3. Coast : float
│ │ │ ├ 26e4. Preload : float
│ │ │ ├ 26e5. Front Share : float
│ │ │ ├ 26e6. Torque Bias Ratio Power : float
│ │ │ ├ 26e7. Torque Bias Ratio Coast : float
│ │ │ ├ 26e8. Thermal Capacity : float
│ │ │ ├ 26e9. Surface : float
│ │ │ ├ 26e10. Heat Transfer Coeff : float
│ │ │ ├ 26e11. Wear Factor : float
│ │ │ ├ 26e12. Friction Reduction With T : float
│ │ │ └ 26e13. Friction Ref T : float
│ │ ├ 26f2. Center Diff : object
│ │ │ ├ 26e1. Type : enum
│ │ │ ├ 26e2. Power : float
│ │ │ ├ 26e3. Coast : float
│ │ │ ├ 26e4. Preload : float
│ │ │ ├ 26e5. Front Share : float
│ │ │ ├ 26e6. Torque Bias Ratio Power : float
│ │ │ ├ 26e7. Torque Bias Ratio Coast : float
│ │ │ ├ 26e8. Thermal Capacity : float
│ │ │ ├ 26e9. Surface : float
│ │ │ ├ 26e10. Heat Transfer Coeff : float
│ │ │ ├ 26e11. Wear Factor : float
│ │ │ ├ 26e12. Friction Reduction With T : float
│ │ │ └ 26e13. Friction Ref T : float
│ │ ├ 26f3. Rear Diff : object
│ │ │ ├ 26e1. Type : enum
│ │ │ ├ 26e2. Power : float
│ │ │ ├ 26e3. Coast : float
│ │ │ ├ 26e4. Preload : float
│ │ │ ├ 26e5. Front Share : float
│ │ │ ├ 26e6. Torque Bias Ratio Power : float
│ │ │ ├ 26e7. Torque Bias Ratio Coast : float
│ │ │ ├ 26e8. Thermal Capacity : float
│ │ │ ├ 26e9. Surface : float
│ │ │ ├ 26e10. Heat Transfer Coeff : float
│ │ │ ├ 26e11. Wear Factor : float
│ │ │ ├ 26e12. Friction Reduction With T : float
│ │ └ └ 26e13. Friction Ref T : float
│ ├ 26g. Front Lock Controllers : object
│ │ ├ 4l9a. Name : string
│ │ ├ 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
│ │ │ ├ 4l9b1. Input Var : enum
│ │ │ ├ 4l9b2. Combinator Mode : enum
│ │ │ ├ 4l9b3. Lut : string - path
```

```
│  │  │  ├ 4l9b4. Filter Gain : float
│  │  │  ├ 4l9b5. Up Limit : float
│  │  │  ├ 4l9b6. Down Limit : float
│  │  │  ├ 4l9b7. Current Value : float
│  │  └ └ 4l9b8. Const Value : float
│  ├ 26h. Center Lock Controllers : object
│  │  ├ 4l9a. Name : string
│  │  ├ 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
│  │  │  ├ 4l9b1. Input Var : enum
│  │  │  ├ 4l9b2. Combinator Mode : enum
│  │  │  ├ 4l9b3. Lut : string - path
│  │  │  ├ 4l9b4. Filter Gain : float
│  │  │  ├ 4l9b5. Up Limit : float
│  │  │  ├ 4l9b6. Down Limit : float
│  │  │  ├ 4l9b7. Current Value : float
│  │  └ └ 4l9b8. Const Value : float
│  ├ 26i. Rear Lock Controllers : object
│  │  ├ 4l9a. Name : string
│  │  ├ 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
│  │  │  ├ 4l9b1. Input Var : enum
│  │  │  ├ 4l9b2. Combinator Mode : enum
│  │  │  ├ 4l9b3. Lut : string - path
│  │  │  ├ 4l9b4. Filter Gain : float
│  │  │  ├ 4l9b5. Up Limit : float
│  │  │  ├ 4l9b6. Down Limit : float
│  │  │  ├ 4l9b7. Current Value : float
│  │  └ └ 4l9b8. Const Value : float
│  ├ 26j. Awd Clutches [x] : object | can have multiple Awd Clutches
│  │  ├ 26j1. Position : integer
│  │  └ 26j2. Preload : float
│  ├ 26k. Turbo Controllers [x] : object with an array of stages within |
can have multiple Turbo Controllers
│  │  ├ 4l9a. Name : string
│  │  ├ 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
│  │  │  ├ 4l9b1. Input Var : enum
│  │  │  ├ 4l9b2. Combinator Mode : enum
│  │  │  ├ 4l9b3. Lut : string - path
│  │  │  ├ 4l9b4. Filter Gain : float
│  │  │  ├ 4l9b5. Up Limit : float
│  │  │  ├ 4l9b6. Down Limit : float
│  │  │  ├ 4l9b7. Current Value : float
│  │  └ └ 4l9b8. Const Value : float
│  ├ 26l. Turbo Settings : object
│  └ └ 26l1. Boost Lv : float
├ 27. Ai Car Data : string - path
└ 28. mm : integer
```

Descriptions - Car Data

Key	Description
1	…..

Enum list - Car Engine

Id	Enum	Values
4l9b1	Input Var	UndefinedInput, Brake, Gas, LatG, LonG, Steer, Speed, Gear, SlipRatioFrontAVG, SlipRatioRearAVG, SlipRatioFrontMAX, SlipRatioRearMAX, SlipAngleFrontAVG, SlipAngleRearAVG, SlipAngleFrontMAX, SlipAngleRearMAX, OversteerFactor, RearSpeedRatio, SteerDEG, Const, RPMS, WheelSteerDEG, LoadSpreadLF, LoadSpreadRF, AvgTravelRear, SusTravelLR, SusTravelRR, SteerYawDeltaLeft, SteerYawDeltaRight, ErsChargeLevel, ErsCoastTorque
4l9b2	CombinatorMode	UndefinedMode, Add, Mult
4q5a1	Type	<None>, Poly3, Poly5, Piece Wise, Damper Lut Data
18c1	Mode	UseEffect, UseAngle
26e1	Type	LSD, Spool, Torsen, EpicyclicTorsen, EpicyclicLSD, TorqueVectoring

C. Example data

I. Chosen Car Data for Example

- Ferrari 296 GTB (slug : ks_ferrari_286_gtb)
- Audi R8 LMS GT3 Evo II (slug : ks_audi_r8_lms_gt3_evo_2)
- Renault 5 GT Turbo (slug : ks_renault_5_gt_turbo)

II. Example

Ferrari 296 GTB

```
├ 1. Screen Name : None
├ 2. General
│ ├ 2a. Screen Name : Ferrari 296 GTB
│ ├ 2b. Total Mass : 1750.00000
│ ├ 2c. Tank Position : 0.00000, -0.16715, -032949
│ ├ 2d. Fuel : 60.00000
│ ├ 2e. Max Fuel : 120.00000
│ ├ 2f. Efficiency : 0.00000
│ ├ 2g. Kg Per Liter : 0.75500
│ ├ 2h. Body Box Sizes : 2.00000, 0.70000, 4.22149
│ ├ 2i. Pickup Front Height : -0.35200
│ ├ 2j. Pickup Rear Height : -0.35200
```

```
│ ├ 2k. Check Rules : false
│ ├ 2l. Minimum Height : 0.00000
│ ├ 2m. Torsional Stiffness : 30000.00000
│ ├ 2n. Torsional Damping : 500.00000
│ ├ 2o. Body Mesh Offset
│ │ ├ 2o1. Position : 0.000, 0.000, 0.000
│ │ ├ 2o2. Rotation : 0.000, 0.000, 0.000
│ └ └ 2o3. Scale : 0.000, 0.000, 0.000
├ 3. General Path : None
├ 4. Suspensions
│ ├ 4a. Wheel Base : 2.60000
│ ├ 4b. Longitudinal Cg Location : 0.39500
│ ├ 4c. Base Y Front : -0.11000
│ ├ 4d. Base Y Rear : -0.08200
│ ├ 4e. Track Front : 1.72600
│ ├ 4f. Track Rear : 1.71000
│ ├ 4g. Damage
│ │ ├ 4g1. Min Velocity : 40.00000
│ │ ├ 4g2. Gain : 0.00040
│ │ ├ 4g3. Max Damage : 0.05000
│ │ └ 4g4. Debug Log : true
│ ├ 4h. Coilover Front path :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb_front.coilover
│ ├ 4i. Coilover Rear Path :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb_rear.coilover
│ ├ 4j. Front Suspension Path :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb_front.suspension
│ ├ 4k. Rear Suspension Path :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb_front.suspension
│ ├ 4l. Heavy Springs : None
│ ├ 4m. Arb Front
│ │ ├ 4m1. Stiffness : 26000.00000
│ │ ├ 4m2. Controller
│ │ │ ├ 4l9a. Name : None
│ │ │ └ 4l9b. Stages : None
│ ├ 4n. Arb Rear
│ │ ├ 4n1. Stiffness : 20000.00000
│ │ ├ 4n2. Controller
│ │ │ ├ 4l9a. Name : None
│ │ │ ├ 4l9b. Stages : None
│ ├ 4o. Flex Bar Front
│ │ ├ 4o1. Stiffness : 0.00000
│ │ ├ 4o2. Controller
│ │ │ ├ 4l9a. Name : None
│ │ │ ├ 4l9b. Stages : None
│ ├ 4p. Flex Bar Rear
│ │ ├ 4p1. Stiffness 0.00000
│ │ ├ 4p2. Controller
│ │ │ ├ 4l9a. Name : None
│ │ │ ├ 4l9b. Stages : None
│ ├ 4q. Dampers Controller : None
│ └ 4r. Has Dampers Cockpit Settings : false
├ 5. Drivetrain Path :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.drivetrain
├ 6. Gearbox Path :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.gearbox
```

```
├ 7. Clutch Path :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.clutch
├ 8. Engine Path :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.carengine
├ 9. Brakes Path :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.brakesystem
├ 10. Steering System
│ ├ 10a. Four W S Controllers
│ │ ├ 4l9a. Name : None
│ │ ├ 4l9b. Stages : None
├ 11. Electronics
│ ├ 11a. T C
│ │ ├ 11a1. Has T C2 : false
│ │ ├ 11a2. Frequency Hz : 333.00000
│ │ ├ 11a3. Min Speed Kmh : 35.00000
│ │ ├ 11a4. Gear Change Time : 0.0800
│ │ ├ 11a5. Min Cut Level : 8.00000
│ │ ├ 11a6. Max Cut Level : 0.90000
│ │ ├ 11a7. Settings 1
│ │ │ ├ 11a7a. Min Slip Ratio : 0.00000
│ │ │ ├ 11a7b. Max Slip Ratio : 0.00000
│ │ │ ├ 11a7c. Ref Slip Angle Deg : 0.00000
│ │ │ ├ 11a7d. Engine Cut Level : 0.00000
│ │ │ ├ 11a7e. Angular A C Cgain : 0.00000
│ │ │ ├ 11a7f. Oversteer Gain : 0.00000
│ │ │ ├ 11a7g. Slip Angle Activation Deg : 0.00000
│ │ ├ 11a7. Settings 2
│ │ │ ├ 11a7a. Min Slip Ratio : 0.09000
│ │ │ ├ 11a7b. Max Slip Ratio : 0.35000
│ │ │ ├ 11a7c. Ref Slip Angle Deg : 15.00000
│ │ │ ├ 11a7d. Engine Cut Level : 1.80000
│ │ │ ├ 11a7e. Angular A C Cgain : 4.50000
│ │ │ ├ 11a7f. Oversteer Gain : 3.00000
│ │ │ ├ 11a7g. Slip Angle Activation Deg : 6.00000
│ │ ├ 11a7. Settings 3
│ │ │ ├ 11a7a. Min Slip Ratio : 0.07000
│ │ │ ├ 11a7b. Max Slip Ratio : 0.20000
│ │ │ ├ 11a7c. Ref Slip Angle Deg : 12.00000
│ │ │ ├ 11a7d. Engine Cut Level : 1.50000
│ │ │ ├ 11a7e. Angular A C Cgain : 6.00000
│ │ │ ├ 11a7f. Oversteer Gain : 5.00000
│ │ │ ├ 11a7g. Slip Angle Activation Deg : 5.00000
│ │ ├ 11a7. Settings 4
│ │ │ ├ 11a7a. Min Slip Ratio : 0.04500
│ │ │ ├ 11a7b. Max Slip Ratio : 0.15000
│ │ │ ├ 11a7c. Ref Slip Angle Deg : 7.00000
│ │ │ ├ 11a7d. Engine Cut Level : 1.0000
│ │ │ ├ 11a7e. Angular A C Cgain : 10.00000
│ │ │ ├ 11a7f. Oversteer Gain : 7.00000
│ │ │ ├ 11a7g. Slip Angle Activation Deg : 3.00000
│ ├ 11b. A B S
│ │ ├ 11b1. Settings 1
│ │ │ ├ 11b1a. Min Slip Ratio : -1.00000
│ │ │ ├ 11b1b. Max Slip Ratio : -1.00000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 0.00000
│ │ │ ├ 11b1d. Cut Level : 0.00000
```

```
│ │ │ ├ 11b1f. Max Torque Variation : 0.70000
│ │ ├ 11b1. Settings 2
│ │ │ ├ 11b1a. Min Slip Ratio : 0.07000
│ │ │ ├ 11b1b. Max Slip Ratio : 0.12000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 9.00000
│ │ │ ├ 11b1d. Cut Level : 0.05000
│ │ │ ├ 11b1f. Max Torque Variation : 1.00000

│ │ ├ Frequency : 40.00000
│ │ ├ Channels : 4
│ │ ├ Min Speed Kmh : 0.00000
│ ├ 11c. E D L : None
│ ├ 11d. E S P
│ │ ├ 11d1. Frequency Hz : 100.00000
│ │ ├ 11d2. Min Speed Kmh : 20.00000
│ │ ├ 11d3. Settings 1
│ │ │ ├ 11d3a. Gain : 0.00000
│ │ │ ├ 11d3b. Steer Gain : 0.00000
│ │ │ ├ 11d3c. Min Steer Gain : 0.00000
│ │ │ ├ 11d3d. Steer Gain Max Speed : 0.00000
│ │ │ ├ 11d3e. Oversteer Gain : 0.00000
│ │ │ ├ 11d3f. Understeer Gain : 0.00000
│ │ │ ├ 11d3g. Max Slip Ratio : 0.00000
│ │ │ ├ 11d3h. Dead Zone : 0.00000
│ │ │ ├ 11d3i. Filter Gain : 0.00000
│ │ │ ├ 11d3j. Brake Perc : 0.00000
│ │ │ └ 11d3k. Brake Perc Activation : 0.00000
│ │ ├ 11d3. Settings 2
│ │ │ ├ 11d3a. Gain : 0.50000
│ │ │ ├ 11d3b. Steer Gain : 0.50000
│ │ │ ├ 11d3c. Min Steer Gain : 0.50000
│ │ │ ├ 11d3d. Steer Gain Max Speed : 0.00000
│ │ │ ├ 11d3e. Oversteer Gain : 0.50000
│ │ │ ├ 11d3f. Understeer Gain : 0.50000
│ │ │ ├ 11d3g. Max Slip Ratio : 0.19000
│ │ │ ├ 11d3h. Dead Zone : 0.30000
│ │ │ ├ 11d3i. Filter Gain : 0.95000
│ │ │ ├ 11d3j. Brake Perc : 0.10000
│ └ └ └ 11d3k. Brake Perc Activation : 0.90000
├ 12. Electronics Path : None
├ 13. Controls
│ ├ 13a. Ff Mult : 1.10000
│ ├ 13b. Steer Lock : 342.00000
│ ├ 13c. Steer Ratio : 12.00000
│ ├ 13d. Linear Steer Rod Ratio : 0.00200
│ └ 13e. Steer Assist : 1.00000
├ 14. Box Colliders 1
│ ├ 14a. Center : 0.00000, -0.26000, 0.33000
│ ├ 14b. Size : 1.78000, 0.15000, 3.94000
│ └ 14c. Pitch Rotation Deg : 0.200000
├ 15. Front Tyre Compounds 1 :
content\cars\common_phsx\tyres\hypercar\hypercar_245_35_20.tyre
├ 16. Rear Tyre Compounds 1 :
content\cars\common_phsx\tyres\hypercar\hypercar_305_35_20.tyre
├ 17. Aero
│ ├ 17a. Slip Gain Multiple : 1.00000
```

```
│ ├ 17b. Speed Factor Mult : 2.00000
│ ├ 17c. Downforces : None
│ ├ 17d. Front Lift : None
│ ├ 17e. Rear Lift : None
│ ├ 17f. Drag : None
│ └ 17g. Wings Path 1 :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb0.wing
│ └ 17g. Wings Path 2 :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb1.wing
│ └ 17g. Wings Path 3 :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb2.wing
│ └ 17g. Wings Path 4 :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb3.wing
├ 18. Drs : None
├ 19. Ers
│ ├ 19a. Torque Lut :
content\cars\ks_ferrari_296_gtb\data\kers\KERS_TORQUE.curve
│ ├ 19b. Coast Lut :
content\cars\ks_ferrari_296_gtb\data\kers\KERS_TORQUE.curve
│ ├ 19c. Battery Charge Kj : 26820.00000
│ ├ 19d. Has Button Override : false
│ ├ 19e. Max Kj Per Lap : 0.00000
│ ├ 19f. Max Charge Kj Per Lap : 0.00000
│ ├ 19g. Heat Charge Perc : 0.00000
│ ├ 19h. Heat Power Kw : 0.00000
│ ├ 19i. Default Power Controller Index : 0
│ ├ 19j. Power Controllers Front : None
│ ├ 19k. Power Controllers Rear 1
│ │ ├ 4l9a. Name : MAP Q
│ │ ├ 4l9b. Stages 1
│ │ │ ├ 4l9b1. Input Var : Gas
│ │ │ ├ 4l9b2. Combinator Mode : Add
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kers_Q_CONTROLLER_G
AS.curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 2
│ │ │ ├ 4l9b1. Input Var : Gear
│ │ │ ├ 4l9b2. Combinator Mode : Mult
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_GEAR
.curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 3
│ │ │ ├ 4l9b1. Input Var : SlipRatioRearAVG
│ │ │ ├ 4l9b2. Combinator Mode : Mult
```

```
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_SLIP
.curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 4
│ │ │ ├ 4l9b1. Input Var : RPMS
│ │ │ ├ 4l9b2. Combinator Mode : Mult
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_RPM.
curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 5
│ │ │ ├ 4l9b1. Input Var : Gas
│ │ │ ├ 4l9b2. Combinator Mode : Add
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersTHROTTLE_coast.
curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 6
│ │ │ ├ 4l9b1. Input Var : Brake
│ │ │ ├ 4l9b2. Combinator Mode : Mult
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_BRAK
E.curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ └ └ 4l9b8. Const Value : 0.00000
│ ├ 19k. Power Controllers Rear 2
│ │ ├ 4l9a. Name : MAP P
│ │ ├ 4l9b. Stages 1
│ │ │ ├ 4l9b1. Input Var : Gas
│ │ │ ├ 4l9b2. Combinator Mode : Add
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kers_P_CONTROLLER_G
AS.curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 2
│ │ │ ├ 4l9b1. Input Var : Gear
```

```
│ │ │ ├ 4l9b2. Combinator Mode : Mult
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_GEAR
.curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 3
│ │ │ ├ 4l9b1. Input Var : SlipRatioRearAVG
│ │ │ ├ 4l9b2. Combinator Mode : Mult
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_SLIP
.curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 4
│ │ │ ├ 4l9b1. Input Var : RPMS
│ │ │ ├ 4l9b2. Combinator Mode : Mult
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_RPM.
curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 5
│ │ │ ├ 4l9b1. Input Var : Gas
│ │ │ ├ 4l9b2. Combinator Mode : Add
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersTHROTTLE_coast.
curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 6
│ │ │ ├ 4l9b1. Input Var : Brake
│ │ │ ├ 4l9b2. Combinator Mode : Mult
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_BRAK
E.curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ └ └ 4l9b8. Const Value : 0.00000
│ ├ 19k. Power Controllers Rear 3
│ │ ├ 4l9a. Name : MAP H
│ │ ├ 4l9b. Stages 1
```

```
│ │ │ ├ 4l9b1. Input Var : Gas
│ │ │ ├ 4l9b2. Combinator Mode : Add
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kers_H_CONTROLLER_G
AS.curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 2
│ │ │ ├ 4l9b1. Input Var : Gear
│ │ │ ├ 4l9b2. Combinator Mode : Mult
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_GEAR
.curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 3
│ │ │ ├ 4l9b1. Input Var : SlipRatioRearAVG
│ │ │ ├ 4l9b2. Combinator Mode : Mult
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_SLIP
.curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 4
│ │ │ ├ 4l9b1. Input Var : RPMS
│ │ │ ├ 4l9b2. Combinator Mode : Mult
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_RPM.
curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 5
│ │ │ ├ 4l9b1. Input Var : Gas
│ │ │ ├ 4l9b2. Combinator Mode : Add
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersTHROTTLE_coast.
curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ │ └ 4l9b8. Const Value : 0.00000
│ │ ├ 4l9b. Stages 6
│ │ │ ├ 4l9b1. Input Var : Brake
```

```
│ │ │ ├ 4l9b2. Combinator Mode : Mult
│ │ │ ├ 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_BRAK
E.curve
│ │ │ ├ 4l9b4. Filter Gain : 0.96000
│ │ │ ├ 4l9b5. Up Limit : 1.00000
│ │ │ ├ 4l9b6. Down Limit : -1.00000
│ │ │ ├ 4l9b7. Current Value : 0.00000
│ │ └ └ 4l9b8. Const Value : 0.00000
│ ├ 19l. Brake Rear Correction : 5.00000
│ ├ 19m. Has Cockpit Controls : false
│ ├ 19n. Cockpit Controls
│ │ ├ 19n1. Delivery Profile : false
│ │ ├ 19n2. Mgu H Mode : false
│ │ └ 19n3. Recovery : false
│ ├ 19o. Has Front Motors : false
│ ├ 19p. Front Motors
│ │ ├ 19p1. Torque Lut : None
│ │ ├ 19p2 Discharge Time : 0.00000
│ │ └ 19p3. Torque Vectoring Bias : 0.00000
├ 20. Setup Limits :
content\cars\ks_ferrari_296_gtb\data\setup\limits_f296gtb.carsetuplimits
├ 21. Collider Mesh :
content\cars\ks_ferrari_296_gtb\collider\ferrari_296_gtb_collider.mesh
├ 22 Body Mesh Offset
│ ├ 22a. Position : 0.000, -0.482, 0.247
│ ├ 22b. Rotation: 0.100, 0.000, 0.000
│ └ 22c. Scale : 1.000, 1.000, 1.000
├ 23. Stock Setup : content/cars/
ks_ferrari_296_gtb\data\setup\f296gtb_stock.carsetup
├ 24. Ai Setup : None
├ 25. Wet Setup : None
├ 26. Performance Modes 1
│ ├ 26a. Performance Mode Name : QUAL
│ ├ 26b. Electronics Settings
│ │ ├ 26b1. Tc1 : 2.00000
│ │ ├ 26b2. Tc2 : 1.00000
│ │ ├ 26b3. Abs : 1.00000
│ │ ├ 26b4. Esc : 0.00000
│ │ ├ 26b5. Ebb : 0.000
│ │ ├ 26b6. Engine Map : 0.00000
│ │ ├ 26b7. Telemetry laps To Record : 0.00000
│ │ ├ 26b8. Turbo Boost Lv : 0.00000
│ │ ├ 26b9. Ers Deployment Map : 0.000
│ │ ├ 26b10. Ers Recharge Lv : 100.000
│ │ └ 26b11. Ers Heat Charging : 0.000
│ ├ 26c. Brakes Settings : None
│ ├ 26d. Damper Settings : None
│ ├ 26e. Differential Data : None
│ ├ 26f. Four W D Differentials : None
│ ├ 26g. Front Lock Controllers : None
│ ├ 26h. Center Lock Controllers : None
│ ├ 26i. Rear Lock Controllers : None
│ ├ 26j. Awd Clutches : None
│ ├ 26k. Turbo Controllers : None
│ ├ 26l. Turbo Settings : None
```

```
├ 26. Performance Modes 2
│ ├ 26a. Performance Mode Name : PERF
│ ├ 26b. Electronics Settings
│ │ ├ 26b1. Tc1 : 2.00000
│ │ ├ 26b2. Tc2 : 1.00000
│ │ ├ 26b3. Abs : 1.00000
│ │ ├ 26b4. Esc : 1.00000
│ │ ├ 26b5. Ebb : 0.000
│ │ ├ 26b6. Engine Map : 0.00000
│ │ ├ 26b7. Telemetry laps To Record : 0.00000
│ │ ├ 26b8. Turbo Boost Lv : 0.00000
│ │ ├ 26b9. Ers Deployment Map : 1.000
│ │ ├ 26b10. Ers Recharge Lv : 100.000
│ │ └ 26b11. Ers Heat Charging : 0.000
│ ├ 26c. Brakes Settings : None
│ ├ 26d. Damper Settings : None
│ ├ 26e. Differential Data : None
│ ├ 26f. Four W D Differentials : None
│ ├ 26g. Front Lock Controllers : None
│ ├ 26h. Center Lock Controllers : None
│ ├ 26i. Rear Lock Controllers : None
│ ├ 26j. Awd Clutches : None
│ ├ 26k. Turbo Controllers : None
│ ├ 26l. Turbo Settings : None
├ 26. Performance Modes 3
│ ├ 26a. Performance Mode Name : HYBRID
│ ├ 26b. Electronics Settings
│ │ ├ 26b1. Tc1 : 2.00000
│ │ ├ 26b2. Tc2 : 1.00000
│ │ ├ 26b3. Abs : 1.00000
│ │ ├ 26b4. Esc : 1.00000
│ │ ├ 26b5. Ebb : 0.000
│ │ ├ 26b6. Engine Map : 1.00000
│ │ ├ 26b7. Telemetry laps To Record : 0.00000
│ │ ├ 26b8. Turbo Boost Lv : 0.00000
│ │ ├ 26b9. Ers Deployment Map : 2.000
│ │ ├ 26b10. Ers Recharge Lv : 100.000
│ │ └ 26b11. Ers Heat Charging : 0.000
│ ├ 26c. Brakes Settings : None
│ ├ 26d. Damper Settings : None
│ ├ 26e. Differential Data : None
│ ├ 26f. Four W D Differentials : None
│ ├ 26g. Front Lock Controllers : None
│ ├ 26h. Center Lock Controllers : None
│ ├ 26i. Rear Lock Controllers : None
│ ├ 26j. Awd Clutches : None
│ ├ 26k. Turbo Controllers : None
│ ├ 26l. Turbo Settings : None
├ 27. Ai Car Data :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.aicardata
└ 28. mm : 1
```

Audi R8 LMS GT3 Evo II

```
├ 1. Screen Name : Audi R8 LMS GT3 Evo II
├ 2. General
│ ├ 2a. Screen Name : Audi R8 LMS GT3 Evo II
│ ├ 2b. Total Mass : 1355
│ ├ 2c. Tank Position : 0.00000, -0.02900, -0.25600
│ ├ 2d. Fuel : 0.00000
│ ├ 2e. Max Fuel : 120.00000
│ ├ 2f. Efficiency : 0.40000
│ ├ 2g. Kg Per Liter : 0.75500
│ ├ 2h. Body Box Sizes : 1.98000, 1.19000, 4.46000
│ ├ 2i. Pickup Front Height : -0.34900
│ ├ 2j. Pickup Rear Height : -0.31800
│ ├ 2k. Check Rules : true
│ ├ 2l. Minimum Height : 450.00000
│ ├ 2m. Torsional Stiffness : 40000.00000
│ ├ 2n. Torsional Damping : 400.00000
│ ├ 2o. Body Mesh Offset
│ │ ├ 2o1. Position : 0.000, 0.000, 0.000
│ │ ├ 2o2. Rotation : 0.000, 0.000, 0.000
│ └ └ 2o3. Scale : 0.000, 0.000, 0.000
├ 3. General Path : None
├ 4. Suspensions
│ ├ 4a. Wheel Base : 2.70000
│ ├ 4b. Longitudinal Cg Location : 0.42000
│ ├ 4c. Base Y Front : -0.03500
│ ├ 4d. Base Y Rear : -0.03500
│ ├ 4e. Track Front : 1.66700
│ ├ 4f. Track Rear : 1.67000
│ ├ 4g. Damage
│ │ ├ 4g1. Min Velocity : 40.00000
│ │ ├ 4g2. Gain : 0.00040
│ │ ├ 4g3. Max Damage : 0.05000
│ │ └ 4g4. Debug Log : true
│ ├ 4h. Coilover Front path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2_fron
t.coilover
│ ├ 4i. Coilover Rear Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2_rear
.coilover
│ ├ 4j. Front Suspension Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2_fron
t.suspension
│ ├ 4k. Rear Suspension Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2_rear
.suspension
│ ├ 4l. Heavy Springs : None
│ ├ 4m. Arb Front
│ │ ├ 4m1. Stiffness : 56800.00000
│ │ ├ 4m2. Controller
│ │ │ ├ 4l9a. Name : None
│ │ │ ├ 4l9b. Stages : None
│ ├ 4n. Arb Rear
│ │ ├ 4n1. Stiffness : 39900.00000
│ │ ├ 4n2. Controller
│ │ │ ├ 4l9a. Name : None
```

```
│   │   │ ├ 4l9b. Stages : None
│   ├ 4o. Flex Bar Front
│   │ ├ 4o1. Stiffness : 0.00000
│   │ ├ 4o2. Controller
│   │ │ ├ 4l9a. Name : None
│   │ │ ├ 4l9b. Stages : None
│   ├ 4p. Flex Bar Rear
│   │ ├ 4p1. Stiffness : 0.00000
│   │ ├ 4p2. Controller
│   │ │ ├ 4l9a. Name : None
│   │ │ ├ 4l9b. Stages : None
│   ├ 4q. Dampers Controller : None
│   └ 4r. Has Dampers Cockpit Settings : false
├ 5. Drivetrain Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.driv
etrain
├ 6. Gearbox Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.gear
box
├ 7. Clutch Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.clut
ch
├ 8. Engine Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.care
ngine
├ 9. Brakes Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.brak
esystem
├ 10. Steering System
│ ├ 10a. Four W S Controllers
│ │ ├ 4l9a. Name : None
│ │ ├ 4l9b. Stages : None
├ 11. Electronics
│ ├ 11a. T C
│ │ ├ 11a1. Has T C2 : true
│ │ ├ 11a2. Frequency Hz : 333.00000
│ │ ├ 11a3. Min Speed Kmh : 30.00000
│ │ ├ 11a4. Gear Change Time : 0.07000
│ │ ├ 11a5. Min Cut Level : 0.00000
│ │ ├ 11a6. Max Cut Level : 6.00000
│ │ ├ 11a7. Settings 1
│ │ │ ├ 11a7a. Min Slip Ratio : 0.00000
│ │ │ ├ 11a7b. Max Slip Ratio : 0.00000
│ │ │ ├ 11a7c. Ref Slip Angle Deg : 0.00000
│ │ │ ├ 11a7d. Engine Cut Level : 0.00000
│ │ │ ├ 11a7e. Angular A C Cgain : 0.00000
│ │ │ ├ 11a7f. Oversteer Gain : 0.00000
│ │ │ ├ 11a7g. Slip Angle Activation Deg : 0.00000
│ │ ├ 11a7. Settings 2
│ │ │ ├ 11a7a. Min Slip Ratio : 0.18000
│ │ │ ├ 11a7b. Max Slip Ratio : 0.35000
│ │ │ ├ 11a7c. Ref Slip Angle Deg : 15.00000
│ │ │ ├ 11a7d. Engine Cut Level : 1.50000
│ │ │ ├ 11a7e. Angular A C Cgain : 1.00000
│ │ │ ├ 11a7f. Oversteer Gain : 1.00000
│ │ │ ├ 11a7g. Slip Angle Activation Deg : 5.00000
```

```
│ │ │ ├ 11a7. Settings 3
│ │ │ │ ├ 11a7a. Min Slip Ratio : 0.15000
│ │ │ │ ├ 11a7b. Max Slip Ratio : 0.30000
│ │ │ │ ├ 11a7c. Ref Slip Angle Deg : 13.00000
│ │ │ │ ├ 11a7d. Engine Cut Level : 1.50000
│ │ │ │ ├ 11a7e. Angular A C Cgain : 1.50000
│ │ │ │ ├ 11a7f. Oversteer Gain : 1.50000
│ │ │ │ ├ 11a7g. Slip Angle Activation Deg : 5.00000
│ │ │ ├ 11a7. Settings 4
│ │ │ │ ├ 11a7a. Min Slip Ratio : 0.12000
│ │ │ │ ├ 11a7b. Max Slip Ratio : 0.28000
│ │ │ │ ├ 11a7c. Ref Slip Angle Deg : 11.00000
│ │ │ │ ├ 11a7d. Engine Cut Level : 1.25000
│ │ │ │ ├ 11a7e. Angular A C Cgain : 2.00000
│ │ │ │ ├ 11a7f. Oversteer Gain : 2.00000
│ │ │ │ ├ 11a7g. Slip Angle Activation Deg : 4.50000
│ │ │ ├ 11a7. Settings 5
│ │ │ │ ├ 11a7a. Min Slip Ratio : 0.10000
│ │ │ │ ├ 11a7b. Max Slip Ratio : 0.25000
│ │ │ │ ├ 11a7c. Ref Slip Angle Deg : 10.00000
│ │ │ │ ├ 11a7d. Engine Cut Level : 1.00000
│ │ │ │ ├ 11a7e. Angular A C Cgain : 2.50000
│ │ │ │ ├ 11a7f. Oversteer Gain : 2.50000
│ │ │ │ ├ 11a7g. Slip Angle Activation Deg : 4.00000
│ │ │ ├ 11a7. Settings 6
│ │ │ │ ├ 11a7a. Min Slip Ratio : 0.08000
│ │ │ │ ├ 11a7b. Max Slip Ratio : 0.22000
│ │ │ │ ├ 11a7c. Ref Slip Angle Deg : 8.50000
│ │ │ │ ├ 11a7d. Engine Cut Level : 1.00000
│ │ │ │ ├ 11a7e. Angular A C Cgain : 3.00000
│ │ │ │ ├ 11a7f. Oversteer Gain : 3.00000
│ │ │ │ ├ 11a7g. Slip Angle Activation Deg : 4.00000
│ │ │ ├ 11a7. Settings 7
│ │ │ │ ├ 11a7a. Min Slip Ratio : 0.07000
│ │ │ │ ├ 11a7b. Max Slip Ratio : 0.22000
│ │ │ │ ├ 11a7c. Ref Slip Angle Deg : 7.50000
│ │ │ │ ├ 11a7d. Engine Cut Level : 1.00000
│ │ │ │ ├ 11a7e. Angular A C Cgain : 3.50000
│ │ │ │ ├ 11a7f. Oversteer Gain : 3.50000
│ │ │ │ ├ 11a7g. Slip Angle Activation Deg : 3.50000
│ │ │ ├ 11a7. Settings 8
│ │ │ │ ├ 11a7a. Min Slip Ratio : 0.06000
│ │ │ │ ├ 11a7b. Max Slip Ratio : 0.22000
│ │ │ │ ├ 11a7c. Ref Slip Angle Deg : 7.00000
│ │ │ │ ├ 11a7d. Engine Cut Level : 0.50000
│ │ │ │ ├ 11a7e. Angular A C Cgain : 4.00000
│ │ │ │ ├ 11a7f. Oversteer Gain : 4.00000
│ │ │ │ ├ 11a7g. Slip Angle Activation Deg : 3.00000
│ │ │ ├ 11a7. Settings 9
│ │ │ │ ├ 11a7a. Min Slip Ratio : 0.05000
│ │ │ │ ├ 11a7b. Max Slip Ratio : 0.19000
│ │ │ │ ├ 11a7c. Ref Slip Angle Deg : 7.00000
│ │ │ │ ├ 11a7d. Engine Cut Level : 0.45000
│ │ │ │ ├ 11a7e. Angular A C Cgain : 5.00000
│ │ │ │ ├ 11a7f. Oversteer Gain : 5.00000
│ │ │ │ ├ 11a7g. Slip Angle Activation Deg : 3.00000
```

```
│ │ ├ 11a7. Settings 10
│ │ │ ├ 11a7a. Min Slip Ratio : 0.05000
│ │ │ ├ 11a7b. Max Slip Ratio : 0.17000
│ │ │ ├ 11a7c. Ref Slip Angle Deg : 7.00000
│ │ │ ├ 11a7d. Engine Cut Level : 0.40000
│ │ │ ├ 11a7e. Angular A C Cgain : 7.50000
│ │ │ ├ 11a7f. Oversteer Gain : 6.00000
│ │ │ ├ 11a7g. Slip Angle Activation Deg : 3.00000
│ │ ├ 11a7. Settings 11
│ │ │ ├ 11a7a. Min Slip Ratio : 0.04000
│ │ │ ├ 11a7b. Max Slip Ratio : 0.15000
│ │ │ ├ 11a7c. Ref Slip Angle Deg : 6.00000
│ │ │ ├ 11a7d. Engine Cut Level : 0.35000
│ │ │ ├ 11a7e. Angular A C Cgain : 8.00000
│ │ │ ├ 11a7f. Oversteer Gain : 6.50000
│ │ │ ├ 11a7g. Slip Angle Activation Deg : 2.50000
│ │ ├ 11a7. Settings 12
│ │ │ ├ 11a7a. Min Slip Ratio : 0.03000
│ │ │ ├ 11a7b. Max Slip Ratio : 0.15000
│ │ │ ├ 11a7c. Ref Slip Angle Deg : 6.00000
│ │ │ ├ 11a7d. Engine Cut Level : 0.30000
│ │ │ ├ 11a7e. Angular A C Cgain : 8.00000
│ │ │ ├ 11a7f. Oversteer Gain : 7.00000
│ │ │ ├ 11a7g. Slip Angle Activation Deg : 2.50000
│ │ ├ 11a7. Settings 13
│ │ │ ├ 11a7a. Min Slip Ratio : 0.03000
│ │ │ ├ 11a7b. Max Slip Ratio : 0.14000
│ │ │ ├ 11a7c. Ref Slip Angle Deg : 5.00000
│ │ │ ├ 11a7d. Engine Cut Level : 0.30000
│ │ │ ├ 11a7e. Angular A C Cgain : 8.50000
│ │ │ ├ 11a7f. Oversteer Gain : 7.50000
│ │ │ ├ 11a7g. Slip Angle Activation Deg : 3.00000
│ ├ 11b. A B S
│ │ ├ 11b1. Settings 1
│ │ │ ├ 11b1a. Min Slip Ratio : -1.00000
│ │ │ ├ 11b1b. Max Slip Ratio : -1.00000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 0.00000
│ │ │ ├ 11b1d. Cut Level : 0.00000
│ │ │ ├ 11b1f. Max Torque Variation : 0.00000
│ │ ├ 11b1. Settings 2
│ │ │ ├ 11b1a. Min Slip Ratio : 0.12000
│ │ │ ├ 11b1b. Max Slip Ratio : 0.14000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 7.00000
│ │ │ ├ 11b1d. Cut Level : 0.20000
│ │ │ ├ 11b1f. Max Torque Variation : 1.00000
│ │ ├ 11b1. Settings 3
│ │ │ ├ 11b1a. Min Slip Ratio : 0.11000
│ │ │ ├ 11b1b. Max Slip Ratio : 0.14000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 7.00000
│ │ │ ├ 11b1d. Cut Level : 0.20000
│ │ │ ├ 11b1f. Max Torque Variation : 1.00000
│ │ ├ 11b1. Settings 4
│ │ │ ├ 11b1a. Min Slip Ratio : 0.10000
│ │ │ ├ 11b1b. Max Slip Ratio : 0.12000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 7.00000
│ │ │ ├ 11b1d. Cut Level : 0.20000
```

```
│ │ │ ├ 11b1f. Max Torque Variation : 1.00000
│ │ ├ 11b1. Settings 5
│ │ │ ├ 11b1a. Min Slip Ratio : 0.08000
│ │ │ ├ 11b1b. Max Slip Ratio : 0.10000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 7.00000
│ │ │ ├ 11b1d. Cut Level : 0.20000
│ │ │ ├ 11b1f. Max Torque Variation : 1.00000
│ │ ├ 11b1. Settings 6
│ │ │ ├ 11b1a. Min Slip Ratio : 0.07000
│ │ │ ├ 11b1b. Max Slip Ratio : 0.08000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 7.00000
│ │ │ ├ 11b1d. Cut Level : 0.20000
│ │ │ ├ 11b1f. Max Torque Variation : 1.00000
│ │ ├ 11b1. Settings 7
│ │ │ ├ 11b1a. Min Slip Ratio : 0.06000
│ │ │ ├ 11b1b. Max Slip Ratio : 0.08000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 7.00000
│ │ │ ├ 11b1d. Cut Level : 0.20000
│ │ │ ├ 11b1f. Max Torque Variation : 1.00000
│ │ ├ 11b1. Settings 8
│ │ │ ├ 11b1a. Min Slip Ratio : 0.05000
│ │ │ ├ 11b1b. Max Slip Ratio : 0.07000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 7.00000
│ │ │ ├ 11b1d. Cut Level : 0.20000
│ │ │ ├ 11b1f. Max Torque Variation : 1.00000
│ │ ├ 11b1. Settings 9
│ │ │ ├ 11b1a. Min Slip Ratio : 0.05000
│ │ │ ├ 11b1b. Max Slip Ratio : 0.06000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 7.00000
│ │ │ ├ 11b1d. Cut Level : 0.20000
│ │ │ ├ 11b1f. Max Torque Variation : 1.00000
│ │ ├ 11b1. Settings 10
│ │ │ ├ 11b1a. Min Slip Ratio : 0.04000
│ │ │ ├ 11b1b. Max Slip Ratio : 0.05000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 7.00000
│ │ │ ├ 11b1d. Cut Level : 0.20000
│ │ │ ├ 11b1f. Max Torque Variation : 1.00000
│ │ ├ 11b1. Settings 11
│ │ │ ├ 11b1a. Min Slip Ratio : 0.02500
│ │ │ ├ 11b1b. Max Slip Ratio : 0.03500
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 7.00000
│ │ │ ├ 11b1d. Cut Level : 0.20000
│ │ │ ├ 11b1f. Max Torque Variation : 1.00000
│ │ ├ 11b1. Settings 12
│ │ │ ├ 11b1a. Min Slip Ratio : 0.01000
│ │ │ ├ 11b1b. Max Slip Ratio : 0.02000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 7.00000
│ │ │ ├ 11b1d. Cut Level : 0.20000
│ │ │ ├ 11b1f. Max Torque Variation : 1.00000
│ │ ├ Frequency : 40.00000
│ │ ├ Channels : 4
│ │ ├ Min Speed Kmh : 20.00000
│ ├ 11c. E D L : None
│ ├ 11d. E S P
│ │ ├ 11d1. Frequency Hz : 0.00000
│ │ ├ 11d2. Min Speed Kmh : 0.00000
```

```
│ │ ├ 11d3. Settings : None
├ 12. Electronics Path : None
├ 13. Controls
│ ├ 13a. Ff Mult : 1.50000
│ ├ 13b. Steer Lock : 240.00000
│ ├ 13c. Steer Ratio : 14.30000
│ ├ 13d. Linear Steer Rod Ratio : 0.00235
│ └ 13e. Steer Assist : 1.00000
├ 14. Box Colliders 1
│ ├ 14a. Center : 0.00000, -0.23500, 0.30000
│ ├ 14b. Size : 1.78000, 0.20000, 4.07000
│ └ 14c. Pitch Rotation Deg : 0.65000
├ 15. Front Tyre Compounds 1 :
content\cars\common_phsx\tyres\racing_slicks\slick_325_680_18.tyre
├ 16. Rear Tyre Compounds 1 :
ontent\cars\common_phsx\tyres\racing_slicks\slick_325_705_18.tyre
├ 17. Aero
│ ├ 17a. Slip Gain Multiple : 1.00000
│ ├ 17b. Speed Factor Mult : 2.00000
│ ├ 17c. Downforces 1
│ │ ├ 17c1. Position : 0.82140, -0.34900, 1.57000
│ │ ├ 17c2. Cl Gain : 1.00000
│ │ ├ 17c3. Cd Gain : 1.00000
│ │ ├ 17c4. Yaw Gain : 0.00000
│ │ ├ 17c5. Drag Per Cool Transfer : 0.01500
│ │ ├ 17c6. Damage C L 1 : 0.01000
│ │ ├ 17c6. Damage C L 2 : 0.00500
│ │ ├ 17c6. Damage C L 3 : 0.00300
│ │ ├ 17c6. Damage C L 4 : 0.00300
│ │ ├ 17c7. Damage C D 1 : 0.01500
│ │ ├ 17c7. Damage C D 2 : 0.00500
│ │ ├ 17c7. Damage C D 3 : 0.00300
│ │ ├ 17c7. Damage C D 4 : 0.00300
│ │ ├ 17c8. Downforce Controllers : None
│ │ ├ 17c9. Lift Per Front Angle : 0.00000
│ │ ├ 17c10. Lift Per Rear Angle : 0.00000
│ │ ├ 17c11. Drag Per Front Angle : 0.00000
│ │ ├ 17c12. Drag Per Rear Angle : 0.00000
│ │ ├ 17c13. Default Front Angle : 0.00000
│ │ └ 17c14. Default Rear Angle : 0.00000
│ ├ 17c. Downforces 2
│ │ ├ 17c1. Position : 0.82140, -0.34900, 1.57000
│ │ ├ 17c2. Cl Gain : 1.00000
│ │ ├ 17c3. Cd Gain : 1.00000
│ │ ├ 17c4. Yaw Gain : 0.00000
│ │ ├ 17c5. Drag Per Cool Transfer : 0.01500
│ │ ├ 17c6. Damage C L 1 : 0.01000
│ │ ├ 17c6. Damage C L 2 : 0.00500
│ │ ├ 17c6. Damage C L 3 : 0.00300
│ │ ├ 17c6. Damage C L 4 : 0.00300
│ │ ├ 17c7. Damage C D 1 : 0.01500
│ │ ├ 17c7. Damage C D 2 : 0.00500
│ │ ├ 17c7. Damage C D 3 : 0.00300
│ │ ├ 17c7. Damage C D 4 : 0.00300
│ │ ├ 17c8. Downforce Controllers : None
│ │ ├ 17c9. Lift Per Front Angle : 0.00000
```

```
│ │ ├ 17c10. Lift Per Rear Angle : 0.00000
│ │ ├ 17c11. Drag Per Front Angle : 0.00000
│ │ ├ 17c12. Drag Per Rear Angle : 0.00000
│ │ ├ 17c13. Default Front Angle : 0.00000
│ │ └ 17c14. Default Rear Angle : 0.00000
│ ├ 17c. Downforces 3
│ │ ├ 17c1. Position : 0.82140, -0.31800, -1.13500
│ │ ├ 17c2. Cl Gain : 1.00000
│ │ ├ 17c3. Cd Gain : 1.00000
│ │ ├ 17c4. Yaw Gain : -0.10000
│ │ ├ 17c5. Drag Per Cool Transfer : 0.01000
│ │ ├ 17c6. Damage C L 1 : 0.00500
│ │ ├ 17c6. Damage C L 2 : 0.01000
│ │ ├ 17c6. Damage C L 3 : 0.00300
│ │ ├ 17c6. Damage C L 4 : 0.00300
│ │ ├ 17c7. Damage C D 1 : 0.00500
│ │ ├ 17c7. Damage C D 2 : 0.01500
│ │ ├ 17c7. Damage C D 3 : 0.00300
│ │ ├ 17c7. Damage C D 4 : 0.00300
│ │ ├ 17c8. Downforce Controllers : None
│ │ ├ 17c9. Lift Per Front Angle : 0.00000
│ │ ├ 17c10. Lift Per Rear Angle : 0.00000
│ │ ├ 17c11. Drag Per Front Angle : 0.00000
│ │ ├ 17c12. Drag Per Rear Angle : 0.00000
│ │ ├ 17c13. Default Front Angle : 0.00000
│ │ └ 17c14. Default Rear Angle : 0.00000
│ ├ 17c. Downforces 4
│ │ ├ 17c1. Position : 0.82140, -0.31800, -1.13500
│ │ ├ 17c2. Cl Gain : 1.00000
│ │ ├ 17c3. Cd Gain : 1.00000
│ │ ├ 17c4. Yaw Gain : -0.10000
│ │ ├ 17c5. Drag Per Cool Transfer : 0.01000
│ │ ├ 17c6. Damage C L 1 : 0.00500
│ │ ├ 17c6. Damage C L 2 : 0.01000
│ │ ├ 17c6. Damage C L 3 : 0.00300
│ │ ├ 17c6. Damage C L 4 : 0.00300
│ │ ├ 17c7. Damage C D 1 : 0.00500
│ │ ├ 17c7. Damage C D 2 : 0.01500
│ │ ├ 17c7. Damage C D 3 : 0.00300
│ │ ├ 17c7. Damage C D 4 : 0.00300
│ │ ├ 17c8. Downforce Controllers : None
│ │ ├ 17c9. Lift Per Front Angle : 0.00000
│ │ ├ 17c10. Lift Per Rear Angle : 0.00000
│ │ ├ 17c11. Drag Per Front Angle : 0.00000
│ │ ├ 17c12. Drag Per Rear Angle : 0.00000
│ │ ├ 17c13. Default Front Angle : 0.00000
│ │ └ 17c14. Default Rear Angle : 0.00000
│ ├ 17d. Front Lift :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\frontczmap.surface3d
│ ├ 17e. Rear Lift :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\rearczmap.surface3d
│ ├ 17f. Drag :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\cxmap.surface3d
│ └ 17g. Wings Path 1 :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\Audi_R8_GT3_rear_wing_re
ar_modifier.wing
```

```
│ └ 17g. Wings Path 2 :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\Audi_R8_GT3_rear_wing_fr
ont_modifier.wing
├ 18. Drs : None
├ 19. Ers : None
├ 20. Setup Limits :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\Audi_R8_GT3_rear_wing_fr
ont_modifier.wing
├ 21. Collider Mesh :
content\cars\ks_audi_r8_lms_gt3_evo_2\collider\audi_r8_gt3_evo2_collider
.mesh
├ 22. Body Mesh Offset
│ ├ 22a. Position : 0.000, -0.419, 0.230
│ ├ 22b. Rotation: 0.650, 0.000, 0.000
│ └ 22c. Scale : 1.000, 1.000, 1.000
├ 23. Stock Setup :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\setup\ks_audi_r8_lms_gt3_evo_
2_safe.carsetup
├ 24. Ai Setup : None
├ 25. Wet Setup :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\setup\ks_audi_r8_lms_gt3_evo_
2_wet.carsetup
├ 26. Performance Modes : None
├ 27. Ai Car Data :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.aica
rdata
└ 28. mm : 1
```

Renault 5 GT Turbo

```
├ 1. Screen Name : None
├ 2. General
│ ├ 2a. Screen Name : Renault 5 GT Turbo
│ ├ 2b. Total Mass : 910.00000
│ ├ 2c. Tank Position : 0.00000, -0.20000, -1.56750
│ ├ 2d. Fuel : 30.00000
│ ├ 2e. Max Fuel : 50.00000
│ ├ 2f. Efficiency : 0.00000
│ ├ 2g. Kg Per Liter : 0.75500
│ ├ 2h. Body Box Sizes : 1.60000, 1.10000, 3.30000
│ ├ 2i. Pickup Front Height : -0.49000
│ ├ 2j. Pickup Rear Height : -0.48000
│ ├ 2k. Check Rules : false
│ ├ 2l. Minimum Height : 0.00000
│ ├ 2m. Torsional Stiffness : 11000.00000
│ ├ 2n. Torsional Damping : 50.00000
│ ├ 2o. Body Mesh Offset
│ │ ├ 2o1. Position : 0.000, 0.000, 0.000
│ │ ├ 2o2. Rotation : 0.000, 0.000, 0.000
│ └ └ 2o3. Scale : 0.000, 0.000, 0.000
├ 3. General Path : None
├ 4. Suspensions :
│ ├ 4a. Wheel Base : 2.40700
│ ├ 4b. Longitudinal Cg Location : 0.64000
```

```
│ ├ 4c. Base Y Front : -0.42000
│ ├ 4d. Base Y Rear : -0.43000
│ ├ 4e. Track Front : 1.34000
│ ├ 4f. Track Rear : 1.31500
│ ├ 4g. Damage
│ │ ├ 4g1. Min Velocity : 40.00000
│ │ ├ 4g2. Gain : 0.00040
│ │ ├ 4g3. Max Damage : 0.05000
│ │ └ 4g4. Debug Log : true
│ ├ 4h. Coilover Front path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo_front.coil
over
│ ├ 4i. Coilover Rear Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo_rear.coilo
ver
│ ├ 4j. Front Suspension Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo_front.susp
ension
│ ├ 4k. Rear Suspension Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo_rear.coilo
ver
│ ├ 4l. Heavy Springs : None
│ ├ 4m. Arb Front
│ │ ├ 4m1. Stiffness : 8000.00000
│ │ ├ 4m2. Controller
│ │ │ ├ 4l9a. Name : None
│ │ │ ├ 4l9b. Stages : None
│ ├ 4n. Arb Rear
│ │ ├ 4n1. Stiffness : 6000.00000
│ │ ├ 4n2. Controller
│ │ │ ├ 4l9a. Name : None
│ │ │ ├ 4l9b. Stages : None
│ ├ 4o. Flex Bar Front
│ │ ├ 4o1. Stiffness : 0.00000
│ │ ├ 4o2. Controller
│ │ │ ├ 4l9a. Name : None
│ │ │ ├ 4l9b. Stages : None
│ ├ 4p. Flex Bar Rear
│ │ ├ 4p1. Stiffness : 10000.00000
│ │ ├ 4p2. Controller
│ │ │ ├ 4l9a. Name : None
│ │ │ ├ 4l9b. Stages : None
│ ├ 4q. Dampers Controller : None
│ └ 4r. Has Dampers Cockpit Settings : false
├ 5. Drivetrain Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.drivetrain
├ 6. Gearbox Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.gearbox
├ 7. Clutch Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.clutch
├ 8. Engine Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.carengine
├ 9. Brakes Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.brakesystem
├ 10. Steering System
│ ├ 10a. Four W S Controllers
```

```
│ │ ├ 4l9a. Name : None
│ │ ├ 4l9b. Stages : None
├ 11. Electronics
│ ├ 11a. T C
│ │ ├ 11a1. Has T C2 : false
│ │ ├ 11a2. Frequency Hz : 150.00000
│ │ ├ 11a3. Min Speed Kmh : 20.00000
│ │ ├ 11a4. Gear Change Time : 0.08000
│ │ ├ 11a5. Min Cut Level : 10.00000
│ │ ├ 11a6. Max Cut Level : 1.00000
│ │ ├ 11a7. Settings 1 :
│ │ │ ├ 11a7a. Min Slip Ratio : -1.00000
│ │ │ ├ 11a7b. Max Slip Ratio : -1.00000
│ │ │ ├ 11a7c. Ref Slip Angle Deg : 0.00000
│ │ │ ├ 11a7d. Engine Cut Level : 0.00000
│ │ │ ├ 11a7e. Angular A C Cgain : 0.00000
│ │ │ ├ 11a7f. Oversteer Gain : 0.00000
│ │ │ ├ 11a7g. Slip Angle Activation Deg : 0.00000
│ ├ 11b. A B S
│ │ ├ 11b1. Settings 1
│ │ │ ├ 11b1a. Min Slip Ratio : -1.00000
│ │ │ ├ 11b1b. Max Slip Ratio : -1.00000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 0.00000
│ │ │ ├ 11b1d. Cut Level : 0.00000
│ │ │ ├ 11b1f. Max Torque Variation : 0.70000
│ │ ├ 11b1. Settings 2
│ │ │ ├ 11b1a. Min Slip Ratio : 0.06500
│ │ │ ├ 11b1b. Max Slip Ratio : 0.12000
│ │ │ ├ 11b1c. Ref Slip Angle Deg : 8.00000
│ │ │ ├ 11b1d. Cut Level : 0.40000
│ │ │ ├ 11b1f. Max Torque Variation : 0.75000
│ │ ├ Frequency : 40.00000
│ │ ├ Channels : 4
│ │ ├ Min Speed Kmh : 0.00000
│ ├ 11c. E D L
│ │ ├ 11c1. Active : true
│ │ ├ 11c2. Brake Torque Power : 40.00000
│ │ ├ 11c3. Brake Torque Coast : 200.00000
│ │ ├ 11c4. Dead Zone Coast : 0.05000
│ │ ├ 11c5. Dead Zone Power : 0.10000
│ │ ├ 11c6. Max Spin Power : 0.40000
│ │ ├ 11c7. Max Spin Coaster : 0.20000
│ │ ├ 11c8. Min Speed : 0.00000
│ ├ 11d. E S P
│ │ ├ 11d1. Frequency Hz : 0.00000
│ │ ├ 11d2. Min Speed Kmh : 0.00000
│ │ ├ 11d3. Settings : None
├ 12. Electronics Path : None
├ 13. Controls
│ ├ 13a. Ff Mult : 2.90000
│ ├ 13b. Steer Lock : 630.00000
│ ├ 13c. Steer Ratio : -19.60000
│ ├ 13d. Linear Steer Rod Ratio : 0.00220
│ └ 13e. Steer Assist : 1.00000
├ 14. Box Colliders 1
│ ├ 14a. Center : 0.00000, -0.24000, -0.29000
```

```
│ ├ 14b. Size : 1.40000, 0.10000, 3.23000
│ └ 14c. Pitch Rotation Deg : 0.00000
├ 15. Front Tyre Compounds 1 :
content\cars\common_phsx\tyres\road\road_195_55_13.tyre
├ 16. Rear Tyre Compounds 1 :
content\cars\common_phsx\tyres\road\road_195_55_13.tyre
├ 17. Aero
│ ├ 17a. Slip Gain Multiple : 1.00000
│ ├ 17b. Speed Factor Mult : 2.00000
│ ├ 17c. Downforces : None
│ ├ 17d. Front Lift : None
│ ├ 17e. Rear Lift : None
│ ├ 17f. Drag : None
│ └ 17g. Wings Path 1 :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo0.wing
├ 18. Drs : None
├ 19. Ers : None
├ 20. Setup Limits :
content\cars\ks_renault_5_gt_turbo\data\setup\renault_5_gt_turbo.carsetu
plimits
├ 21. Collider Mesh :
content\cars\ks_renault_5_gt_turbo\collider\renault_5_gt_turbo_collider.
mesh
├ 22. Body Mesh Offset
│ ├ 22a. Position : 0.000, -0.632, -0.340
│ ├ 22b. Rotation : 0.700, 0.000, 0.000
│ └ 22c. Scale : 1.000, 1.000, 1.000
├ 23. Stock Setup :
content\cars\ks_renault_5_gt_turbo\data\setup\renault_5_gt_turbo.carsetu
p
├ 24. Ai Setup : None
├ 25. Wet Setup : None
├ 26. Performance Modes : None
├ 27. Ai Car Data :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo.aicardata
└ 28. mm : 1
```

4. Car Engine [.carengine]

A. Description

I. General Description

The **Car Engine** asset is the heart of the vehicle's propulsion system within the simulation engine. While previous assets focused on chassis dimensions (Car Data) or deceleration (BrakeSystem), the Car Engine file dictates how the vehicle generates mechanical energy.

It maps the complete thermodynamic and rotational behavior of the internal combustion engine (ICE), electric motor, or hybrid powertrain. It calculates how much raw power is produced at any given RPM, how the engine responds to the driver's throttle inputs, how it builds or sheds rotational speed (inertia), and how its performance is impacted by thermal limits and turbocharging.

II. Area of Influence / Impact on Vehicle Dynamics

The parameters configured in the Car Engine asset heavily influence the vehicle's straight-line speed, drivability, and endurance:

- **Acceleration & Top Speed:** Driven entirely by the torque output across the rev range. The shape of the power curve dictates how fast the car pulls out of corners and its absolute top speed on straights.
- **Traction Control & Drivability:** How smoothly or aggressively the engine delivers torque when the driver steps on the gas pedal. A snappy engine makes throttle modulation difficult, increasing wheelspin.
- **Engine Braking & Corner Entry**: When the driver lifts off the throttle, internal engine friction and compression create a drag torque (Engine Brake). This slows down the driven wheels, acting as a natural brake balance shift toward the drive axle.
- **Fuel Consumption & Thermal Management:** Dictates the fuel burn rate relative to RPM and throttle positioning, directly affecting pit-stop strategy. It also manages engine temperatures, where overheating can cause component failure or power loss.

III. Key Architecture & Data Fields Explained

The data within a Car Engine schema is typically divided into **Power Generation**, **Rotational Dynamics**, and **Turbo/Aspiration Systems**.

1 - POWER GENERATION & POWER CURVES

- **Torque Curve / Power Curve:** The foundational lookup table mapping **Engine RPM (X-axis)** to **Torque Output (Y-axis, usually in Nm)**. The engine's raw horsepower is directly calculated from this relationship.

- M**ax RPM / Limiter:** The absolute maximum rotational speed the engine can achieve before a fuel/ignition cut-off is triggered to prevent damage.
- **Idle RPM:** The baseline rotational speed at which the engine runs when there is zero driver throttle input.

2 - ROTATIONAL DYNAMICS & THROTTLE RESPONSE

- **Engine Inertia:** The rotational mass of the flywheel, crankshaft, and pistons. A low inertia value allows the engine to rev up and drop RPMs incredibly fast (like a Formula 1 car), while high inertia creates a slower, heavier revving characteristic (like a heavy V8 or diesel truck).
- **Engine Brake Torque:** The negative torque (resistance) generated by the engine when the throttle is fully released (0%). Crucial for modulating car balance during weight transfers.
- **Throttle Map / Curve:** A lookup table correlating physical pedal position to actual throttle plate opening. This can be linear, aggressive (for immediate bite), or progressive (for easier throttle control in wet weather).

3 - ASPIRATION, TURBOCHARGING & THERMAL BEHAVIOR

- **Turbo Boost / Wastegate Settings:** Configures turbocharger behavior, including maximum boost pressure, spool-up time (turbo lag), and wastegate limits.
- **Fuel Consumption Factor:** The efficiency coefficient of fuel burn per unit of power generated, used to compute real-time fuel depletion.
- **Optimal Temperature & Damage Thresholds:** The ideal operational thermal window for water and oil. Running the engine above these critical thresholds triggers progressive performance degradation or catastrophic engine failure.

IV. Short Interpretation of Asset Implementation

When analyzing an engine's physics profile, the data layout reveals the distinct mechanical identity of the powertrain:

- **The High-Revving Naturally Aspirated Profile (e.g., V10 GT3 / Vintage Race Cars):** Characterized by an ultra-low Engine Inertia and a linear Torque Curve that peaks very high up in the rev range (8000 + RPM). These engines have a sharp, instantaneous response to the throttle and feature high Engine Brake Torque, meaning lifting off the gas sharply stabilizes or upsets the rear axle depending on vehicle balance.
- **The Modern Turbocharged Profile (e.g., Modern GT3 / Le Mans Hypercars):** Features a flat, wide plateau in the Torque Curve across the mid-range RPMs rather than a steep peak. It includes complex turbocharger data blocks defining spool rates. The throttle map is often configured progressively to mask "turbo kick"—the sudden surge of torque that happens when boost pressure builds up—ensuring the tires aren't instantly overwhelmed.
- **The Hybrid/EV Assist Profile:** Includes overlaying curves where an electric motor delivers instant maximum torque at 0 RPM, seamlessly tapering off as the internal combustion engine reaches its optimal power band higher up.

B. Schema

```
├ 1. Engine Type : enum
├ 2. Inertia : float
├ 3. Power Curve : string - path
├ 4. Coast Curve : string - path
├ 5. Maps [x] : object | can have multiple Maps
│ ├ 5a. Type : enum
│ ├ 5b. Power Mult : float
│ ├ 5c. Consumption Mult : float
│ ├ 5d. Throttle Response Curve : string - path
│ ├ 5e. Throttle Gain K R P M : float
│ ├ 5f. Throttle Ref R P M Move : float
│ ├ 5g. Throttle Lag Up : float
│ ├ 5h. Throttle Lag Dn : float
│ └ 5i. Throttling Factor : float
├ 6. Minimum : integer
├ 7. Limiter : integer
```

Enum - Car Engine

1	Engine Type	motor, combustion
5a	Type	gamma, cervone
16b1	Input Var	UndefinedInput, Brake, Gas, LatG, LonG, Steer, Speed, Gear, SlipRatioFrontAVG, SlipRatioRearAVG, SlipRatioFrontMAX, SlipRatioRearMAX, SlipAngleFrontAVG, SlipAngleRearAVG, SlipAngleFrontMAX, SlipAngleRearMAX, OversteerFactor, RearSpeedRatio, SteerDEG, Const, RPMS, WheelSteerDEG, LoadSpreadLF, LoadSpreadRF, AvgTravelRear, SusTravelLR, SusTravelRR, SteerYawDeltaLeft, SteerYawDeltaRight, ErsChargeLevel, ErsCoastTorque
16b2	CombinatorMode	UndefinedMode, Add, Mult

```
├ 8. Limiter Cycles : integer
├ 9. Throttle Response Curve : string - path
├ 10. Throttle Lag Up : float
├ 11. Throttle Lag Dn : float
├ 12. Throttle Rev Chocking : float
├ 13. Ignition Time S : float
├ 14. Starter Engine Torque : float
├ 15. Start E C U Assist : object
│ ├ 15a. Rpm Range : float
│ ├ 15b. Gain : float
│ ├ 15c. Speed Range K H M : float
│ ├ 15d. Rpm Limiter : float
│ ├ 15e. Limiter Cycles : integer
│ ├ 15f. Use Clutch : boolean
│ ├ 15g. Slip Ratio Target : float
│ ├ 15h. Clutch Gain : float
│ └ 15i. Gas Gain : float
├ 16. Turbo Controllers [x] : object with an array of stages within |
can have multiple Turbo Controllers
│ ├ 16a. Name : string
```

```
│  ├ 16b. Stages [x] | object | can have multiple Stage
│  │ ├ 16b1. Input Var : enum
│  │ ├ 16b2. Combinator Mode : enum
│  │ ├ 16b3. Lut : string - path
│  │ ├ 16b4. Filter Gain : float
│  │ ├ 16b5. Up Limit : float
│  │ ├ 16b6. Down Limit : float
│  │ ├ 16b7. Current Value : float
│  └ └ 16b8. Const Value : float
├ 17. Waste Gate Controllers [x] : object with an array of stages within
│ can have multiple Waste Gate Controllers
│  ├ 16a. Name : string
│  ├ 16b. Stages [x] | object | can have multiple Stage
│  │ ├ 16b1. Input Var : enum
│  │ ├ 16b2. Combinator Mode : enum
│  │ ├ 16b3. Lut : string - path
│  │ ├ 16b4. Filter Gain : float
│  │ ├ 16b5. Up Limit : float
│  │ ├ 16b6. Down Limit : float
│  │ ├ 16b7. Current Value : float
│  └ └ 16b8. Const Value : float
├ 18. Max Turbo Boost : float
├ 19. Bov Threshold : float
├ 20. Turbos To Load [x] : string - path | can have multiple Turbos To
Load
├ 21. Battery Data : object
│  ├ 21a. Capacity Kwh : float
│  ├ 21b. Dischage Efficiency : float
│  ├ 21c. Charge Efficiency : float
│  ├ 21d. Temp Eff Loss Per Deg : float
│  ├ 21e. Convection K : float
│  ├ 21f. Convection Forced K : float
│  ├ 21g. Thermal Capacity : float
│  ├ 21h. Max Temp : float
└  └ 21i. Min Temp : float
```

C. Example data

I. Chosen Car Engine for Example

- Alpine A290 b (slug : ks_alpine_a290_b)
- Ferrari SF 25 (slug : ks_ferrari_sf_25)
- Chevrolet Camaro ZL1 (slug : ks_renault_5_gt_turbo)
- Datsun 240z Fairlady (slug : ks_datsun_240z_fairlady)

II. Example

Alpine A290 b

```
├ 1. Engine Type : motor
├ 2. Inertia : 0.06000
├ 3. Power Curve : content\cars\ks_alpine_a290_b\data\HEADER_POWER.curve
├ 4. Coast Curve : content\cars\ks_alpine_a290_b\data\coast.curve
├ 5. Maps : None
├ 6. Minimum : 0
├ 7. Limiter : 15200
├ 8. Limiter Cycles : 20
├ 9. Throttle Response Curve :
content\cars\ks_alpine_a290_b\data\throttle.curve
├ 10. Throttle Lag Up : 0.00000
├ 11. Throttle Lag Dn : 0.00000
├ 12. Throttle Rev Chocking : 0.00000
├ 13. Ignition Time S : 0.00000
├ 14. Starter Engine Torque : 0.00000
├ 15. Start E C U Assist
│ ├ 15a. Rpm Range : 0.00000
│ ├ 15b. Gain : 0.00000
│ ├ 15c. Speed Range K H M : 0.00000
│ ├ 15d. Rpm Limiter : 0.00000
│ ├ 15e. Limiter Cycles : 0
│ ├ 15f. Use Clutch : false
│ ├ 15g. Slip Ratio Target : 0.00000
│ ├ 15h. Clutch Gain : 0.00000
│ └ 15i. Gas Gain : 0.00000
├ 16. Turbo Controllers : None
├ 17. Waste Gate Controllers : None
├ 18. Max Turbo Boost : 0.00000
├ 19. Bov Threshold : 0.00000
├ 20. Turbos To Load : None
├ 21. Battery Data
│ ├ 21a. Capacity Kwh : 52.000
│ ├ 21b. Dischage Efficiency : 0.800
│ ├ 21c. Charge Efficiency : 0.600
│ ├ 21d. Temp Eff Loss Per Deg : 0.01000
│ ├ 21e. Convection K : 0.00100
│ ├ 21f. Convection Forced K : 0.00050
│ ├ 21g. Thermal Capacity : 100.000
│ ├ 21h. Max Temp : 35.000
└ └ 21i. Min Temp : 15.000
```

Ferrari SF 25

```
├ 1. Engine Type : combustion
├ 2. Inertia : None
├ 3. Power Curve : content\cars\ks_ferrari_sf_25\data\HEADER_POWER.curve
├ 4. Coast Curve : content\cars\ks_ferrari_sf_25\data\coast_smooth.curve
├ 5. Maps 1
│ ├ 5a. Type : gamma
│ ├ 5b. Power Mult : 1.00000
│ ├ 5c. Consumption Mult : 0.60000
│ ├ 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
```

```
│ ├ 5f. Throttle Ref R P M Move : 0.00000
│ ├ 5g. Throttle Lag Up : 0.00000
│ ├ 5h. Throttle Lag Dn : 0.00000
│ └ 5i. Throttling Factor : 0.00000
├ 5. Maps 2
│ ├ 5a. Type : gamma
│ ├ 5b. Power Mult : 1.00000
│ ├ 5c. Consumption Mult : 0.60000
│ ├ 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
│ ├ 5f. Throttle Ref R P M Move : 0.00000
│ ├ 5g. Throttle Lag Up : 0.00000
│ ├ 5h. Throttle Lag Dn : 0.00000
│ └ 5i. Throttling Factor : 0.00000
├ 5. Maps 3
│ ├ 5a. Type : gamma
│ ├ 5b. Power Mult : 0.89000
│ ├ 5c. Consumption Mult : 0.56000
│ ├ 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
│ ├ 5f. Throttle Ref R P M Move : 0.00000
│ ├ 5g. Throttle Lag Up : 0.00000
│ ├ 5h. Throttle Lag Dn : 0.00000
│ └ 5i. Throttling Factor : 0.00000
├ 5. Maps 4
│ ├ 5a. Type : gamma
│ ├ 5b. Power Mult : 0.89000
│ ├ 5c. Consumption Mult : 0.56000
│ ├ 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
│ ├ 5f. Throttle Ref R P M Move : 0.00000
│ ├ 5g. Throttle Lag Up : 0.00000
│ ├ 5h. Throttle Lag Dn : 0.00000
│ └ 5i. Throttling Factor : 0.00000
├ 5. Maps 5
│ ├ 5a. Type : gamma
│ ├ 5b. Power Mult : 0.82000
│ ├ 5c. Consumption Mult : 0.55000
│ ├ 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
│ ├ 5f. Throttle Ref R P M Move : 0.00000
│ ├ 5g. Throttle Lag Up : 0.00000
│ ├ 5h. Throttle Lag Dn : 0.00000
│ └ 5i. Throttling Factor : 0.00000
├ 5. Maps 6
│ ├ 5a. Type : gamma
│ ├ 5b. Power Mult : 0.82000
│ ├ 5c. Consumption Mult : 0.55000
│ ├ 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
│ ├ 5f. Throttle Ref R P M Move : 0.00000
```

```
│ ├ 5g. Throttle Lag Up : 0.00000
│ ├ 5h. Throttle Lag Dn : 0.00000
│ └ 5i. Throttling Factor : 0.00000
├ 5. Maps 7
│ ├ 5a. Type : gamma
│ ├ 5b. Power Mult : 0.72000
│ ├ 5c. Consumption Mult : 0.50000
│ ├ 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
│ ├ 5f. Throttle Ref R P M Move : 0.00000
│ ├ 5g. Throttle Lag Up : 0.00000
│ ├ 5h. Throttle Lag Dn : 0.00000
│ └ 5i. Throttling Factor : 0.00000
├ 5. Maps 8
│ ├ 5a. Type : gamma
│ ├ 5b. Power Mult : 0.72000
│ ├ 5c. Consumption Mult : 0.50000
│ ├ 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
│ ├ 5f. Throttle Ref R P M Move : 0.00000
│ ├ 5g. Throttle Lag Up : 0.00000
│ ├ 5h. Throttle Lag Dn : 0.00000
│ └ 5i. Throttling Factor : 0.00000
├ 5. Maps 9
│ ├ 5a. Type : gamma
│ ├ 5b. Power Mult : 0.95000
│ ├ 5c. Consumption Mult : 0.58500
│ ├ 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
│ ├ 5f. Throttle Ref R P M Move : 0.00000
│ ├ 5g. Throttle Lag Up : 0.00000
│ ├ 5h. Throttle Lag Dn : 0.00000
│ └ 5i. Throttling Factor : 0.00000
├ 5. Maps 10
│ ├ 5a. Type : gamma
│ ├ 5b. Power Mult : 0.95000
│ ├ 5c. Consumption Mult : 0.58500
│ ├ 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
│ ├ 5f. Throttle Ref R P M Move : 0.00000
│ ├ 5g. Throttle Lag Up : 0.00000
│ ├ 5h. Throttle Lag Dn : 0.00000
│ └ 5i. Throttling Factor : 0.00000
├ 5. Maps 11
│ ├ 5a. Type : gamma
│ ├ 5b. Power Mult : 1.00000
│ ├ 5c. Consumption Mult : 0.60000
│ ├ 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_progressive.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
│ ├ 5f. Throttle Ref R P M Move : 0.00000
│ ├ 5g. Throttle Lag Up : 0.00000
```

```
│ ├ 5h. Throttle Lag Dn : 0.00000
│ └ 5i. Throttling Factor : 0.00000
├ 5. Maps 12
│ ├ 5a. Type : gamma
│ ├ 5b. Power Mult : 1.00000
│ ├ 5c. Consumption Mult : 0.60000
│ ├ 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_wet.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
│ ├ 5f. Throttle Ref R P M Move : 0.00000
│ ├ 5g. Throttle Lag Up : 0.00000
│ ├ 5h. Throttle Lag Dn : 0.00000
│ └ 5i. Throttling Factor : 0.00000
├ 6. Minimum : 2000
├ 7. Limiter : 15000
├ 8. Limiter Cycles : 60
├ 9. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
├ 10. Throttle Lag Up : 0.00000
├ 11. Throttle Lag Dn : 0.00000
├ 12. Throttle Rev Chocking : 0.00000
├ 13. Ignition Time S : 0.00000
├ 14. Starter Engine Torque : 40.00000
├ 15. Start E C U Assist
│ ├ 15a. Rpm Range : 0.00000
│ ├ 15b. Gain : 0.00000
│ ├ 15c. Speed Range K H M : 0.00000
│ ├ 15d. Rpm Limiter : 0.00000
│ ├ 15e. Limiter Cycles : 0
│ ├ 15f. Use Clutch : false
│ ├ 15g. Slip Ratio Target : 0.00000
│ ├ 15h. Clutch Gain : 0.00000
│ └ 15i. Gas Gain : 0.00000
├ 16. Turbo Controllers 1
│ ├ 16a. Name : None
│ ├ 16b. Stages 1
│ │ ├ 16b1. Input Var : RPMS
│ │ ├ 16b2. Combinator Mode : Add
│ │ ├ 16b3. Lut :
content\cars\ks_ferrari_sf_25\data\ctrl_turbo0CONTROLLER_0.curve
│ │ ├ 16b4. Filter Gain : 0.99000
│ │ ├ 16b5. Up Limit : 10000.00000
│ │ ├ 16b6. Down Limit : 0.00000
│ │ ├ 16b7. Current Value : 0.00000
│ └ └ 16b8. Const Value : 0.00000
├ 17. Waste Gate Controllers : None
├ 18. Max Turbo Boost : 2.20000
├ 19. Bov Threshold : 0.00000
├ 20. Turbos To Load 1 :
content\cars\ks_ferrari_sf_25\data\ks_ferrari_sf_250.turbo
├ 21. Battery Data
│ ├ 21a. Capacity Kwh : 0.000
│ ├ 21b. Dischage Efficiency : 0.000
│ ├ 21c. Charge Efficiency : 0.000
│ ├ 21d. Temp Eff Loss Per Deg : 0.00000
│ ├ 21e. Convection K : 0.00000
```

```
│ ├ 21f. Convection Forced K : 0.00000
│ ├ 21g. Thermal Capacity : 0.000
│ ├ 21h. Max Temp : 0.000
└ └ 21i. Min Temp : 0.000
```

Chevrolet Camaro ZL1

```
├ 1. Engine Type : combustion
├ 2. Inertia : 0.45000
├ 3. Power Curve :
content\cars\ks_chevrolet_camaro_zl1\data\HEADER_POWER.curve
├ 4. Coast Curve : content\cars\ks_chevrolet_camaro_zl1\data\coast.curve
├ 5. Maps 1
│ ├ 5a. Type : gamma
│ ├ 5b. Power Mult : 1.00000
│ ├ 5c. Consumption Mult : 1.00000
│ ├ 5d. Throttle Response Curve :
content\cars\ks_chevrolet_camaro_zl1\data\throttlemap_response.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
│ ├ 5f. Throttle Ref R P M Move : 0.00000
│ ├ 5g. Throttle Lag Up : 0.80000
│ ├ 5h. Throttle Lag Dn : 0.92000
│ └ 5i. Throttling Factor : 0.00000
├ 6. Minimum : 900
├ 7. Limiter : 6600
├ 8. Limiter Cycles : 30
├ 9. Throttle Response Curve :
content\cars\ks_chevrolet_camaro_zl1\data\throttlemap_response.curve
├ 10. Throttle Lag Up : 0.80000
├ 11. Throttle Lag Dn : 0.90000
├ 12. Throttle Rev Chocking : 0.00000
├ 13. Ignition Time S : 0.00000
├ 14. Starter Engine Torque : 60.00000
├ 15. Start E C U Assist
│ ├ 15a. Rpm Range : 1500.00000
│ ├ 15b. Gain : 0.10000
│ ├ 15c. Speed Range K H M : 120.00000
│ ├ 15d. Rpm Limiter : 3800.00000
│ ├ 15e. Limiter Cycles : 0
│ ├ 15f. Use Clutch : true
│ ├ 15g. Slip Ratio Target : 0.14000
│ ├ 15h. Clutch Gain : 5.00000
│ └ 15i. Gas Gain : 1.75000
├ 16. Turbo Controllers 1
│ ├ 16a. Name : None
│ ├ 16b. Stages 1
│ │ ├ 16b1. Input Var : RPMS
│ │ ├ 16b2. Combinator Mode : Add
│ │ ├ 16b3. Lut :
content\cars\ks_chevrolet_camaro_zl1\data\rpms_boost.curve
│ │ ├ 16b4. Filter Gain : 0.00000
│ │ ├ 16b5. Up Limit : 1.20000
│ │ ├ 16b6. Down Limit : 0.00000
│ │ ├ 16b7. Current Value : 0.00000
```

```
│ └ └ 16b8. Const Value : 0.00000
├ 17. Waste Gate Controllers : None
├ 18. Max Turbo Boost : 0.60000
├ 19. Bov Threshold : 0.00000
├ 20. Turbos To Load 1 :
content\cars\ks_chevrolet_camaro_zl1\data\ks_chevrolet_camaro_zl1_compre
ssor.turbo
├ 21. Battery Data
│ ├ 21a. Capacity Kwh : 0.000
│ ├ 21b. Dischage Efficiency : 0.000
│ ├ 21c. Charge Efficiency : 0.000
│ ├ 21d. Temp Eff Loss Per Deg : 0.00000
│ ├ 21e. Convection K : 0.00000
│ ├ 21f. Convection Forced K : 0.00000
│ ├ 21g. Thermal Capacity : 0.000
│ ├ 21h. Max Temp : 0.000
└ └ 21i. Min Temp : 0.000
```

Datsun 240z Fairlady

```
├ 1. Engine Type : combustion
├ 2. Inertia : 0.18000
├ 3. Power Curve :
content\cars\ks_datsun_240z_fairlady\data\HEADER_POWER.curve
├ 4. Coast Curve :
content\cars\ks_datsun_240z_fairlady\data\ks_datsun_240z_fairlady_coast1
.curve
├ 5. Maps 1
│ ├ 5a. Type : cervone
│ ├ 5b. Power Mult : 1.00000
│ ├ 5c. Consumption Mult : 1.00000
│ ├ 5d. Throttle Response Curve :
content\cars\ks_datsun_240z_fairlady\data\throttle.curve
│ ├ 5e. Throttle Gain K R P M : 0.00000
│ ├ 5f. Throttle Ref R P M Move : 0.00000
│ ├ 5g. Throttle Lag Up : 0.00000
│ ├ 5h. Throttle Lag Dn : 0.00000
│ └ 5i. Throttling Factor : 0.00000
├ 6. Minimum : 600
├ 7. Limiter : 7000
├ 8. Limiter Cycles : 8
├ 9. Throttle Response Curve :
content\cars\ks_datsun_240z_fairlady\data\throttle.curve
├ 10. Throttle Lag Up : 0.00000
├ 11. Throttle Lag Dn : 0.00000
├ 12. Throttle Rev Chocking : 0.00000
├ 13. Ignition Time S : 0.00000
├ 14. Starter Engine Torque : 30.00000
├ 15. Start E C U Assist
│ ├ 15a. Rpm Range : 0.00000
│ ├ 15b. Gain : 0.00000
│ ├ 15c. Speed Range K H M : 0.00000
│ ├ 15d. Rpm Limiter : 0.00000
│ ├ 15e. Limiter Cycles : 0
│ ├ 15f. Use Clutch : false
```

```
│ ├ 15g. Slip Ratio Target : 0.00000
│ ├ 15h. Clutch Gain : 0.00000
│ └ 15i. Gas Gain : 0.00000
├ 16. Turbo Controllers : None
├ 17. Waste Gate Controllers : None
├ 18. Max Turbo Boost : 0.00000
├ 19. Bov Threshold : 0.00000
├ 20. Turbos To Load : None
├ 21. Battery Data
│ ├ 21a. Capacity Kwh : 0.000
│ ├ 21b. Dischage Efficiency : 0.000
│ ├ 21c. Charge Efficiency : 0.000
│ ├ 21d. Temp Eff Loss Per Deg : 0.00000
│ ├ 21e. Convection K : 0.00000
│ ├ 21f. Convection Forced K : 0.00000
│ ├ 21g. Thermal Capacity : 0.000
│ ├ 21h. Max Temp : 0.000
└ └ 21i. Min Temp : 0.000
```

5. Car Setup [.carsetup]

A. Description

I. General Description

The **Car Setup** asset is the primary adjustment interface that translates mechanical physics constants into highly customizable track settings. While other files define the immovable physical constraints of a vehicle (like the frame weight or engine potential), the CAR SETUP file acts as the pit garage dashboard.

It stores all adjustable parameters that engineers and drivers modify to adapt the vehicle to specific racetracks, weather conditions, or driving styles. It acts as a set of instructions applied on top of the base suspension, aerodynamics, and drivetrain geometries to maximize grip, control stability, and balance the car's handling.

II. Area of Influence / Impact on Vehicle Dynamics

The variables configured in the CAR SETUP asset have an immediate and dramatic impact on every dynamic phase of lap execution:

- **Mechanical Grip & Tire Patch Optimization:** By altering alignment and tire pressures, this asset controls how flat the tire stays against the tarmac during high-load cornering, directly maximizing lateral adhesion.
- **Aerodynamic Platform Control:** Ride heights and rake (the angle of the car's floor relative to the track) dictate how much downforce wings and underbody diffusers generate, dramatically shifting high-speed stability and drag.
- **Transient Phase Handling (Weight Transfer):** Springs and dampers regulate *how fast* and *how far* the body rolls when cornering, pitches under braking, or squats under acceleration, giving the driver a predictable or snappy platform.
- **Corner Exit Propulsion (Power Delivery):** Differential settings govern how power shifts between the drive wheels, dictating whether a car neatly clips an exit apex or snaps into wheelspin/snap-oversteer.

III. Key Architecture & Data Fields Explained

The parameters in a CAR SETUP file are traditionally compartmentalized into five core operational groups: **Tyres**, **Aerodynamics**, **Suspension (Springs/Alignment)**, **Dampers (Shock Absorbers)**, and **Drivetrain (Differential)**.

1 - TYRES (THE CONTACT PATCH)

- **Cold Pressure:** The initial inflating pressure of the tire before leaving the pitlane. It directly influences the hot operating pressure and optimal carcass temperature window.

- **Camber (Front/Rear):** The inward or outward tilt of the wheels when viewed from the front. Negative camber ensures that when the car rolls into a corner, the outer tire leans flat against the asphalt for maximum cornering grip.

- **Toe (Front/Rear):** The angle of the wheels relative to the vehicle's longitudinal centerline. Front *toe-out* increases turn-in agility, while rear *toe-in* stabilizes the rear end during straight-line acceleration and heavy braking.

2 - AERODYNAMICS (THE AIRFLOW PLATFORM)

- **Wing Angles (Front/Rear Spoiler):** The mechanical inclination of aero planes. Higher rear wing angles increase downforce (stability in fast corners) at the expense of higher straight-line drag (lower top speed).

- **Ride Height (Front/Rear):** The clearance between the car's floor and the ground. Lowering the car increases venturi/ground-effect downforce, but going too low can cause the floor to bottom out on bumps, destroying all grip instantly.

3 - SUSPENSION GEOMETRY & RATES

- **Wheel Rate / Spring Stiffness:** The stiffness of the main suspension coil springs. Stiffer springs stabilize the aero platform and stop body roll, but reduce compliance on bumpy circuits or kerbs.

- **Anti-Roll Bar (ARB - Front/Rear):** A torsion bar linking the left and right suspension sides. It purely resists body roll in corners. Adjusting the stiffness ratio between front and rear ARBs is the primary tool for shifting handling balance between understeer and oversteer.

4 - DAMPERS (TRANSIENT SHOCK ABSORPTION)

Dampers control the speed of suspension movement and are split into travel directions and shaft speeds:

- **Bump / Compression (Slow & Fast):** Resists the suspension compressing. *Slow Bump* manages body roll from driver inputs; *Fast Bump* absorbs high-speed impacts like hitting a track kerb or bump.

- **Rebound / Extension (Slow & Fast):** Resists the suspension extending back out. *Slow Rebound* regulates how the car settles as it exits a corner or releases the brakes; *Fast Rebound* controls how fast the tire snaps back down to touch the track after a kerb strike.

5 - DRIVETRAIN & DIFFERENTIAL

- **Diff Preload:** The static friction locking the differential when zero throttle is applied. Higher preload stabilizes the car during snappy, mid-corner lift-offs.

- **Diff Power (Lock):** The locking percentage under acceleration. High lock forces both tires to spin together for maximum exit drive, but can cause power-understeer or sudden power-oversteer if traction breaks.

- **Diff Coast (Lock):** The locking percentage under deceleration/braking. High coast lock keeps the rear stable under heavy braking into a corner but resists initial turn-in agility.

IV. Interpretation of Setup Configuration Strategies

When reading a car's exported .SETUP file, the values map out a very specific racing engineering intent based on the target circuit layout:

- **The High-Downforce Monza/Spa vs. Monaco Profile:** For high-speed tracks like Spa-Francorchamps or Monza, you will notice ultra-low wing angles to minimize drag, stiffer springs to support the high aero load compressing the suspension, and lower ride heights. For Monaco, the wings are cranked to the absolute maximum, springs are softened to absorb bumpy street surfaces, and toe angles are aggressive to force the car around tight hairpins.

- **The Endurance Balance Management:** In high-tier setups, engineers deliberately leave a margin in the ride heights and tire pressures. This accommodates changing ambient temperatures over a multi-hour race stint and accounts for structural components wearing out, ensuring the aerodynamic floor platform stays stable regardless of fuel burn or tyre degradation.

B. Schema

```
├ 1. Import Setup : string - path
├ 2. Mechanical Balance : object
│ ├ 2a. Arbs [x] : float | can have multiple Arbs
│ ├ 2b. Steer Ratio : float
│ ├ 2c. Brakes : object
│ │ ├ 2c1. Front Bias : float
│ │ ├ 2c2. Torque Multiplier : float
│ │ └ 2c3. Brake Ducts [x] : float | can have multiple Brake Ducts
│ ├ 2d. Differential : object
│ │ ├ 2d1. Power : float
│ │ ├ 2d2. Coast : float
│ └ └ 2d3. Preload : float
├ 3. Suspensions [x] : object
│ ├ 3a. Wheel Rate : float
│ ├ 3b. Bump Stop Up : object
│ │ ├ 3b1. Range : float
│ │ └ 3b2. Rate : float
│ ├ 3c. Bump Stop Down : object
│ │ ├ 3c1. Range : float
│ │ └ 3c2. Rate : float
│ ├ 3d. Helper Rate : float
│ └ 3e. Helper Range : float
├ 4. Dampers [x] : object | can have multiple Dampers
│ ├ 4a. Slow Bump : float
│ ├ 4b. Fast Bump : float
│ ├ 4c. Slow Rebound : float
│ └ 4e. Fast Rebound : float
├ 5. Alignements [x] : object | can have multiple Alignements
│ ├ 5a. Pressure : float
│ ├ 5b. Camber : float
│ ├ 5c. Toe : float
│ ├ 5d. Caster : float
│ ├ 5e. Static Camber : float
```

```
│ ├ 5f. Toe Out Linear : float
│ └ 5g. Compound : float
├ 6. Electronics : object
│ ├ 6a. Tc1 : float
│ ├ 6b. Tc2 : float
│ ├ 6c. Abs : float
│ ├ 6d. Esc : float
│ ├ 6e. Ebb : float
│ ├ 6f. Engine Map : float
│ ├ 6g. Telemetry Laps to Record : float
│ ├ 6h. Turbo Boost Lv : float
│ ├ 6i. Ers Deployment Map : float
│ ├ 6j. Ers Recharge Lv : float
│ └ 6k. Ers Heat Charging : float
├ 7. Aero : object
│ ├ 7a. Collar Positions Mm [x] : float | can have multiple Collar
Positions Mm
│ ├ 7b. Front Target Height : float
│ ├ 7c. Rear Target Height : float
│ ├ 7d. Front Wing Angle : float
│ └ 7e. Rear Wing Angle : float
├ 8. Fuel Strategy : object
│ └ 8a. Fuel : float
├ 9. Final State Name : string – path
├ 10. Version : integer
├ 11. Is Setup Shared : boolean
```

C. Example data

I. Chosen Car Engine for Example

- Ausi Sport Quattro (slug : ks_audi_sport_quattro)
- Alfa Romeo Junior (slug : ks_alfa_romeo_junior)
- Ferrari 488 Challenge Evo (slug : ks_ferrari_488_challenge_evo [preset : safe_1]

II. Example

Audi Sport Quattro

```
├ 1. Import Setup : None
├ 2. Mechanical Balance
│ ├ 2a. Arbs 1 : 25000.00000
│ ├ 2a. Arbs 2 : 20000.00000
│ ├ 2b. Steer Ratio : 18.00000
│ ├ 2c. Brakes
│ │ ├ 2c1. Front Bias : 72.00000
│ │ ├ 2c2. Torque Multiplier : 100.00000
│ │ └ 2c3. Brake Ducts : None
│ ├ 2d. Differential
│ │ ├ 2d1. Power : 0.00000
│ │ ├ 2d2. Coast : 0.00000
│ │ └ 2d3. Preload : 0.00000
```

```
├ 3. Suspensions 1
│ ├ 3a. Wheel Rate : 50000.00000
│ ├ 3b. Bump Stop Up
│ │ ├ 3b1. Range : 0.02645
│ │ └ 3b2. Rate : 650.00000
│ ├ 3c. Bump Stop Down
│ │ ├ 3c1. Range : 0.11855
│ │ └ 3c2. Rate : 650.00000
│ ├ 3d. Helper Rate : 0.00000
│ └ 3e. Helper Range : 0.00000
├ 3. Suspensions 2
│ ├ 3a. Wheel Rate : 50000.00000
│ ├ 3b. Bump Stop Up
│ │ ├ 3b1. Range : 0.02645
│ │ └ 3b2. Rate : 650.00000
│ ├ 3c. Bump Stop Down
│ │ ├ 3c1. Range : 0.11855
│ │ └ 3c2. Rate : 650.00000
│ ├ 3d. Helper Rate : 0.00000
│ └ 3e. Helper Range : 0.00000
├ 3. Suspensions 3
│ ├ 3a. Wheel Rate : 42750.00000
│ ├ 3b. Bump Stop Up
│ │ ├ 3b1. Range : 0.04487
│ │ └ 3b2. Rate : 400.00000
│ ├ 3c. Bump Stop Down
│ │ ├ 3c1. Range : 0.04513
│ └ └ 3c2. Rate : 400.00000
├ 3. Suspensions 4
│ ├ 3a. Wheel Rate : 42750.00000
│ ├ 3b. Bump Stop Up
│ │ ├ 3b1. Range : 0.04487
│ │ └ 3b2. Rate : 400.00000
│ ├ 3c. Bump Stop Down
│ │ ├ 3c1. Range : 0.04513
│ │ └ 3c2. Rate : 400.00000
│ ├ 3d. Helper Rate : 0.00000
│ └ 3e. Helper Range : 0.00000
├ 4. Dampers 1
│ ├ 4a. Slow Bump : 6000.00000
│ ├ 4b. Fast Bump : 1500.00000
│ ├ 4c. Slow Rebound : 8000.00000
│ └ 4e. Fast Rebound : 2300.00000
├ 4. Dampers 2
│ ├ 4a. Slow Bump : 6000.00000
│ ├ 4b. Fast Bump : 1500.00000
│ ├ 4c. Slow Rebound : 8000.00000
│ └ 4e. Fast Rebound : 2300.00000
├ 4. Dampers 3
│ ├ 4a. Slow Bump : 5000.00000
│ ├ 4b. Fast Bump : 1200.00000
│ ├ 4c. Slow Rebound : 7000.00000
│ └ 4e. Fast Rebound : 1900.00000
├ 4. Dampers 4
│ ├ 4a. Slow Bump : 5000.00000
│ ├ 4b. Fast Bump : 1200.00000
```

```
│ ├ 4c. Slow Rebound : 7000.00000
│ └ 4e. Fast Rebound : 1900.00000
├ 5. Alignements 1
│ ├ 5a. Pressure : 28.00000
│ ├ 5b. Camber : -1.00000
│ ├ 5c. Toe : -0.03000
│ ├ 5d. Caster : -0.04510
│ ├ 5e. Static Camber : -1.69155
│ ├ 5f. Toe Out Linear : 0.00098
│ └ 5g. Compound : 0.00000
├ 5. Alignements 2
│ ├ 5a. Pressure : 28.00000
│ ├ 5b. Camber : -1.00000
│ ├ 5c. Toe : -0.03000
│ ├ 5d. Caster : -0.04510
│ ├ 5e. Static Camber : -1.76592
│ ├ 5f. Toe Out Linear : 0.00095
│ └ 5g. Compound : 0.00000
├ 5. Alignements 3
│ ├ 5a. Pressure : 28.00000
│ ├ 5b. Camber : -0.60000
│ ├ 5c. Toe : 0.05000
│ ├ 5d. Caster : 0.00000
│ ├ 5e. Static Camber : -0.66797
│ ├ 5f. Toe Out Linear : 0.00022
│ └ 5g. Compound : 0.00000
├ 5. Alignements 4
│ ├ 5a. Pressure : 28.00000
│ ├ 5b. Camber : -0.60000
│ ├ 5c. Toe : 0.05000
│ ├ 5d. Caster : 0.00000
│ ├ 5e. Static Camber : -0.75145
│ ├ 5f. Toe Out Linear : 0.00022
│ └ 5g. Compound : 0.00000
├ 6. Electronics
│ ├ 6a. Tc1 : 0.00000
│ ├ 6b. Tc2 : 0.00000
│ ├ 6c. Abs : 1.00000
│ ├ 6d. Esc : 0.00000
│ ├ 6e. Ebb : 0.000
│ ├ 6f. Engine Map : 0.00000
│ ├ 6g. Telemetry Laps to Record : 0.00000
│ ├ 6h. Turbo Boost Lv : 0.00000
│ ├ 6i. Ers Deployment Map : 0.000
│ ├ 6j. Ers Recharge Lv : 0.000
│ └ 6k. Ers Heat Charging : 0.000
├ 7. Aero
│ ├ 7a. Collar Positions Mm 1 : 112.59116
│ ├ 7a. Collar Positions Mm 2 : 107.20738
│ ├ 7a. Collar Positions Mm 3 : 76.21446
│ ├ 7a. Collar Positions Mm 4 : 75.33346
│ ├ 7b. Front Target Height : 200.00000
│ ├ 7c. Rear Target Height : 200.00000
│ ├ 7d. Front Wing Angle : 0.00000
│ └ 7e. Rear Wing Angle : 0.00000
├ 8. Fuel Strategy
```

```
│ └ 8a. Fuel : 30.00000
├ 9. Final State Name :
ks_audi_sport_quattro_preset_sq_mech_1_preset_sq_visual_1
├ 10. Version : None
├ 11. Is Setup Shared : false
```

Alfa Romeo Junior

```
├ 1. Import Setup
├ 2. Mechanical Balance
│ ├ 2a. Arbs 1 : 40000.00000
│ ├ 2a. Arbs 1 : 38000.00000
│ ├ 2b. Steer Ratio : -14.60000
│ ├ 2c. Brakes
│ │ ├ 2c1. Front Bias : 80.00000
│ │ ├ 2c2. Torque Multiplier : 100.00000
│ │ └ 2c3. Brake Ducts : None
│ ├ 2d. Differential
│ │ ├ 2d1. Power : 0.00000
│ │ ├ 2d2. Coast : 0.00000
│ └ └ 2d3. Preload : 0.00000
├ 3. Suspensions 1
│ ├ 3a. Wheel Rate : 49000.00000
│ ├ 3b. Bump Stop Up
│ │ ├ 3b1. Range : 0.01194
│ │ └ 3b2. Rate : 400.00000
│ ├ 3c. Bump Stop Down
│ │ ├ 3c1. Range : 0.07306
│ │ └ 3c2. Rate : 400.00000
│ ├ 3d. Helper Rate : 0.00000
│ └ 3e. Helper Range : 0.00000
├ 3. Suspensions 2
│ ├ 3a. Wheel Rate : 49000.00000
│ ├ 3b. Bump Stop Up
│ │ ├ 3b1. Range : 0.01194
│ │ └ 3b2. Rate : 400.00000
│ ├ 3c. Bump Stop Down
│ │ ├ 3c1. Range : 0.07306
│ │ └ 3c2. Rate : 400.00000
│ ├ 3d. Helper Rate : 0.00000
│ └ 3e. Helper Range : 0.00000
├ 3. Suspensions 3
│ ├ 3a. Wheel Rate : 45000.00000
│ ├ 3b. Bump Stop Up
│ │ ├ 3b1. Range : -0.00872
│ │ └ 3b2. Rate : 400.00000
│ ├ 3c. Bump Stop Down
│ │ ├ 3c1. Range : 0.0372
│ │ └ 3c2. Rate : 500.00000
│ ├ 3d. Helper Rate : 0.00000
│ └ 3e. Helper Range : 0.00000
├ 3. Suspensions 4
│ ├ 3a. Wheel Rate : 45000.00000
│ ├ 3b. Bump Stop Up
```

```
│  │ ├ 3b1. Range : -0.00872
│  │ └ 3b2. Rate : 400.00000
│  ├ 3c. Bump Stop Down
│  │ ├ 3c1. Range : 0.09372
│  │ └ 3c2. Rate : 500.00000
│  ├ 3d. Helper Rate : 0.00000
│  └ 3e. Helper Range : 0.00000
├ 4. Dampers 1
│ ├ 4a. Slow Bump : 8200.00000
│ ├ 4b. Fast Bump : 3000.00000
│ ├ 4c. Slow Rebound : 10500.00000
│ └ 4e. Fast Rebound : 3300.00000
├ 4. Dampers 2
│ ├ 4a. Slow Bump : 8200.00000
│ ├ 4b. Fast Bump : 3000.00000
│ ├ 4c. Slow Rebound : 10500.00000
│ └ 4e. Fast Rebound : 3300.00000
├ 4. Dampers 3
│ ├ 4a. Slow Bump : 5000.00000
│ ├ 4b. Fast Bump : 3500.00000
│ ├ 4c. Slow Rebound : 6500.00000
│ └ 4e. Fast Rebound : 3300.00000
├ 4. Dampers 4
│ ├ 4a. Slow Bump : 5000.00000
│ ├ 4b. Fast Bump : 3500.00000
│ ├ 4c. Slow Rebound : 6500.00000
│ └ 4e. Fast Rebound : 3300.00000
├ 5. Alignements 1
│ ├ 5a. Pressure : 26.00000
│ ├ 5b. Camber : -1.30000
│ ├ 5c. Toe : -0.05000
│ ├ 5d. Caster : -0.04980
│ ├ 5e. Static Camber : -1.57347
│ ├ 5f. Toe Out Linear : 0.00018
│ └ 5g. Compound : 0.00000
├ 5. Alignements 2
│ ├ 5a. Pressure : 26.00000
│ ├ 5b. Camber : -1.30000
│ ├ 5c. Toe : -0.05000
│ ├ 5d. Caster : -0.04980
│ ├ 5e. Static Camber : -1.56779
│ ├ 5f. Toe Out Linear : 0.00016
│ └ 5g. Compound : 0.00000
├ 5. Alignements 3
│ ├ 5a. Pressure : 27.00000
│ ├ 5b. Camber : -1.40000
│ ├ 5c. Toe : 0.05000
│ ├ 5d. Caster : -0.12000
│ ├ 5e. Static Camber : -1.39969
│ ├ 5f. Toe Out Linear : -0.00142
│ └ 5g. Compound : 0.00000
├ 5. Alignements 4
│ ├ 5a. Pressure : 27.00000
│ ├ 5b. Camber : -1.40000
│ ├ 5c. Toe : 0.05000
│ ├ 5d. Caster : -0.12000
```

```
│ ├ 5e. Static Camber : -1.39560
│ ├ 5f. Toe Out Linear : -0.00143
│ └ 5g. Compound : 0.00000
├ 6. Electronics
│ ├ 6a. Tc1 : 1.00000
│ ├ 6b. Tc2 : 1.00000
│ ├ 6c. Abs : 1.00000
│ ├ 6d. Esc : 0.00000
│ ├ 6e. Ebb : 0.000
│ ├ 6f. Engine Map : 0.00000
│ ├ 6g. Telemetry Laps to Record : 0.00000
│ ├ 6h. Turbo Boost Lv : 0.00000
│ ├ 6i. Ers Deployment Map : 0.000
│ ├ 6j. Ers Recharge Lv : 0.000
│ └ 6k. Ers Heat Charging : 0.000
├ 7. Aero
│ ├ 7a. Collar Positions Mm 1 : 102.44676
│ ├ 7a. Collar Positions Mm 2 : 102.44756
│ ├ 7a. Collar Positions Mm 3 : 105.31527
│ ├ 7a. Collar Positions Mm 4 : 105.31618
│ ├ 7b. Front Target Height : 185.00000
│ ├ 7c. Rear Target Height : 190.00000
│ ├ 7d. Front Wing Angle : 0.00000
│ └ 7e. Rear Wing Angle : 0.00000
├ 8. Fuel Strategy
│ └ 8a. Fuel : 3.00000
├ 9. Final State Name :
ks_alfa_romeo_junior_preset_mln_mech_1_preset_mln_visual_1
├ 10. Version : 0
├ 11. Is Setup Shared : false
```

Ferrari 488 Challenge Evo [preset : safe_1]

```
├ 1. Import Setup : None
├ 2. Mechanical Balance
│ ├ 2a. Arbs 1 : 34000.00000
│ ├ 2a. Arbs 2 : 17000.00000
│ ├ 2b. Steer Ratio : 15.00000
│ ├ 2c. Brakes
│ │ ├ 2c1. Front Bias : 64.00000
│ │ ├ 2c2. Torque Multiplier : 100.00000
│ │ └ 2c3. Brake Ducts : None
│ ├ 2d. Differential
│ │ ├ 2d1. Power : 0.00000
│ │ ├ 2d2. Coast : 0.30000
│ └ └ 2d3. Preload : 10.00000
├ 3. Suspensions 1
│ ├ 3a. Wheel Rate : 160000.00000
│ ├ 3b. Bump Stop Up
│ │ ├ 3b1. Range : 0.02748
│ │ └ 3b2. Rate : 500.00000
│ ├ 3c. Bump Stop Down
│ │ ├ 3c1. Range : 0.01252
```

```
│ │ └ 3c2. Rate : 300.00000
│ ├ 3d. Helper Rate : 0.00000
│ └ 3e. Helper Range : 0.00000
├ 3. Suspensions 2
│ ├ 3a. Wheel Rate : 160000.00000
│ ├ 3b. Bump Stop Up
│ │ ├ 3b1. Range : 0.02748
│ │ └ 3b2. Rate : 500.00000
│ ├ 3c. Bump Stop Down
│ │ ├ 3c1. Range : 0.01252
│ │ └ 3c2. Rate : 300.00000
│ ├ 3d. Helper Rate : 0.00000
│ └ 3e. Helper Range : 0.00000
├ 3. Suspensions 3
│ ├ 3a. Wheel Rate : 150000.00000
│ ├ 3b. Bump Stop Up
│ │ ├ 3b1. Range : 0.01958
│ │ └ 3b2. Rate : 300.00000
│ ├ 3c. Bump Stop Down
│ │ ├ 3c1. Range : 0.04042
│ │ └ 3c2. Rate : 300.00000
│ ├ 3d. Helper Rate : 0.00000
│ └ 3e. Helper Range : 0.00000
├ 3. Suspensions 4
│ ├ 3a. Wheel Rate : 150000.00000
│ ├ 3b. Bump Stop Up
│ │ ├ 3b1. Range : 0.01958
│ │ └ 3b2. Rate : 300.00000
│ ├ 3c. Bump Stop Down
│ │ ├ 3c1. Range : 0.04042
│ │ └ 3c2. Rate : 300.00000
│ ├ 3d. Helper Rate : 0.00000
│ └ 3e. Helper Range : 0.00000
├ 4. Dampers 1
│ ├ 4a. Slow Bump : 8.00000
│ ├ 4b. Fast Bump : 5000.00000
│ ├ 4c. Slow Rebound : 9.00000
│ └ 4e. Fast Rebound : 5000.00000
├ 4. Dampers 2
│ ├ 4a. Slow Bump : 8.00000
│ ├ 4b. Fast Bump : 5000.00000
│ ├ 4c. Slow Rebound : 9.00000
│ └ 4e. Fast Rebound : 5000.00000
├ 4. Dampers 3
│ ├ 4a. Slow Bump : 9.00000
│ ├ 4b. Fast Bump : 5000.00000
│ ├ 4c. Slow Rebound : 10.00000
│ └ 4e. Fast Rebound : 5000.00000
├ 4. Dampers 4
│ ├ 4a. Slow Bump : 9.00000
│ ├ 4b. Fast Bump : 5000.00000
│ ├ 4c. Slow Rebound : 10.00000
│ └ 4e. Fast Rebound : 5000.00000
├ 5. Alignements 1
│ ├ 5a. Pressure : 24.00000
│ ├ 5b. Camber : -3.50000
```

```
│ ├ 5c. Toe : -0.10000
│ ├ 5d. Caster : -0.07000
│ ├ 5e. Static Camber : -3.09347
│ ├ 5f. Toe Out Linear : 0.00055
│ └ 5g. Compound : 0.00000
├ 5. Alignements 2
│ ├ 5a. Pressure : 24.00000
│ ├ 5b. Camber : -3.50000
│ ├ 5c. Toe : -0.10000
│ ├ 5d. Caster : -0.07000
│ ├ 5e. Static Camber : -3.09532
│ ├ 5f. Toe Out Linear : 0.00056
│ └ 5g. Compound : 0.00000
├ 5. Alignements 3
│ ├ 5a. Pressure : 23.50000
│ ├ 5b. Camber : -3.00000
│ ├ 5c. Toe : 0.10000
│ ├ 5d. Caster : 0.00000
│ ├ 5e. Static Camber : -3.21468
│ ├ 5f. Toe Out Linear : -0.00023
│ └ 5g. Compound : 0.00000
├ 5. Alignements 4
│ ├ 5a. Pressure : 23.50000
│ ├ 5b. Camber : -3.00000
│ ├ 5c. Toe : 0.10000
│ ├ 5d. Caster : 0.00000
│ ├ 5e. Static Camber : -3.21772
│ ├ 5f. Toe Out Linear : -0.00023
│ └ 5g. Compound : 0.00000
├ 6. Electronics
│ ├ 6a. Tc1 : 3.00000
│ ├ 6b. Tc2 : 3.00000
│ ├ 6c. Abs : 3.00000
│ ├ 6d. Esc : 0.00000
│ ├ 6e. Ebb : 0.000
│ ├ 6f. Engine Map : 0.00000
│ ├ 6g. Telemetry Laps to Record : 5.00000
│ ├ 6h. Turbo Boost Lv : 1.00000
│ ├ 6i. Ers Deployment Map : 0.000
│ ├ 6j. Ers Recharge Lv : 0.000
│ └ 6k. Ers Heat Charging : 0.000
├ 7. Aero
│ ├ 7a. Collar Positions Mm 1 : 0.88076
│ ├ 7a. Collar Positions Mm 2 : 0.88185
│ ├ 7a. Collar Positions Mm 3 : 29.68013
│ ├ 7a. Collar Positions Mm 4 : 29.68015
│ ├ 7b. Front Target Height : 75.00000
│ ├ 7c. Rear Target Height : 90.00000
│ ├ 7d. Front Wing Angle : 0.00000
│ └ 7e. Rear Wing Angle : 12.00000
├ 8. Fuel Strategy
│ └ 8a. Fuel : 30.00000
├ 9. Final State Name :
ks_ferrari_488_challenge_evo_preset_f488ce_mech_1_preset_f488ce_visual_1
├ 10. Version : 0
├ 11. Is Setup Shared : false
```

6. Car Setup Limits [.carsetuplimits]

A. Description

I. General Description

The Car Setup Limits asset is the structural and regulatory counterpart to the vehicle configuration file (Car Setup). While a setup file dictates the specific mechanical settings active on a vehicle at any given moment, the Car Setup Limits file defines the rigid boundaries, rules, and user interface properties governing those adjustments.

It serves as the master blueprint for the garage interface and simulation engine. It specifies which mechanical components can be altered by the user, the absolute minimum and maximum allowable physics values, the precise increments (steps) for tuning, and how data fields should be formatted, displayed, converted, or masked within the user interface.

II. Area of Influence / Impact on Vehicle Dynamics

The parameters configured within the Car Setup Limits file act as a critical control filter over a vehicle's performance envelope and user accessibility:

- **Preservation of Vehicle Identity (Road vs. Race):** It prevents unrealistic modifications, ensuring a standard production street car cannot be fitted with fully adjustable racing spoilers, bespoke motorsport suspension systems, or pure open-wheel steering ratios

- **Exploration of the Performance Envelope:** It defines how far a race engineer can push tuning boundaries. If the minimum front ride height threshold is restricted too high, the vehicle will be fundamentally blocked from exploiting ground-effect underbody aerodynamics.

- **Garage User Interface and User Experience:** By setting precise incremental adjustments, it determines whether a value changes smoothly millimeter by millimeter or clicks through coarse mechanical steps. It also translates raw backend physics numbers into legible cockpit metrics (e.g., clicks, degrees, bars, or PSI).

- **Performance Balancing & Fair Play (BOP):** In competitive multiplayer and esports environments, locking down or narrowing these limits is the primary method used to standardize hardware components across different vehicle models, ensuring a balanced grid without rewriting core chassis physics meshes..

III. Key Architecture & Data Fields Explained

The schema for this asset mirrors the organizational layout of the standard setup file (Tyres, Aerodynamics, Suspension, Dampers, Drivetrain). However, instead of storing a single static setting, every individual parameter is transformed into a **nested object** containing regulatory and interface variables:

1 - THE ANATOMY OF A PARAMETER LIMIT OBJECT

Every adjustable mechanical setting utilizes the following structural parameters:

- **Min / Max:** The absolute lower and upper physical boundaries allowed by the simulation engine for this component.
- **Step:** The exact mechanical increment or decrement applied per single "click" inside the garage menu.
- **Is Modifiable:** A boolean flag (TRUE/FALSE). When set to FALSE, the adjustment slider is locked and greyed out in the user interface.
- **Hide Value:** A toggle allowing developers to obscure the raw physical value from the user screen (often used in high-level motorsport to simulate proprietary team data secrecy).
- **Unit:** The textual unit label rendered on screen to contextualize the value (e.g., MM, NM, HZ, PSI, DEG).
- **Fractional Digits:** Specifies the decimal point precision displayed on the user interface screen.
- **Lut (Look-Up Table):** Maps the adjustment steps to an external curve, allowing linear menu clicks to translate into non-linear physical changes.

2 - CORE CATEGORIES OF APPLICATION

- **Tyres:** Enforces minimum safe cold inflation pressures to prevent tire carcass deflation/delamination, alongside maximum ranges for wheel alignment geometry (camber and toe).
- **Aerodynamics:** Locks down the operational angles of front splitters and rear wings (often constrained to 1-degree or 0.5-degree steps) and limits the safe compression clearance of the underbody floor.
- **Suspension Geometry & Rates:** Dictates the stiffest and softest allowable coil spring options (WHEEL RATE) and limits the structural thickness options for front and rear anti-roll bars (ARBS).
- **Dampers:** Defines the total number of valving adjustments ("clics") available for shock absorber damping across low-speed and high-speed compression (BUMP) and extension (REBOUND) cycles.
- **Drivetrain & Differential:** Sets boundaries on differential clutch locking percentages under acceleration (POWER), deceleration (COAST), and static resistance (PRELOAD).

IV. Interpretation of Setup Configuration Strategies

By reading a vehicle's specific limits file, developers and players can immediately identify its structural classification and engineering intent:

- **The Stock Production Profile:** In standard road vehicles, the vast majority of IS MODIFIABLE flags are hard-locked to FALSE. Parameters like spring rates, dampers, differential behaviors, and aerodynamics are frozen to factory settings. Only basic tyre pressures and minor alignment tolerances remain adjustable for track days.

- **The Pure Motorsport Profile (GT3 / Prototypes / Open-Wheel):** Almost all mechanical and aerodynamic flags are set to TRUE. However, the window between MIN and MAX is deliberately narrow, engineered strictly to keep the car operating within its designed aerodynamic or suspension kinematic sweet spot. Increments match real-world race hardware (e.g., fixed damper click values).

- **The BOP (Balance of Performance) Regulation Strategy:** Rather than altering a vehicle's engine power or weight mesh directly, series organizers can adjust this file to restrict tuning freedom. For instance, raising the MIN value of a rear wing angle forces a car to run higher minimum downforce—intentionally increasing straight-line drag to slow it down on high-speed circuits.

B. Schema

```
├ 1. Car Data : string – path
├ 2. Mechanical Balance : object
│ ├ 2a. Arbs [x] : object | can have multiple Arbs
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string – path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 2b. Steer Ratio : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string – path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 2c. Brakes : object
│ │ ├ 2c1. Front Bias : object
│ │ │ ├ 2a1. Step : float
│ │ │ ├ 2a2. Min : float
│ │ │ ├ 2a3. Max : float
│ │ │ ├ 2a4. Lut : string – path
│ │ │ ├ 2a5. Is Modifiable : boolean
│ │ │ ├ 2a6. Hide Value : boolean
│ │ │ ├ 2a7. Is Negative : boolean
│ │ │ ├ 2a8. Unit : string
│ │ │ ├ 2a9. Fractional Digits : integer
│ │ │ └ 2a10. Treat As Boolean : boolean
│ │ ├ 2c2. Torque Multiplier : object
```

```
│   │   │   ├ 2a1. Step : float
│   │   │   ├ 2a2. Min : float
│   │   │   ├ 2a3. Max : float
│   │   │   ├ 2a4. Lut : string - path
│   │   │   ├ 2a5. Is Modifiable : boolean
│   │   │   ├ 2a6. Hide Value : boolean
│   │   │   ├ 2a7. Is Negative : boolean
│   │   │   ├ 2a8. Unit : string
│   │   │   ├ 2a9. Fractional Digits : integer
│   │   │   └ 2a10. Treat As Boolean : boolean
│   │   ├ 2c3. Brake Ducts [x] : object | can have multiple Brake Ducts
│   │   │   ├ 2a1. Step : float
│   │   │   ├ 2a2. Min : float
│   │   │   ├ 2a3. Max : float
│   │   │   ├ 2a4. Lut : string - path
│   │   │   ├ 2a5. Is Modifiable : boolean
│   │   │   ├ 2a6. Hide Value : boolean
│   │   │   ├ 2a7. Is Negative : boolean
│   │   │   ├ 2a8. Unit : string
│   │   │   ├ 2a9. Fractional Digits : integer
│   │   └   └ 2a10. Treat As Boolean : boolean
│   ├ 2d. Differential : object
│   │   ├ 2d1. Power : object
│   │   │   ├ 2a1. Step : float
│   │   │   ├ 2a2. Min : float
│   │   │   ├ 2a3. Max : float
│   │   │   ├ 2a4. Lut : string - path
│   │   │   ├ 2a5. Is Modifiable : boolean
│   │   │   ├ 2a6. Hide Value : boolean
│   │   │   ├ 2a7. Is Negative : boolean
│   │   │   ├ 2a8. Unit : string
│   │   │   ├ 2a9. Fractional Digits : integer
│   │   │   └ 2a10. Treat As Boolean : boolean
│   │   ├ 2d2. Coast : object
│   │   │   ├ 2a1. Step : float
│   │   │   ├ 2a2. Min : float
│   │   │   ├ 2a3. Max : float
│   │   │   ├ 2a4. Lut : string - path
│   │   │   ├ 2a5. Is Modifiable : boolean
│   │   │   ├ 2a6. Hide Value : boolean
│   │   │   ├ 2a7. Is Negative : boolean
│   │   │   ├ 2a8. Unit : string
│   │   │   ├ 2a9. Fractional Digits : integer
│   │   │   └ 2a10. Treat As Boolean : boolean
│   │   ├ 2d3. Preload : object
│   │   │   ├ 2a1. Step : float
│   │   │   ├ 2a2. Min : float
│   │   │   ├ 2a3. Max : float
│   │   │   ├ 2a4. Lut : string - path
│   │   │   ├ 2a5. Is Modifiable : boolean
│   │   │   ├ 2a6. Hide Value : boolean
│   │   │   ├ 2a7. Is Negative : boolean
│   │   │   ├ 2a8. Unit : string
│   │   │   ├ 2a9. Fractional Digits : integer
│   └   └   └ 2a10. Treat As Boolean : boolean
├ 3. Suspensions [x] : object | can have multiple Suspensions
```

```
│ ├ 3a. Wheel Rate : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string - path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 3b. Bump Stop Up : object
│ │ ├ 3b1. Range : object
│ │ │ ├ 2a1. Step : float
│ │ │ ├ 2a2. Min : float
│ │ │ ├ 2a3. Max : float
│ │ │ ├ 2a4. Lut : string - path
│ │ │ ├ 2a5. Is Modifiable : boolean
│ │ │ ├ 2a6. Hide Value : boolean
│ │ │ ├ 2a7. Is Negative : boolean
│ │ │ ├ 2a8. Unit : string
│ │ │ ├ 2a9. Fractional Digits : integer
│ │ │ └ 2a10. Treat As Boolean : boolean
│ │ ├ 3b2. Rate : object
│ │ │ ├ 2a1. Step : float
│ │ │ ├ 2a2. Min : float
│ │ │ ├ 2a3. Max : float
│ │ │ ├ 2a4. Lut : string - path
│ │ │ ├ 2a5. Is Modifiable : boolean
│ │ │ ├ 2a6. Hide Value : boolean
│ │ │ ├ 2a7. Is Negative : boolean
│ │ │ ├ 2a8. Unit : string
│ │ │ ├ 2a9. Fractional Digits : integer
│ │ └ └ 2a10. Treat As Boolean : boolean
│ ├ 3c. Bump Stop Down : object
│ │ ├ 3b1. Range : object
│ │ │ ├ 2a1. Step : float
│ │ │ ├ 2a2. Min : float
│ │ │ ├ 2a3. Max : float
│ │ │ ├ 2a4. Lut : string - path
│ │ │ ├ 2a5. Is Modifiable : boolean
│ │ │ ├ 2a6. Hide Value : boolean
│ │ │ ├ 2a7. Is Negative : boolean
│ │ │ ├ 2a8. Unit : string
│ │ │ ├ 2a9. Fractional Digits : integer
│ │ │ └ 2a10. Treat As Boolean : boolean
│ │ ├ 3b2. Range : object
│ │ │ ├ 2a1. Step : float
│ │ │ ├ 2a2. Min : float
│ │ │ ├ 2a3. Max : float
│ │ │ ├ 2a4. Lut : string - path
│ │ │ ├ 2a5. Is Modifiable : boolean
│ │ │ ├ 2a6. Hide Value : boolean
│ │ │ ├ 2a7. Is Negative : boolean
│ │ │ ├ 2a8. Unit : string
│ │ │ ├ 2a9. Fractional Digits : integer
```

```
│ │ └ └ 2a10. Treat As Boolean : boolean
│ ├ 3d. Helper Rate : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string - path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 3e. Helper Range : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string - path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ │ └ 2a10. Treat As Boolean : boolean
├ 4. Dampers [x] : object | can have multiple Dampers
│ ├ 4a. Slow Bump : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string - path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 4b. Fast Bump : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string - path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 4c. Slow Rebound : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string - path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
```

```
│  │  └ 2a10. Treat As Boolean : boolean
│  ├ 4d. Fast Rebound : object
│  │  ├ 2a1. Step : float
│  │  ├ 2a2. Min : float
│  │  ├ 2a3. Max : float
│  │  ├ 2a4. Lut : string – path
│  │  ├ 2a5. Is Modifiable : boolean
│  │  ├ 2a6. Hide Value : boolean
│  │  ├ 2a7. Is Negative : boolean
│  │  ├ 2a8. Unit : string
│  │  ├ 2a9. Fractional Digits : integer
│  │  └ 2a10. Treat As Boolean : boolean
├ 5. Alignments [x] : object | can have multiple Alignments
│  ├ 5a. Pressure : object
│  │  ├ 2a1. Step : float
│  │  ├ 2a2. Min : float
│  │  ├ 2a3. Max : float
│  │  ├ 2a4. Lut : string – path
│  │  ├ 2a5. Is Modifiable : boolean
│  │  ├ 2a6. Hide Value : boolean
│  │  ├ 2a7. Is Negative : boolean
│  │  ├ 2a8. Unit : string
│  │  ├ 2a9. Fractional Digits : integer
│  │  └ 2a10. Treat As Boolean : boolean
│  ├ 5b. Camber : object
│  │  ├ 2a1. Step : float
│  │  ├ 2a2. Min : float
│  │  ├ 2a3. Max : float
│  │  ├ 2a4. Lut : string – path
│  │  ├ 2a5. Is Modifiable : boolean
│  │  ├ 2a6. Hide Value : boolean
│  │  ├ 2a7. Is Negative : boolean
│  │  ├ 2a8. Unit : string
│  │  ├ 2a9. Fractional Digits : integer
│  │  └ 2a10. Treat As Boolean : boolean
│  ├ 5c. Toe : object
│  │  ├ 2a1. Step : float
│  │  ├ 2a2. Min : float
│  │  ├ 2a3. Max : float
│  │  ├ 2a4. Lut : string – path
│  │  ├ 2a5. Is Modifiable : boolean
│  │  ├ 2a6. Hide Value : boolean
│  │  ├ 2a7. Is Negative : boolean
│  │  ├ 2a8. Unit : string
│  │  ├ 2a9. Fractional Digits : integer
│  │  └ 2a10. Treat As Boolean : boolean
│  ├ 5d. Caster : object
│  │  ├ 2a1. Step : float
│  │  ├ 2a2. Min : float
│  │  ├ 2a3. Max : float
│  │  ├ 2a4. Lut : string – path
│  │  ├ 2a5. Is Modifiable : boolean
│  │  ├ 2a6. Hide Value : boolean
│  │  ├ 2a7. Is Negative : boolean
│  │  ├ 2a8. Unit : string
│  │  ├ 2a9. Fractional Digits : integer
```

```
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 5e. Static Camber : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string – path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 5f. Toe Out Linear : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string – path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 5g. Compound : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string – path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ └ └ 2a10. Treat As Boolean : boolean
├ 6. Electronics : object
│ ├ 6a. Tc1 : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string – path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 6b. Tc2 : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string – path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
```

```
│  │  └ 2a10. Treat As Boolean : boolean
│  ├ 6c. Abs : object
│  │  ├ 2a1. Step : float
│  │  ├ 2a2. Min : float
│  │  ├ 2a3. Max : float
│  │  ├ 2a4. Lut : string – path
│  │  ├ 2a5. Is Modifiable : boolean
│  │  ├ 2a6. Hide Value : boolean
│  │  ├ 2a7. Is Negative : boolean
│  │  ├ 2a8. Unit : string
│  │  ├ 2a9. Fractional Digits : integer
│  │  └ 2a10. Treat As Boolean : boolean
│  ├ 6d. Esc : object
│  │  ├ 2a1. Step : float
│  │  ├ 2a2. Min : float
│  │  ├ 2a3. Max : float
│  │  ├ 2a4. Lut : string – path
│  │  ├ 2a5. Is Modifiable : boolean
│  │  ├ 2a6. Hide Value : boolean
│  │  ├ 2a7. Is Negative : boolean
│  │  ├ 2a8. Unit : string
│  │  ├ 2a9. Fractional Digits : integer
│  │  └ 2a10. Treat As Boolean : boolean
│  ├ 6e. Ebb : object
│  │  ├ 2a1. Step : float
│  │  ├ 2a2. Min : float
│  │  ├ 2a3. Max : float
│  │  ├ 2a4. Lut : string – path
│  │  ├ 2a5. Is Modifiable : boolean
│  │  ├ 2a6. Hide Value : boolean
│  │  ├ 2a7. Is Negative : boolean
│  │  ├ 2a8. Unit : string
│  │  ├ 2a9. Fractional Digits : integer
│  │  └ 2a10. Treat As Boolean : boolean
│  ├ 6f. Engine Map : object
│  │  ├ 2a1. Step : float
│  │  ├ 2a2. Min : float
│  │  ├ 2a3. Max : float
│  │  ├ 2a4. Lut : string – path
│  │  ├ 2a5. Is Modifiable : boolean
│  │  ├ 2a6. Hide Value : boolean
│  │  ├ 2a7. Is Negative : boolean
│  │  ├ 2a8. Unit : string
│  │  ├ 2a9. Fractional Digits : integer
│  │  └ 2a10. Treat As Boolean : boolean
│  ├ 6g. Telemetry Laps To Record : object
│  │  ├ 2a1. Step : float
│  │  ├ 2a2. Min : float
│  │  ├ 2a3. Max : float
│  │  ├ 2a4. Lut : string – path
│  │  ├ 2a5. Is Modifiable : boolean
│  │  ├ 2a6. Hide Value : boolean
│  │  ├ 2a7. Is Negative : boolean
│  │  ├ 2a8. Unit : string
│  │  ├ 2a9. Fractional Digits : integer
│  │  └ 2a10. Treat As Boolean : boolean
```

```
│  ├ 6h. Turbo Boost Lv : object
│  │ ├ 2a1. Step : float
│  │ ├ 2a2. Min : float
│  │ ├ 2a3. Max : float
│  │ ├ 2a4. Lut : string – path
│  │ ├ 2a5. Is Modifiable : boolean
│  │ ├ 2a6. Hide Value : boolean
│  │ ├ 2a7. Is Negative : boolean
│  │ ├ 2a8. Unit : string
│  │ ├ 2a9. Fractional Digits : integer
│  │ └ 2a10. Treat As Boolean : boolean
│  ├ 6i. Ers Deployment Map : object
│  │ ├ 2a1. Step : float
│  │ ├ 2a2. Min : float
│  │ ├ 2a3. Max : float
│  │ ├ 2a4. Lut : string – path
│  │ ├ 2a5. Is Modifiable : boolean
│  │ ├ 2a6. Hide Value : boolean
│  │ ├ 2a7. Is Negative : boolean
│  │ ├ 2a8. Unit : string
│  │ ├ 2a9. Fractional Digits : integer
│  │ └ 2a10. Treat As Boolean : boolean
│  ├ 6j. Ers Recharge Lv : object
│  │ ├ 2a1. Step : float
│  │ ├ 2a2. Min : float
│  │ ├ 2a3. Max : float
│  │ ├ 2a4. Lut : string – path
│  │ ├ 2a5. Is Modifiable : boolean
│  │ ├ 2a6. Hide Value : boolean
│  │ ├ 2a7. Is Negative : boolean
│  │ ├ 2a8. Unit : string
│  │ ├ 2a9. Fractional Digits : integer
│  │ └ 2a10. Treat As Boolean : boolean
│  ├ 6k. Ers Heat Charging : object
│  │ ├ 2a1. Step : float
│  │ ├ 2a2. Min : float
│  │ ├ 2a3. Max : float
│  │ ├ 2a4. Lut : string – path
│  │ ├ 2a5. Is Modifiable : boolean
│  │ ├ 2a6. Hide Value : boolean
│  │ ├ 2a7. Is Negative : boolean
│  │ ├ 2a8. Unit : string
│  │ ├ 2a9. Fractional Digits : integer
│  └ └ 2a10. Treat As Boolean : boolean
├ 7. Aero : object
│  ├ 7a. Collar Positions [x] : object | can have multiple Collar
Positions
│  │ ├ 2a1. Step : float
│  │ ├ 2a2. Min : float
│  │ ├ 2a3. Max : float
│  │ ├ 2a4. Lut : string – path
│  │ ├ 2a5. Is Modifiable : boolean
│  │ ├ 2a6. Hide Value : boolean
│  │ ├ 2a7. Is Negative : boolean
│  │ ├ 2a8. Unit : string
│  │ ├ 2a9. Fractional Digits : integer
```

```
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 7b. Front Target Height : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string – path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 7c. Rear Target Height : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string – path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 7d. Front Wing Angle : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string – path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ │ └ 2a10. Treat As Boolean : boolean
│ ├ 7e. Rear Wing Angle : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string – path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
│ └ └ 2a10. Treat As Boolean : boolean
├ 8. Fuel Strategy : object
│ ├ 7a. Fuel : object
│ │ ├ 2a1. Step : float
│ │ ├ 2a2. Min : float
│ │ ├ 2a3. Max : float
│ │ ├ 2a4. Lut : string – path
│ │ ├ 2a5. Is Modifiable : boolean
│ │ ├ 2a6. Hide Value : boolean
│ │ ├ 2a7. Is Negative : boolean
│ │ ├ 2a8. Unit : string
│ │ ├ 2a9. Fractional Digits : integer
```

```
│ └ └ 2a10. Treat As Boolean : boolean
└ 9. Use Single Compound : boolean
```

C. Example data

I. Chosen Car Engine for Example

- BMW M4 CSL (slug : ks_bmw_m4_csl)
- Lamborghini Countach (slug : ks_lamborghini_countach)

II. Example

BMW M4 CSL

```
├ 1. Car Data : None
├ 2. Mechanical Balance
│ ├ 2a. Arbs 1
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 30000.00000
│ │ ├ 2a3. Max : 30000.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
│ ├ 2a. Arbs 2
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 11000.00000
│ │ ├ 2a3. Max : 11000.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
│ ├ 2b. Steer Ratio
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 15.00000
│ │ ├ 2a3. Max : 15.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : None
│ │ └ 2a10. Treat As Boolean : 0
│ ├ 2c. Brakes
│ │ ├ 2c1. Front Bias
│ │ │ ├ 2a1. Step : 1.00000
```

```
│ │ │ ├ 2a2. Min : 0.00000
│ │ │ ├ 2a3. Max : 100.00000
│ │ │ ├ 2a4. Lut : None
│ │ │ ├ 2a5. Is Modifiable : false
│ │ │ ├ 2a6. Hide Value : false
│ │ │ ├ 2a7. Is Negative : false
│ │ │ ├ 2a8. Unit : None
│ │ │ ├ 2a9. Fractional Digits : None
│ │ │ └ 2a10. Treat As Boolean : false
│ │ ├ 2c2. Torque Multiplier
│ │ │ ├ 2a1. Step : 1.00000
│ │ │ ├ 2a2. Min : 100.00000
│ │ │ ├ 2a3. Max : 100.00000
│ │ │ ├ 2a4. Lut : None
│ │ │ ├ 2a5. Is Modifiable : false
│ │ │ ├ 2a6. Hide Value : false
│ │ │ ├ 2a7. Is Negative : false
│ │ │ ├ 2a8. Unit : None
│ │ │ ├ 2a9. Fractional Digits : 0
│ │ │ └ 2a10. Treat As Boolean : false
│ ├ 2d. Differential
│ │ ├ 2d1. Power
│ │ │ ├ 2a1. Step : 1.00000
│ │ │ ├ 2a2. Min : 0.25000
│ │ │ ├ 2a3. Max : 0.25000
│ │ │ ├ 2a4. Lut : None
│ │ │ ├ 2a5. Is Modifiable : false
│ │ │ ├ 2a6. Hide Value : false
│ │ │ ├ 2a7. Is Negative : false
│ │ │ ├ 2a8. Unit : None
│ │ │ ├ 2a9. Fractional Digits : 0
│ │ │ └ 2a10. Treat As Boolean : false
│ │ ├ 2d2. Coast
│ │ │ ├ 2a1. Step : 1.00000
│ │ │ ├ 2a2. Min : 0.55000
│ │ │ ├ 2a3. Max : 0.55000
│ │ │ ├ 2a4. Lut : None
│ │ │ ├ 2a5. Is Modifiable : false
│ │ │ ├ 2a6. Hide Value : false
│ │ │ ├ 2a7. Is Negative : false
│ │ │ ├ 2a8. Unit : None
│ │ │ ├ 2a9. Fractional Digits : 0
│ │ │ └ 2a10. Treat As Boolean : false
│ │ ├ 2d3. Preload
│ │ │ ├ 2a1. Step : 1.00000
│ │ │ ├ 2a2. Min : 1.00000
│ │ │ ├ 2a3. Max : 1.00000
│ │ │ ├ 2a4. Lut : None
│ │ │ ├ 2a5. Is Modifiable : false
│ │ │ ├ 2a6. Hide Value : false
│ │ │ ├ 2a7. Is Negative : false
│ │ │ ├ 2a8. Unit : None
│ │ │ ├ 2a9. Fractional Digits : 0
│ │ │ └ 2a10. Treat As Boolean : false
├ 3. Suspensions 1
│ ├ 3a. Wheel Rate
```

```
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 47000.00000
│ │ ├ 2a3. Max : 47000.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
│ ├ 3b. Bump Stop Up
│ │ ├ 3b1. Range
│ │ │ ├ 2a1. Step : 0.01000
│ │ │ ├ 2a2. Min : -0.09036
│ │ │ ├ 2a3. Max : 0.15964
│ │ │ ├ 2a4. Lut : None
│ │ │ ├ 2a5. Is Modifiable : false
│ │ │ ├ 2a6. Hide Value : false
│ │ │ ├ 2a7. Is Negative : false
│ │ │ ├ 2a8. Unit : None
│ │ │ ├ 2a9. Fractional Digits : 0
│ │ │ └ 2a10. Treat As Boolean : false
│ │ ├ 3b2. Rate
│ │ │ ├ 2a1. Step : 1.00000
│ │ │ ├ 2a2. Min : 600.00000
│ │ │ ├ 2a3. Max : 600.00000
│ │ │ ├ 2a4. Lut : None
│ │ │ ├ 2a5. Is Modifiable : false
│ │ │ ├ 2a6. Hide Value : false
│ │ │ ├ 2a7. Is Negative : false
│ │ │ ├ 2a8. Unit : None
│ │ │ ├ 2a9. Fractional Digits : 0
│ │ └ └ 2a10. Treat As Boolean : false
│ ├ 3c. Bump Stop Down
│ │ ├ 3b1. Range
│ │ │ ├ 2a1. Step : 1.00000
│ │ │ ├ 2a2. Min : 0.07036
│ │ │ ├ 2a3. Max : 0.07036
│ │ │ ├ 2a4. Lut : None
│ │ │ ├ 2a5. Is Modifiable : false
│ │ │ ├ 2a6. Hide Value : false
│ │ │ ├ 2a7. Is Negative : false
│ │ │ ├ 2a8. Unit : None
│ │ │ ├ 2a9. Fractional Digits : 0
│ │ │ └ 2a10. Treat As Boolean : false
│ │ ├ 3b2. Rate
│ │ │ ├ 2a1. Step : 1.00000
│ │ │ ├ 2a2. Min : 1000.00000
│ │ │ ├ 2a3. Max : 1000.00000
│ │ │ ├ 2a4. Lut : None
│ │ │ ├ 2a5. Is Modifiable : false
│ │ │ ├ 2a6. Hide Value : false
│ │ │ ├ 2a7. Is Negative : false
│ │ │ ├ 2a8. Unit : None
│ │ │ ├ 2a9. Fractional Digits : 0
│ │ └ └ 2a10. Treat As Boolean : false
```

```
│ ├ 3d. Helper Rate
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 5000.00000
│ │ ├ 2a3. Max : 5000.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
│ ├ 3e. Helper Range
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 0.03000
│ │ ├ 2a3. Max : 0.03000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
├ 3. Suspensions 2
│ ├ 3a. Wheel Rate
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 47000.00000
│ │ ├ 2a3. Max : 47000.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
│ ├ 3b. Bump Stop Up
│ │ ├ 3b1. Range
│ │ │ ├ 2a1. Step : 0.01000
│ │ │ ├ 2a2. Min : -0.09036
│ │ │ ├ 2a3. Max : 0.15964
│ │ │ ├ 2a4. Lut : None
│ │ │ ├ 2a5. Is Modifiable : false
│ │ │ ├ 2a6. Hide Value : false
│ │ │ ├ 2a7. Is Negative : false
│ │ │ ├ 2a8. Unit : None
│ │ │ ├ 2a9. Fractional Digits : 0
│ │ │ └ 2a10. Treat As Boolean : false
│ │ ├ 3b2. Rate
│ │ │ ├ 2a1. Step : 1.00000
│ │ │ ├ 2a2. Min : 600.00000
│ │ │ ├ 2a3. Max : 600.00000
│ │ │ ├ 2a4. Lut : None
│ │ │ ├ 2a5. Is Modifiable : false
│ │ │ ├ 2a6. Hide Value : false
│ │ │ ├ 2a7. Is Negative : false
│ │ │ ├ 2a8. Unit : None
│ │ │ ├ 2a9. Fractional Digits : 0
```

```
│  │  └ └ 2a10. Treat As Boolean : false
│  ├ 3c. Bump Stop Down
│  │  ├ 3b1. Range
│  │  │  ├ 2a1. Step : 1.00000
│  │  │  ├ 2a2. Min : 0.07036
│  │  │  ├ 2a3. Max : 0.07036
│  │  │  ├ 2a4. Lut : None
│  │  │  ├ 2a5. Is Modifiable : false
│  │  │  ├ 2a6. Hide Value : false
│  │  │  ├ 2a7. Is Negative : false
│  │  │  ├ 2a8. Unit : None
│  │  │  ├ 2a9. Fractional Digits : 0
│  │  │  └ 2a10. Treat As Boolean : false
│  │  ├ 3b2. Rate
│  │  │  ├ 2a1. Step : 1.00000
│  │  │  ├ 2a2. Min : 1000.00000
│  │  │  ├ 2a3. Max : 1000.00000
│  │  │  ├ 2a4. Lut : None
│  │  │  ├ 2a5. Is Modifiable : false
│  │  │  ├ 2a6. Hide Value : false
│  │  │  ├ 2a7. Is Negative : false
│  │  │  ├ 2a8. Unit : None
│  │  │  ├ 2a9. Fractional Digits : 0
│  │  └ └ 2a10. Treat As Boolean : false
│  ├ 3d. Helper Rate
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 5000.00000
│  │  ├ 2a3. Max : 5000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 3e. Helper Range
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.03000
│  │  ├ 2a3. Max : 0.03000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
├ 3. Suspensions 3
│  ├ 3a. Wheel Rate
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 33000.00000
│  │  ├ 2a3. Max : 33000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
```

```
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 3b. Bump Stop Up
│   │   ├ 3b1. Range
│   │   │   ├ 2a1. Step : 1.00000
│   │   │   ├ 2a2. Min : 0.13069
│   │   │   ├ 2a3. Max : 0.13069
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   │   └ 2a10. Treat As Boolean : false
│   │   ├ 3b2. Rate
│   │   │   ├ 2a1. Step : 1.00000
│   │   │   ├ 2a2. Min : 500.00000
│   │   │   ├ 2a3. Max : 500.00000
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   └ └ 2a10. Treat As Boolean : false
│   ├ 3c. Bump Stop Down
│   │   ├ 3b1. Range
│   │   │   ├ 2a1. Step : 1.00000
│   │   │   ├ 2a2. Min : 0.10431
│   │   │   ├ 2a3. Max : 0.10431
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   │   └ 2a10. Treat As Boolean : false
│   │   ├ 3b2. Rate
│   │   │   ├ 2a1. Step : 1.00000
│   │   │   ├ 2a2. Min : 1000.00000
│   │   │   ├ 2a3. Max : 1000.00000
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   └ └ 2a10. Treat As Boolean : false
│   ├ 3d. Helper Rate
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 5000.00000
│   │   ├ 2a3. Max : 5000.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
```

```
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 3e. Helper Range
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 0.03000
│   │   ├ 2a3. Max : 0.03000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
├ 3. Suspensions 4
│   ├ 3a. Wheel Rate
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 33000.00000
│   │   ├ 2a3. Max : 33000.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 3b. Bump Stop Up
│   │   ├ 3b1. Range
│   │   │   ├ 2a1. Step : 1.00000
│   │   │   ├ 2a2. Min : 0.13069
│   │   │   ├ 2a3. Max : 0.13069
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   │   └ 2a10. Treat As Boolean : false
│   │   ├ 3b2. Rate
│   │   │   ├ 2a1. Step : 1.00000
│   │   │   ├ 2a2. Min : 500.00000
│   │   │   ├ 2a3. Max : 500.00000
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   └ └ 2a10. Treat As Boolean : false
│   ├ 3c. Bump Stop Down
│   │   ├ 3b1. Range
│   │   │   ├ 2a1. Step : 1.00000
│   │   │   ├ 2a2. Min : 0.10431
│   │   │   ├ 2a3. Max : 0.10431
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
```

```
│  │  │  ├ 2a6. Hide Value : false
│  │  │  ├ 2a7. Is Negative : false
│  │  │  ├ 2a8. Unit : None
│  │  │  ├ 2a9. Fractional Digits : 0
│  │  │  └ 2a10. Treat As Boolean : false
│  │  ├ 3b2. Rate
│  │  │  ├ 2a1. Step : 1.00000
│  │  │  ├ 2a2. Min : 1000.00000
│  │  │  ├ 2a3. Max : 1000.00000
│  │  │  ├ 2a4. Lut : None
│  │  │  ├ 2a5. Is Modifiable : false
│  │  │  ├ 2a6. Hide Value : false
│  │  │  ├ 2a7. Is Negative : false
│  │  │  ├ 2a8. Unit : None
│  │  │  ├ 2a9. Fractional Digits : 0
│  │  └  └ 2a10. Treat As Boolean : false
│  ├ 3d. Helper Rate
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 5000.00000
│  │  ├ 2a3. Max : 5000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 3e. Helper Range
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.03000
│  │  ├ 2a3. Max : 0.03000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
├ 4. Dampers 1
│  ├ 4a. Slow Bump
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 5500.00000
│  │  ├ 2a3. Max : 5500.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4b. Fast Bump
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 1500.00000
│  │  ├ 2a3. Max : 1500.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
```

```
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
│ ├ 4c. Slow Rebound
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 7500.00000
│ │ ├ 2a3. Max : 7500.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
│ ├ 4d. Fast Rebound
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 4750.00000
│ │ ├ 2a3. Max : 4750.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
├ 4. Dampers 2
│ ├ 4a. Slow Bump
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 5500.00000
│ │ ├ 2a3. Max : 5500.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
│ ├ 4b. Fast Bump
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 1500.00000
│ │ ├ 2a3. Max : 1500.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
│ ├ 4c. Slow Rebound
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 7500.00000
│ │ ├ 2a3. Max : 7500.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
```

```
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4d. Fast Rebound
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 4750.00000
│  │  ├ 2a3. Max : 4750.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
├ 4. Dampers 3
│  ├ 4a. Slow Bump
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 4500.00000
│  │  ├ 2a3. Max : 4500.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4b. Fast Bump
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 2000.00000
│  │  ├ 2a3. Max : 2000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4c. Slow Rebound
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 6500.00000
│  │  ├ 2a3. Max : 6500.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4d. Fast Rebound
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 4250.00000
│  │  ├ 2a3. Max : 4250.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
```

```
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
├ 4. Dampers 4
│   ├ 4a. Slow Bump
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 4500.00000
│   │   ├ 2a3. Max : 4500.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 4b. Fast Bump
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 2000.00000
│   │   ├ 2a3. Max : 2000.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 4c. Slow Rebound
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 6500.00000
│   │   ├ 2a3. Max : 6500.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 4d. Fast Rebound
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 4250.00000
│   │   ├ 2a3. Max : 4250.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
├ 5. Alignments 1
│   ├ 5a. Pressure
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 20.00000
│   │   ├ 2a3. Max : 35.00000
│   │   ├ 2a4. Lut : None
```

```
│   │   ├ 2a5. Is Modifiable : true
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5b. Camber
│   │   ├ 2a1. Step : 0.10000
│   │   ├ 2a2. Min : -2.50000
│   │   ├ 2a3. Max : -1.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : true
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5c. Toe
│   │   ├ 2a1. Step : 0.01000
│   │   ├ 2a2. Min : -0.15000
│   │   ├ 2a3. Max : 0.20000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : true
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5d. Caster
│   │   ├ 2a1. Step : 0.01000
│   │   ├ 2a2. Min : -0.09195
│   │   ├ 2a3. Max : 0.02195
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5e. Static Camber
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : -1.50000
│   │   ├ 2a3. Max : -1.50000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5f. Toe Out Linear
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 0.00050
│   │   ├ 2a3. Max : 0.00050
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
```

```
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5g. Compound
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 2.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  └ └ 2a10. Treat As Boolean : false
├ 5. Alignments 2
│  ├ 5a. Pressure
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 20.00000
│  │  ├ 2a3. Max : 35.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5b. Camber
│  │  ├ 2a1. Step : 0.10000
│  │  ├ 2a2. Min : -2.50000
│  │  ├ 2a3. Max : -1.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5c. Toe
│  │  ├ 2a1. Step : 0.01000
│  │  ├ 2a2. Min : -0.15000
│  │  ├ 2a3. Max : 0.20000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5d. Caster
│  │  ├ 2a1. Step : 0.01000
│  │  ├ 2a2. Min : -0.09195
│  │  ├ 2a3. Max : 0.02195
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
```

```
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5e. Static Camber
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : -1.50000
│  │  ├ 2a3. Max : -1.50000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5f. Toe Out Linear
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00050
│  │  ├ 2a3. Max : 0.00050
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5g. Compound
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 2.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  └ └ 2a10. Treat As Boolean : false
├ 5. Alignments 3
│  ├ 5a. Pressure
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 20.00000
│  │  ├ 2a3. Max : 35.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5b. Camber
│  │  ├ 2a1. Step : 0.10000
│  │  ├ 2a2. Min : -3.00000
│  │  ├ 2a3. Max : -0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
```

```
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5c. Toe
│   │   ├ 2a1. Step : 0.01000
│   │   ├ 2a2. Min : -0.15000
│   │   ├ 2a3. Max : 0.20000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : true
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5d. Caster
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5e. Static Camber
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : -2.00000
│   │   ├ 2a3. Max : -2.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5f. Toe Out Linear
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 0.00020
│   │   ├ 2a3. Max : 0.00020
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5g. Compound
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 2.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : true
│   │   ├ 2a6. Hide Value : false
```

```
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  └  └ 2a10. Treat As Boolean : false
├ 5. Alignments 4
│  ├ 5a. Pressure
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 20.00000
│  │  ├ 2a3. Max : 35.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5b. Camber
│  │  ├ 2a1. Step : 0.10000
│  │  ├ 2a2. Min : -3.00000
│  │  ├ 2a3. Max : -0.50000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5c. Toe
│  │  ├ 2a1. Step : 0.01000
│  │  ├ 2a2. Min : -0.15000
│  │  ├ 2a3. Max : -0.20000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5d. Caster
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5e. Static Camber
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : -2.00000
│  │  ├ 2a3. Max : -2.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
```

```
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5f. Toe Out Linear
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00020
│  │  ├ 2a3. Max : 0.00020
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5g. Compound
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 2.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  └ └ 2a10. Treat As Boolean : false
├ 6. Electronics
│  ├ 6a. Tc1
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 10.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 6b. Tc2
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 6c. Abs
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
```

```
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 6d. Esc
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 2.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 6e. Ebb
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 6f. Engine Map
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 6g. Telemetry Laps To Record
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 99.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 6h. Turbo Boost Lv
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
```

```
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 6i. Ers Deployment Map
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 6j. Ers Recharge Lv
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 6k. Ers Heat Charging
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   └   └ 2a10. Treat As Boolean : false
├ 7. Aero
│   ├ 7a. Collar Positions 1
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 140.00000
│   │   ├ 2a3. Max : 140.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 7a. Collar Positions 2
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 140.00000
│   │   ├ 2a3. Max : 140.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
```

```
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 7a. Collar Positions 3
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 70.00000
│   │   ├ 2a3. Max : 70.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 7a. Collar Positions 4
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 70.00000
│   │   ├ 2a3. Max : 70.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 7b. Front Target Height
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 50.00000
│   │   ├ 2a3. Max : 200.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 7c. Rear Target Height
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 50.00000
│   │   ├ 2a3. Max : 200.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 7d. Front Wing Angle
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
```

```
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 7e. Rear Wing Angle
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
├ 8. Fuel Strategy
│   ├ 7a. Fuel
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 2.00000
│   │   ├ 2a3. Max : 59.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : true
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   └ └ 2a10. Treat As Boolean : false
└ 9. Use Single Compound : false
```

Lamborghini Countach

```
├ 1. Car Data : None
├ 2. Mechanical Balance
│   ├ 2a. Arbs 1
│   │   ├ 2a1. Step : 4000.00000
│   │   ├ 2a2. Min : 28000.00000
│   │   ├ 2a3. Max : 68000.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : true
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 2a. Arbs 2
│   │   ├ 2a1. Step : 5000.00000
│   │   ├ 2a2. Min : 5000.00000
│   │   ├ 2a3. Max : 50000.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : true
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
```

```
│  ├ 2b. Steer Ratio
│  │ ├ 2a1. Step : 1.45000
│  │ ├ 2a2. Min : -14.50000
│  │ ├ 2a3. Max : -14.50000
│  │ ├ 2a4. Lut : None
│  │ ├ 2a5. Is Modifiable : false
│  │ ├ 2a6. Hide Value : false
│  │ ├ 2a7. Is Negative : false
│  │ ├ 2a8. Unit : None
│  │ ├ 2a9. Fractional Digits : None
│  │ └ 2a10. Treat As Boolean : 0
│  ├ 2c. Brakes
│  │ ├ 2c1. Front Bias
│  │ │ ├ 2a1. Step : 1.00000
│  │ │ ├ 2a2. Min : 0.00000
│  │ │ ├ 2a3. Max : 100.00000
│  │ │ ├ 2a4. Lut : None
│  │ │ ├ 2a5. Is Modifiable : false
│  │ │ ├ 2a6. Hide Value : false
│  │ │ ├ 2a7. Is Negative : false
│  │ │ ├ 2a8. Unit : None
│  │ │ ├ 2a9. Fractional Digits : None
│  │ │ └ 2a10. Treat As Boolean : false
│  │ ├ 2c2. Torque Multiplier
│  │ │ ├ 2a1. Step : 10.00000
│  │ │ ├ 2a2. Min : 100.00000
│  │ │ ├ 2a3. Max : 100.00000
│  │ │ ├ 2a4. Lut : None
│  │ │ ├ 2a5. Is Modifiable : false
│  │ │ ├ 2a6. Hide Value : false
│  │ │ ├ 2a7. Is Negative : false
│  │ │ ├ 2a8. Unit : None
│  │ │ ├ 2a9. Fractional Digits : 0
│  │ │ └ 2a10. Treat As Boolean : false
│  ├ 2d. Differential
│  │ ├ 2d1. Power
│  │ │ ├ 2a1. Step : 0.02000
│  │ │ ├ 2a2. Min : 0.20000
│  │ │ ├ 2a3. Max : 0.20000
│  │ │ ├ 2a4. Lut : None
│  │ │ ├ 2a5. Is Modifiable : false
│  │ │ ├ 2a6. Hide Value : false
│  │ │ ├ 2a7. Is Negative : false
│  │ │ ├ 2a8. Unit : None
│  │ │ ├ 2a9. Fractional Digits : 0
│  │ │ └ 2a10. Treat As Boolean : false
│  │ ├ 2d2. Coast
│  │ │ ├ 2a1. Step : 0.04000
│  │ │ ├ 2a2. Min : 0.40000
│  │ │ ├ 2a3. Max : 0.40000
│  │ │ ├ 2a4. Lut : None
│  │ │ ├ 2a5. Is Modifiable : false
│  │ │ ├ 2a6. Hide Value : false
│  │ │ ├ 2a7. Is Negative : false
│  │ │ ├ 2a8. Unit : None
│  │ │ ├ 2a9. Fractional Digits : 0
```

```
│   │   │   └ 2a10. Treat As Boolean : false
│   │   ├ 2d3. Preload
│   │   │   ├ 2a1. Step : 1.00000
│   │   │   ├ 2a2. Min : 10.00000
│   │   │   ├ 2a3. Max : 10.00000
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   │   └ 2a10. Treat As Boolean : false
├ 3. Suspensions 1
│   ├ 3a. Wheel Rate
│   │   ├ 2a1. Step : 4000.00000
│   │   ├ 2a2. Min : 50000.00000
│   │   ├ 2a3. Max : 100000.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 3b. Bump Stop Up
│   │   ├ 3b1. Range
│   │   │   ├ 2a1. Step : 0.01200
│   │   │   ├ 2a2. Min : 0.02933
│   │   │   ├ 2a3. Max : 0.02933
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   │   └ 2a10. Treat As Boolean : false
│   │   ├ 3b2. Rate
│   │   │   ├ 2a1. Step : 35.00000
│   │   │   ├ 2a2. Min : 350.00000
│   │   │   ├ 2a3. Max : 350.00000
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   └ └ 2a10. Treat As Boolean : false
│   ├ 3c. Bump Stop Down
│   │   ├ 3b1. Range
│   │   │   ├ 2a1. Step : 0.00150
│   │   │   ├ 2a2. Min : 0.10567
│   │   │   ├ 2a3. Max : 0.010567
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
```

```
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   │   └ 2a10. Treat As Boolean : false
│   │   ├ 3b2. Rate
│   │   │   ├ 2a1. Step : 35.00000
│   │   │   ├ 2a2. Min : 350.00000
│   │   │   ├ 2a3. Max : 350.00000
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   └ └ 2a10. Treat As Boolean : false
│   ├ 3d. Helper Rate
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 3e. Helper Range
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
├ 3. Suspensions 2
│   ├ 3a. Wheel Rate
│   │   ├ 2a1. Step : 4000.00000
│   │   ├ 2a2. Min : 50000.00000
│   │   ├ 2a3. Max : 100000.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 3b. Bump Stop Up
│   │   ├ 3b1. Range
│   │   │   ├ 2a1. Step : 0.01200
│   │   │   ├ 2a2. Min : 0.02933
│   │   │   ├ 2a3. Max : 0.02933
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
```

```
│  │  │  ├ 2a7. Is Negative : false
│  │  │  ├ 2a8. Unit : None
│  │  │  ├ 2a9. Fractional Digits : 0
│  │  │  └ 2a10. Treat As Boolean : false
│  │  ├ 3b2. Rate
│  │  │  ├ 2a1. Step : 35.00000
│  │  │  ├ 2a2. Min : 350.00000
│  │  │  ├ 2a3. Max : 350.00000
│  │  │  ├ 2a4. Lut : None
│  │  │  ├ 2a5. Is Modifiable : false
│  │  │  ├ 2a6. Hide Value : false
│  │  │  ├ 2a7. Is Negative : false
│  │  │  ├ 2a8. Unit : None
│  │  │  ├ 2a9. Fractional Digits : 0
│  │  └ └ 2a10. Treat As Boolean : false
│  ├ 3c. Bump Stop Down
│  │  ├ 3b1. Range
│  │  │  ├ 2a1. Step : 0.00150
│  │  │  ├ 2a2. Min : 0.10567
│  │  │  ├ 2a3. Max : 0.010567
│  │  │  ├ 2a4. Lut : None
│  │  │  ├ 2a5. Is Modifiable : false
│  │  │  ├ 2a6. Hide Value : false
│  │  │  ├ 2a7. Is Negative : false
│  │  │  ├ 2a8. Unit : None
│  │  │  ├ 2a9. Fractional Digits : 0
│  │  │  └ 2a10. Treat As Boolean : false
│  │  ├ 3b2. Rate
│  │  │  ├ 2a1. Step : 35.00000
│  │  │  ├ 2a2. Min : 350.00000
│  │  │  ├ 2a3. Max : 350.00000
│  │  │  ├ 2a4. Lut : None
│  │  │  ├ 2a5. Is Modifiable : false
│  │  │  ├ 2a6. Hide Value : false
│  │  │  ├ 2a7. Is Negative : false
│  │  │  ├ 2a8. Unit : None
│  │  │  ├ 2a9. Fractional Digits : 0
│  │  └ └ 2a10. Treat As Boolean : false
│  ├ 3d. Helper Rate
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 3e. Helper Range
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
```

```
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
├ 3. Suspensions 3
│   ├ 3a. Wheel Rate
│   │   ├ 2a1. Step : 4000.00000
│   │   ├ 2a2. Min : 50000.00000
│   │   ├ 2a3. Max : 100000.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 3b. Bump Stop Up
│   │   ├ 3b1. Range
│   │   │   ├ 2a1. Step : 0.01200
│   │   │   ├ 2a2. Min : 0.02911
│   │   │   ├ 2a3. Max : 0.02911
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   │   └ 2a10. Treat As Boolean : false
│   │   ├ 3b2. Rate
│   │   │   ├ 2a1. Step : 35.00000
│   │   │   ├ 2a2. Min : 350.00000
│   │   │   ├ 2a3. Max : 350.00000
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   └ └ 2a10. Treat As Boolean : false
│   ├ 3c. Bump Stop Down
│   │   ├ 3b1. Range
│   │   │   ├ 2a1. Step : 0.00150
│   │   │   ├ 2a2. Min : 0.10589
│   │   │   ├ 2a3. Max : 0.10589
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   │   └ 2a10. Treat As Boolean : false
│   │   ├ 3b2. Rate
│   │   │   ├ 2a1. Step : 35.00000
│   │   │   ├ 2a2. Min : 350.00000
│   │   │   ├ 2a3. Max : 350.00000
│   │   │   ├ 2a4. Lut : None
```

```
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   └ └ 2a10. Treat As Boolean : false
│   ├ 3d. Helper Rate
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 3e. Helper Range
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
├ 3. Suspensions 4
│   ├ 3a. Wheel Rate
│   │   ├ 2a1. Step : 4000.00000
│   │   ├ 2a2. Min : 50000.00000
│   │   ├ 2a3. Max : 100000.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 3b. Bump Stop Up
│   │   ├ 3b1. Range
│   │   │   ├ 2a1. Step : 0.01200
│   │   │   ├ 2a2. Min : 0.02911
│   │   │   ├ 2a3. Max : 0.02911
│   │   │   ├ 2a4. Lut : None
│   │   │   ├ 2a5. Is Modifiable : false
│   │   │   ├ 2a6. Hide Value : false
│   │   │   ├ 2a7. Is Negative : false
│   │   │   ├ 2a8. Unit : None
│   │   │   ├ 2a9. Fractional Digits : 0
│   │   │   └ 2a10. Treat As Boolean : false
│   │   ├ 3b2. Rate
│   │   │   ├ 2a1. Step : 35.00000
│   │   │   ├ 2a2. Min : 350.00000
│   │   │   ├ 2a3. Max : 350.00000
```

```
│  │  │  ├ 2a4. Lut : None
│  │  │  ├ 2a5. Is Modifiable : false
│  │  │  ├ 2a6. Hide Value : false
│  │  │  ├ 2a7. Is Negative : false
│  │  │  ├ 2a8. Unit : None
│  │  │  ├ 2a9. Fractional Digits : 0
│  │  └ └ 2a10. Treat As Boolean : false
│  ├ 3c. Bump Stop Down
│  │  ├ 3b1. Range
│  │  │  ├ 2a1. Step : 0.00150
│  │  │  ├ 2a2. Min : 0.10589
│  │  │  ├ 2a3. Max : 0.10589
│  │  │  ├ 2a4. Lut : None
│  │  │  ├ 2a5. Is Modifiable : false
│  │  │  ├ 2a6. Hide Value : false
│  │  │  ├ 2a7. Is Negative : false
│  │  │  ├ 2a8. Unit : None
│  │  │  ├ 2a9. Fractional Digits : 0
│  │  │  └ 2a10. Treat As Boolean : false
│  │  ├ 3b2. Rate
│  │  │  ├ 2a1. Step : 35.00000
│  │  │  ├ 2a2. Min : 350.00000
│  │  │  ├ 2a3. Max : 350.00000
│  │  │  ├ 2a4. Lut : None
│  │  │  ├ 2a5. Is Modifiable : false
│  │  │  ├ 2a6. Hide Value : false
│  │  │  ├ 2a7. Is Negative : false
│  │  │  ├ 2a8. Unit : None
│  │  │  ├ 2a9. Fractional Digits : 0
│  │  └ └ 2a10. Treat As Boolean : false
│  ├ 3d. Helper Rate
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 3e. Helper Range
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
├ 4. Dampers 1
│  ├ 4a. Slow Bump
│  │  ├ 2a1. Step : 1000.00000
│  │  ├ 2a2. Min : 10000.00000
```

```
│  │  ├ 2a3. Max : 10000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4b. Fast Bump
│  │  ├ 2a1. Step : 100.00000
│  │  ├ 2a2. Min : -2000.00000
│  │  ├ 2a3. Max : 2000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4c. Slow Rebound
│  │  ├ 2a1. Step : 1000.00000
│  │  ├ 2a2. Min : 10000.00000
│  │  ├ 2a3. Max : 10000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4d. Fast Rebound
│  │  ├ 2a1. Step : 100.00000
│  │  ├ 2a2. Min : -2000.00000
│  │  ├ 2a3. Max : 2000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
├ 4. Dampers 2
│  ├ 4a. Slow Bump
│  │  ├ 2a1. Step : 1000.00000
│  │  ├ 2a2. Min : 10000.00000
│  │  ├ 2a3. Max : 10000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4b. Fast Bump
│  │  ├ 2a1. Step : 100.00000
│  │  ├ 2a2. Min : -2000.00000
```

```
│  │  ├ 2a3. Max : 2000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4c. Slow Rebound
│  │  ├ 2a1. Step : 1000.00000
│  │  ├ 2a2. Min : 10000.00000
│  │  ├ 2a3. Max : 10000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4d. Fast Rebound
│  │  ├ 2a1. Step : 100.00000
│  │  ├ 2a2. Min : -2000.00000
│  │  ├ 2a3. Max : 2000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
├ 4. Dampers 3
│  ├ 4a. Slow Bump
│  │  ├ 2a1. Step : 1000.00000
│  │  ├ 2a2. Min : 10000.00000
│  │  ├ 2a3. Max : 10000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4b. Fast Bump
│  │  ├ 2a1. Step : 100.00000
│  │  ├ 2a2. Min : -2000.00000
│  │  ├ 2a3. Max : 2000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4c. Slow Rebound
│  │  ├ 2a1. Step : 1000.00000
│  │  ├ 2a2. Min : 10000.00000
```

```
│  │  ├ 2a3. Max : 10000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4d. Fast Rebound
│  │  ├ 2a1. Step : 100.00000
│  │  ├ 2a2. Min : -2000.00000
│  │  ├ 2a3. Max : 2000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
├ 4. Dampers 4
│  ├ 4a. Slow Bump
│  │  ├ 2a1. Step : 1000.00000
│  │  ├ 2a2. Min : 10000.00000
│  │  ├ 2a3. Max : 10000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4b. Fast Bump
│  │  ├ 2a1. Step : 100.00000
│  │  ├ 2a2. Min : -2000.00000
│  │  ├ 2a3. Max : 2000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4c. Slow Rebound
│  │  ├ 2a1. Step : 1000.00000
│  │  ├ 2a2. Min : 10000.00000
│  │  ├ 2a3. Max : 10000.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 4d. Fast Rebound
│  │  ├ 2a1. Step : 100.00000
│  │  ├ 2a2. Min : -2000.00000
```

```
│   │   ├ 2a3. Max : 2000.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
├ 5. Alignments 1
│   ├ 5a. Pressure
│   │   ├ 2a1. Step : 1.00000
│   │   ├ 2a2. Min : 20.00000
│   │   ├ 2a3. Max : 35.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : true
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : psi
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5b. Camber
│   │   ├ 2a1. Step : 0.10000
│   │   ├ 2a2. Min : -2.50000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : Deg
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5c. Toe
│   │   ├ 2a1. Step : 0.01000
│   │   ├ 2a2. Min : -0.20000
│   │   ├ 2a3. Max : 0.20000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : Deg
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5d. Caster
│   │   ├ 2a1. Step : 0.00166
│   │   ├ 2a2. Min : -0.02000
│   │   ├ 2a3. Max : -0.02000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5e. Static Camber
│   │   ├ 2a1. Step : 0.18000
│   │   ├ 2a2. Min : -1.80000
```

```
│  │  ├ 2a3. Max : -1.80000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5f. Toe Out Linear
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5g. Compound
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 1.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  └ └ 2a10. Treat As Boolean : false
├ 5. Alignments 2
│  ├ 5a. Pressure
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 20.00000
│  │  ├ 2a3. Max : 35.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : psi
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5b. Camber
│  │  ├ 2a1. Step : 0.10000
│  │  ├ 2a2. Min : -2.50000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : Deg
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5c. Toe
│  │  ├ 2a1. Step : 0.01000
│  │  ├ 2a2. Min : -0.20000
```

```
│  │  ├ 2a3. Max : 0.20000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : Deg
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5d. Caster
│  │  ├ 2a1. Step : 0.00166
│  │  ├ 2a2. Min : -0.02000
│  │  ├ 2a3. Max : -0.02000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5e. Static Camber
│  │  ├ 2a1. Step : 0.18000
│  │  ├ 2a2. Min : -1.80000
│  │  ├ 2a3. Max : -1.80000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5f. Toe Out Linear
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5g. Compound
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 1.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  └ └ 2a10. Treat As Boolean : false
├ 5. Alignments 3
│  ├ 5a. Pressure
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 20.00000
```

```
│   │   ├ 2a3. Max : 35.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : true
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : psi
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5b. Camber
│   │   ├ 2a1. Step : 0.10000
│   │   ├ 2a2. Min : -2.50000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : Deg
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5c. Toe
│   │   ├ 2a1. Step : 0.01000
│   │   ├ 2a2. Min : -0.20000
│   │   ├ 2a3. Max : 0.20000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : Deg
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5d. Caster
│   │   ├ 2a1. Step : 0.00166
│   │   ├ 2a2. Min : -0.02000
│   │   ├ 2a3. Max : -0.02000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5e. Static Camber
│   │   ├ 2a1. Step : 0.18000
│   │   ├ 2a2. Min : -1.80000
│   │   ├ 2a3. Max : -1.80000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 5f. Toe Out Linear
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
```

```
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
│ ├ 5g. Compound
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 0.00000
│ │ ├ 2a3. Max : 1.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : true
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : None
│ │ ├ 2a9. Fractional Digits : 0
│ └ └ 2a10. Treat As Boolean : false
├ 5. Alignments 4
│ ├ 5a. Pressure
│ │ ├ 2a1. Step : 1.00000
│ │ ├ 2a2. Min : 20.00000
│ │ ├ 2a3. Max : 35.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : true
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : psi
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
│ ├ 5b. Camber
│ │ ├ 2a1. Step : 0.10000
│ │ ├ 2a2. Min : -2.50000
│ │ ├ 2a3. Max : 0.00000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : Deg
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
│ ├ 5c. Toe
│ │ ├ 2a1. Step : 0.01000
│ │ ├ 2a2. Min : -0.20000
│ │ ├ 2a3. Max : 0.20000
│ │ ├ 2a4. Lut : None
│ │ ├ 2a5. Is Modifiable : false
│ │ ├ 2a6. Hide Value : false
│ │ ├ 2a7. Is Negative : false
│ │ ├ 2a8. Unit : Deg
│ │ ├ 2a9. Fractional Digits : 0
│ │ └ 2a10. Treat As Boolean : false
│ ├ 5d. Caster
│ │ ├ 2a1. Step : 0.00166
│ │ ├ 2a2. Min : -0.02000
│ │ ├ 2a3. Max : -0.02000
```

```
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5e. Static Camber
│  │  ├ 2a1. Step : 0.18000
│  │  ├ 2a2. Min : -1.80000
│  │  ├ 2a3. Max : -1.80000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5f. Toe Out Linear
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 5g. Compound
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 1.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  └ └ 2a10. Treat As Boolean : false
├ 6. Electronics
│  ├ 6a. Tc1
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 1.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 6b. Tc2
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
```

```
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 6c. Abs
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 6d. Esc
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 6e. Ebb
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 6f. Engine Map
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 6g. Telemetry Laps To Record
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 99.00000
│  │  ├ 2a4. Lut : None
```

```
│   │   ├ 2a5. Is Modifiable : true
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 6h. Turbo Boost Lv
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 6i. Ers Deployment Map
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 6j. Ers Recharge Lv
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   │   └ 2a10. Treat As Boolean : false
│   ├ 6k. Ers Heat Charging
│   │   ├ 2a1. Step : 0.00000
│   │   ├ 2a2. Min : 0.00000
│   │   ├ 2a3. Max : 0.00000
│   │   ├ 2a4. Lut : None
│   │   ├ 2a5. Is Modifiable : false
│   │   ├ 2a6. Hide Value : false
│   │   ├ 2a7. Is Negative : false
│   │   ├ 2a8. Unit : None
│   │   ├ 2a9. Fractional Digits : 0
│   └ └ 2a10. Treat As Boolean : false
├ 7. Aero
│   ├ 7a. Collar Positions 1
│   │   ├ 2a1. Step : 7.00000
│   │   ├ 2a2. Min : 70.00000
│   │   ├ 2a3. Max : 70.00000
│   │   ├ 2a4. Lut : None
```

```
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 7a. Collar Positions 2
│  │  ├ 2a1. Step : 7.00000
│  │  ├ 2a2. Min : 70.00000
│  │  ├ 2a3. Max : 70.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 7a. Collar Positions 3
│  │  ├ 2a1. Step : 5.00000
│  │  ├ 2a2. Min : 50.00000
│  │  ├ 2a3. Max : 50.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 7a. Collar Positions 4
│  │  ├ 2a1. Step : 5.00000
│  │  ├ 2a2. Min : 50.00000
│  │  ├ 2a3. Max : 50.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 7b. Front Target Height
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 20.00000
│  │  ├ 2a3. Max : 180.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 7c. Rear Target Height
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 20.00000
│  │  ├ 2a3. Max : 180.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
```

```
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 7d. Front Wing Angle
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
│  ├ 7e. Rear Wing Angle
│  │  ├ 2a1. Step : 0.00000
│  │  ├ 2a2. Min : 0.00000
│  │  ├ 2a3. Max : 0.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : false
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  │  └ 2a10. Treat As Boolean : false
├ 8. Fuel Strategy
│  ├ 7a. Fuel
│  │  ├ 2a1. Step : 1.00000
│  │  ├ 2a2. Min : 1.00000
│  │  ├ 2a3. Max : 120.00000
│  │  ├ 2a4. Lut : None
│  │  ├ 2a5. Is Modifiable : true
│  │  ├ 2a6. Hide Value : false
│  │  ├ 2a7. Is Negative : false
│  │  ├ 2a8. Unit : None
│  │  ├ 2a9. Fractional Digits : 0
│  └ └ 2a10. Treat As Boolean : false
└ 9. Use Single Compound : false
```

7. Car Setup Units [.carsetupunits]

A. Description

I. General Description

The **Car Setup Units** asset is the primary translation matrix that standardizes how physical metrics are rendered visually within the simulation's user interface. While other physics files handle raw mathematical processing (often utilizing raw SI metrics like Newtons, meters, or Kelvin internally), the CAR SETUP UNITS file dictates the exact localization string, data labels, and scaling rules for what the player sees on screen.

It ensures that complex mechanics—such as hydraulic pressure, kinetic stiffness, or geometric angles—are mapped out into readable, intuitive measurements that real-world mechanics and simulation drivers can interact with seamlessly across different regions.

II. Area of Influence / Impact on Vehicle Dynamics

While the CAR SETUP UNITS asset does not modify the vehicle's physics engine computations directly, its architecture heavily influences the configuration pipeline and user experience:

- **Tuning Accuracy and Readability:** It prevents telemetry translation errors. By mapping clear unit tokens (e.g., matching a dampening constant to a specific text string), it guides modders to configure logical tuning increments.
- **Localization & Community Usability:** It serves as the primary system allowing global simulation players to quickly comprehend the vehicle state. It dictates whether a driver works with Metric systems, Imperial defaults, or hardware-specific indicators (like "valving clicks").
- **Telemetry Data Alignment:** It simplifies the correlation between in-game setup adjustments and exported engineering telemetry graphs (e.g., matching MoTeC worksheets to the exact garage presentation units).

III. Key Architecture & Data Fields Explained

The file structure maps directly over the variable names found in the global tuning schemas. Instead of assigning thresholds or steps, it assigns text strings, dimensional values, and contextual formatting markers.

1 - WHEEL AND TYRE TELEMETRY ALIGNMENT

Every adjustable mechanical setting utilizes the following structural parameters:

- **Pressure Units:** Dictates the presentation of pneumatic pressure vectors. Typically configured to display as PSI (Pounds per Square Inch) or BAR, ensuring drivers can balance the thermal expanding footprint accurately.

- **Alignment Angles:** Sets the formatting label for spatial geometric alignment. Variables like Camber, Toe, and Caster are universally assigned the degrees symbol (°) or millimeter targets (MM) to track wheel placement relative to the contact patch.

2 - SUSPENSION COMPONENTS & KINEMATICS

- **Spring Stiffness / Wheel Rates:** Controls the representation of pure compression resistance. These strings translate internal structural forces into readable loads, standardizing on metrics like N/MM (Newtons per millimeter), KG/MM, or specialized frequency descriptors like HZ (Hertz)

- **Damper Damping Coefficients:** Maps out energy dissipation factors. Because raw internal forces use complex scientific units like NS/M (Newton-seconds per meter), this asset often simplifies the interface display to strings like NS/M or redirects the interface to parse numbers as absolute hardware steps labeled as CLICS or CLICKS.

3 - AERODYNAMICS & AEROSTATIC CLEARANCE

- **Ride Heights / Ground Clearance:** Dictates the unit configuration for the aerostatic floor platform tracking. It handles structural frame clearances, mapping them cleanly to localized metric length units, universally defined as MM (millimeters) to allow precise platform rake tracking.

- **Aero Wing Inclinometers:** Configures the visual representation of adjustable wing blades, setting the display string to absolute angular units (such as degrees °) or fixed geometric positions.

IV. Interpretation of Setup Configuration Strategies

When interpreting a vehicle's exported architecture, the units configuration immediately establishes the target simulation profile and intended engineering depth:

- **The Scientific Engineering Profile:** Characterized by pure, unmodified physical engineering notations. Damper configurations read strictly as NS/M, and spring configurations read as N/MM. This gives advanced race engineers absolute mathematical transparency when calculating real-time dynamic weight transfer or suspension frequencies.

- **The Driver-Centric / Click Profile:** Simplifies high-tier internal metrics to match active physical components. Instead of displaying a confusing velocity value like 1500 NS/M, the unit system references the hardware's detent steps, displaying simply as integers. This replicates a real racing driver counting physical clicks on an adjustable Ohlins or Penske shock absorber matrix while sitting in the pitlane.

B. Schema

```
├ 1. Mechanical Balance : object
│ ├ 1a. Arbs [x] : string | can have multiple Arbs
│ ├ 1b Steer Ratio : string
│ ├ 1c. Brakes : object
```

```
│  │  ├ 1c1. Front Bias : string
│  │  ├ 1c2. Torque Multiplier : string
│  │  └ 1c3. Brake Ducts [x] : string | can have multiple Brake Ducts
│  ├ 1d. Differential : object
│  │  ├ 1d1. Power : string
│  │  ├ 1d2. Coast : string
│  └ └ 1d3. Preload : string
├ 2. Suspensions [x] : object | can have multiple Suspensions
│  ├ 2a. Wheel Rate : string
│  ├ 2b. Bump Stop Up : object
│  │  ├ 2b1. Range : string
│  │  └ 2b2. Rate : string
│  ├ 2c. Bump Stop Down : object
│  │  ├ 2b1. Range : string
│  │  └ 2b2. Rate : string
│  ├ 2d. Helper Rate : string
│  └ 2e. Helper Range : string
├ 3. Dampers [x] : object | can have multiple Dampers
│  ├ 3a. Slow Bump : string
│  ├ 3b. Fast Bump : string
│  ├ 3c. Slow Rebound : string
│  └ 3d. Fast Rebound : string
├ 4. Alignments [x] : object | can have multiple Alignments
│  ├ 4a. Pressure : string
│  ├ 4b. Camber : string
│  ├ 4c. Toe : string
│  ├ 4d. Caster : string
│  ├ 4e. Static Camber : string
│  ├ 4f. Toe Out Camber : string
│  └ 4g. Compound : string
├ 5. Electronics : object
│  ├ 5a. Tc1 : string
│  ├ 5b. Tc2 : string
│  ├ 5c. Abs : string
│  ├ 5d. Esc : string
│  ├ 5e. Ebb : string
│  ├ 5f. Engine Map : string
│  ├ 5g. Telemetry Laps To Record : string
│  ├ 5h. Turbo Boost Lv : string
│  ├ 5i. Ers Deployment Map : string
│  ├ 5j. Ers Recharge Lv : string
│  └ 5k. Ers Heat Charging : string
├ 6. Aero : object
│  ├ 6a. Collar Positions [x] : string | can have multiple Collar
Positions
│  ├ 6b. Front Target Height : string
│  ├ 6c. Rear Target Height : string
│  ├ 6e. Front Wing Angle : string
│  └ 6f. Rear Wing Angle : string
├ 7. Fuel Strategy : object
│  └ 7a. Fuel : string
└ 8. Use Single Compound : string
```

C. Example data

I. Chosen Car Engine for Example

- Setup Units (slug : setup_units) [common_phsx]

II. Example

Setup Units

```
├ 1. Mechanical Balance
│ ├ 1a. Arbs 1 : N/m
│ ├ 1a. Arbs 2 : N/m
│ ├ 1b Steer Ratio : None
│ ├ 1c. Brakes
│ │ ├ 1c1. Front Bias : %
│ │ ├ 1c2. Torque Multiplier : %
│ │ └ 1c3. Brake Ducts : None
│ ├ 1d. Differential
│ │ ├ 1d1. Power : None
│ │ ├ 1d2. Coast : None
│ └ └ 1d3. Preload : Nm
├ 2. Suspensions 1
│ ├ 2a. Wheel Rate : N/m
│ ├ 2b. Bump Stop Up
│ │ ├ 2b1. Range : m
│ │ └ 2b2. Rate : N
│ ├ 2c. Bump Stop Down
│ │ ├ 2b1. Range : m
│ │ └ 2b2. Rate : N
│ ├ 2d. Helper Rate : N/m
│ └ 2e. Helper Range : m
├ 2. Suspensions 2
│ ├ 2a. Wheel Rate : N/m
│ ├ 2b. Bump Stop Up
│ │ ├ 2b1. Range : m
│ │ └ 2b2. Rate : N
│ ├ 2c. Bump Stop Down
│ │ ├ 2b1. Range : m
│ │ └ 2b2. Rate : N
│ ├ 2d. Helper Rate : N/m
│ └ 2e. Helper Range : m
├ 2. Suspensions 3
│ ├ 2a. Wheel Rate : N/m
│ ├ 2b. Bump Stop Up
│ │ ├ 2b1. Range : m
│ │ └ 2b2. Rate : N
│ ├ 2c. Bump Stop Down
│ │ ├ 2b1. Range : m
│ │ └ 2b2. Rate : N
│ ├ 2d. Helper Rate : N/m
│ └ 2e. Helper Range : m
```

```
├ 2. Suspensions 4
│ ├ 2a. Wheel Rate : N/m
│ ├ 2b. Bump Stop Up
│ │ ├ 2b1. Range : m
│ │ └ 2b2. Rate : N
│ ├ 2c. Bump Stop Down
│ │ ├ 2b1. Range : m
│ │ └ 2b2. Rate : N
│ ├ 2d. Helper Rate : N/m
│ └ 2e. Helper Range : m
├ 2. Suspensions 5
│ ├ 2a. Wheel Rate : N/m
│ ├ 2b. Bump Stop Up
│ │ ├ 2b1. Range : m
│ │ └ 2b2. Rate : N
│ ├ 2c. Bump Stop Down
│ │ ├ 2b1. Range : m
│ │ └ 2b2. Rate : N
│ ├ 2d. Helper Rate : N/m
│ └ 2e. Helper Range : m
├ 2. Suspensions 6
│ ├ 2a. Wheel Rate : N/m
│ ├ 2b. Bump Stop Up
│ │ ├ 2b1. Range : m
│ │ └ 2b2. Rate : N
│ ├ 2c. Bump Stop Down
│ │ ├ 2b1. Range : m
│ │ └ 2b2. Rate : N
│ ├ 2d. Helper Rate : N/m
│ └ 2e. Helper Range : m
├ 3. Dampers 1
│ ├ 3a. Slow Bump : Ns/m
│ ├ 3b. Fast Bump : Ns/m
│ ├ 3c. Slow Rebound : Ns/m
│ └ 3d. Fast Rebound : Ns/m
├ 3. Dampers 2
│ ├ 3a. Slow Bump : Ns/m
│ ├ 3b. Fast Bump : Ns/m
│ ├ 3c. Slow Rebound : Ns/m
│ └ 3d. Fast Rebound : Ns/m
├ 3. Dampers 3
│ ├ 3a. Slow Bump : Ns/m
│ ├ 3b. Fast Bump : Ns/m
│ ├ 3c. Slow Rebound : Ns/m
│ └ 3d. Fast Rebound : Ns/m
├ 3. Dampers 4
│ ├ 3a. Slow Bump : Ns/m
│ ├ 3b. Fast Bump : Ns/m
│ ├ 3c. Slow Rebound : Ns/m
│ └ 3d. Fast Rebound : Ns/m
├ 3. Dampers 5
│ ├ 3a. Slow Bump : Ns/m
│ ├ 3b. Fast Bump : Ns/m
│ ├ 3c. Slow Rebound : Ns/m
│ └ 3d. Fast Rebound : Ns/m
├ 3. Dampers 6
```

```
│ ├ 3a. Slow Bump : Ns/m
│ ├ 3b. Fast Bump : Ns/m
│ ├ 3c. Slow Rebound : Ns/m
│ └ 3d. Fast Rebound : Ns/m
├ 4. Alignments 1
│ ├ 4a. Pressure : PSI
│ ├ 4b. Camber : °
│ ├ 4c. Toe : °
│ ├ 4d. Caster : None
│ ├ 4e. Static Camber : °
│ ├ 4f. Toe Out Camber : °
│ └ 4g. Compound : None
├ 4. Alignments 2
│ ├ 4a. Pressure : PSI
│ ├ 4b. Camber : °
│ ├ 4c. Toe : °
│ ├ 4d. Caster : None
│ ├ 4e. Static Camber : °
│ ├ 4f. Toe Out Camber : °
│ └ 4g. Compound : None
├ 4. Alignments 3
│ ├ 4a. Pressure : PSI
│ ├ 4b. Camber : °
│ ├ 4c. Toe : °
│ ├ 4d. Caster : None
│ ├ 4e. Static Camber : °
│ ├ 4f. Toe Out Camber : °
│ └ 4g. Compound : None
├ 4. Alignments 4
│ ├ 4a. Pressure : PSI
│ ├ 4b. Camber : °
│ ├ 4c. Toe : °
│ ├ 4d. Caster : None
│ ├ 4e. Static Camber : °
│ ├ 4f. Toe Out Camber : °
│ └ 4g. Compound : None
├ 5. Electronics
│ ├ 5a. Tc1 : None
│ ├ 5b. Tc2 : None
│ ├ 5c. Abs : None
│ ├ 5d. Esc : None
│ ├ 5e. Ebb : None
│ ├ 5f. Engine Map : None
│ ├ 5g. Telemetry Laps To Record : None
│ ├ 5h. Turbo Boost Lv : None
│ ├ 5i. Ers Deployment Map : None
│ ├ 5j. Ers Recharge Lv : None
│ └ 5k. Ers Heat Charging : None
├ 6. Aero
│ ├ 6a. Collar Positions 1 : m
│ ├ 6a. Collar Positions 2 : m
│ ├ 6a. Collar Positions 3 : m
│ ├ 6a. Collar Positions 4 : m
│ ├ 6b. Front Target Height : mm
│ ├ 6c. Rear Target Height : mm
│ ├ 6e. Front Wing Angle : °
```

```
│ └ 6f. Rear Wing Angle : °
├ 7. Fuel Strategy
│ └ 7a. Fuel : L
└ 8. Use Single Compound : None
```